

Accounting Textbook

Chapter Summaries

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Chapter 1: Financial Reporting and Accounting Standards

Executive Summary

Chapter 1 establishes why a single set of high-quality international accounting standards is essential in an increasingly globalized economy, focusing on IFRS as that standard. It traces the objective of financial reporting—to provide decision-useful information to investors and creditors—and identifies the IASB as the primary body responsible for developing and maintaining IFRS. The chapter concludes by acknowledging ongoing challenges: political pressures, the expectations gap, deficiencies in non-financial and forward-looking disclosures, ethical dilemmas, and the unfinished work of IFRS–U.S. GAAP convergence.

Key Topics

1. Global capital markets and IFRS adoption: Over 149 jurisdictions require or permit IFRS; 27,000 of 49,000 companies on the world's 88 largest exchanges use IFRS, enabling cross-border capital raising and investment comparability.
2. Objective of financial reporting: General-purpose financial statements must provide information useful to present and potential equity investors, lenders, and creditors in deciding whether to provide resources to an entity.
3. Financial statements vs. financial reporting: The four core statements are the statement of financial position, income statement (or statement of comprehensive income), statement of cash flows, and statement of changes in equity; broader financial reporting also includes prospectuses, news releases, and management forecasts.
4. Capital allocation role of accounting: Reliable, relevant, and timely financial information directs scarce capital toward productive enterprises, promoting economic efficiency and market liquidity.
5. Standard-setting structure—IASB and IOSCO: The IASB (14 full-time members, London-based) issues IFRS under an open due-process; IOSCO supports IFRS adoption globally without itself setting standards.
6. Hierarchy of IFRS: Companies apply (1) IFRS/IAS standards and IFRS Interpretations Committee interpretations, then (2) the Conceptual Framework, then (3) pronouncements of other bodies using a similar framework (e.g., U.S. GAAP).
7. Challenges in financial reporting: Key unresolved issues include political influence on standard-setting, the expectations gap between public perception and auditor capabilities, lack of non-financial and forward-looking disclosures, and ethical pressures on accountants.
8. IFRS vs. U.S. GAAP convergence: IFRS is principles-based and simpler; U.S. GAAP (FASB) is more rules-based and detailed. The two boards have aligned revenue recognition, financial instruments, and lease standards, but significant differences remain and full U.S. adoption of IFRS is considered politically dead.

Important Definitions

Term	Definition
International Financial Reporting Standards (IFRS)	Financial accounting standards issued by the IASB, used or permitted in over 149 jurisdictions, intended to serve as a single global set of high-quality reporting rules.

Term	Definition
International Accounting Standards Board (IASB)	The independent, London-based private-sector body of 14 full-time members that develops and issues IFRS in the public interest.
Financial Accounting	The process culminating in the preparation of financial reports on an enterprise for use by both internal and external parties, including investors, creditors, and regulators.
Managerial Accounting	The process of identifying, measuring, analyzing, and communicating financial information needed by management to plan, control, and evaluate a company's operations; not shared with external parties.
General-Purpose Financial Statements	Financial statements designed to meet the common information needs of a wide variety of users at the least cost, with investors and creditors as the primary user group.
Decision-Usefulness	The quality of financial information that enables investors and creditors to assess a company's ability to generate future net cash inflows and management's stewardship of economic resources.
Entity Perspective	The concept that a company is viewed as a separate and distinct economic unit from its owners; assets belong to the entity, not to individual shareholders or creditors.
Accrual-Basis Accounting	An accounting method that records revenues when earned and expenses when incurred, regardless of when cash changes hands, providing a more accurate picture of ongoing financial performance than cash-basis accounting.
Expectations Gap	The difference between what the public believes accountants should do and what accountants believe they can do, highlighted by corporate scandals such as Parmalat, Siemens, and Carillion.
Due Process (IASB)	The IASB's transparent, multi-step standard-setting procedure involving public agenda setting, research, preliminary views, exposure drafts, public comment periods, and re-deliberation before a final IFRS is issued.
Economic Consequences	The impact of accounting reports on the wealth positions of issuers and users of financial information and on the decisions and behaviors that follow from that impact.
IFRS Interpretations Committee	The IASB sub-body that resolves newly identified financial reporting issues and conflicting interpretations not specifically addressed by existing IFRS, acting as a 'problem filter' for the IASB.

Accounting Treatments

- Revenue recognition under accrual basis: Revenue is recognized when the entity satisfies its performance obligation (goods delivered or service performed), not when cash is received—matching economic activity to the period it occurs.
- Fair value measurement: IFRS requires or permits fair value for many financial assets (investments, derivatives); however, the 2016 financial instruments standard (IFRS 9) allows amortized cost for held-for-collection assets where the business model is to collect contractual cash flows.

- Hierarchy of authoritative guidance: When a specific IFRS addresses a transaction, it governs; if silent, the Conceptual Framework applies; if still silent, analogous guidance from other standard-setters (e.g., U.S. GAAP) may be used—but the overriding requirement is fair presentation.
- True and fair override: In rare cases where strict compliance with a standard would produce a misleading result, a company may depart from that standard, but must provide extensive disclosure explaining the rationale and the effect of the departure.
- IFRS for SMEs simplifications: Smaller entities may omit topics irrelevant to them, use simplified measurement approaches, and provide substantially fewer disclosures than required under full IFRS, while still applying the same core principles.

Critical Formulas / Calculations

No quantitative formulas are presented in Chapter 1. The chapter is conceptual and institutional in nature.

Capital Allocation Efficiency (conceptual): Reliable Information → Accurate Risk/Return Assessment → Efficient Resource Allocation → Productive Economy.
Example: If Nokia and McDonald's both report under IFRS, an investor can directly compare their return on assets and allocate capital to the higher-performing entity.

Accrual vs. Cash Basis (illustrative): $\text{Net Income (Accrual)} = \text{Revenues Earned} - \text{Expenses Incurred}$. Example: A company earns \$100,000 of services in December but collects cash in January; under accrual, \$100,000 appears in December revenue, not January.

Illustrative Example

Bloom NV (a hypothetical company) prepares financial statements 'in conformity with IFRS.' An auditor is asked which pronouncement applies to a lease transaction. Step 1: Check IFRS 16 Leases (an IASB standard)—it directly addresses the transaction, so it governs. Step 2: If IFRS 16 were silent on a sub-issue, the auditor would consult the Conceptual Framework for Financial Reporting. Step 3: If still unresolved, the auditor could reference analogous U.S. GAAP guidance (ASC 842), provided it does not conflict with existing IFRS. The overriding test is whether the resulting financial statements provide a fair presentation (true and fair view).

Disclosures / Presentation Notes

- Companies claiming IFRS compliance must apply all standards and interpretations in force; selective application is not permitted, and any departure requires a 'true and fair override' disclosure explaining the nature, reason, and financial effect of the departure.
- Note disclosures are an integral part of each of the four primary financial statements; they must include accounting policies, management judgements, and estimation uncertainties that are material to understanding the statements.
- IFRS encourages but does not mandate certain supplementary disclosures (e.g., non-financial key performance measures, forward-looking management commentary); however, regulatory requirements or company custom may require their inclusion in the annual report.
- Companies using XBRL (eXtensible Business Reporting Language) must tag their financial data consistently with the applicable taxonomy, enabling machine-readable, comparable cross-border analysis by investors and regulators.

Potential Pitfalls

- Confusing the Conceptual Framework with an IFRS standard: The Conceptual Framework is not itself an IFRS and does not override any specific standard; it is only applied when no relevant IFRS or interpretation exists, and any IFRS that conflicts with the Framework still takes precedence.
- Assuming IFRS is enforced by the IASB: The IASB has no direct enforcement mechanism—it is a private body. Enforcement is delegated to national or regional regulators (e.g., the EU requires all listed companies to use IFRS; the U.S. SEC has not mandated IFRS for domestic filers), meaning actual compliance varies by jurisdiction.
- Treating the 'objective of financial reporting' as management-focused: The primary user group is investors and creditors, not management, tax authorities, or regulators. Financial statements must be prepared from the entity perspective (company as distinct from its owners), not the proprietary perspective (focused solely on shareholders' interests).

Notes: The chapter text was complete in 1,282 lines with no truncation detected. Numerical data in Illustrations 1.1 (revenue figures for top 20 global companies) and 1.2 (exchange market capitalization and listing counts) were present as table headers and row labels but specific dollar values for most entries were not fully rendered in the extracted text—treat those figures as [VERIFY] against the original PDF. Illustration 1.4 (IASB organizational chart) and 1.5 (IASB due process steps) were referenced by caption only; their visual content was not captured in the text extraction.

Chapter 2: Conceptual Framework for Financial Reporting

Executive Summary

Chapter 2 establishes the IASB's Conceptual Framework for Financial Reporting as the theoretical foundation underlying all accounting standards, organized across three levels: the objective of financial reporting, the qualitative characteristics and elements of financial statements, and the recognition, measurement, and disclosure concepts. The framework concludes that useful financial information must be relevant and faithfully represented, and that companies apply accrual-basis accounting guided by the measurement, revenue recognition, expense recognition, and full disclosure principles to faithfully report economic reality.

Key Topics

1. **Purpose and Need for a Conceptual Framework:** A conceptual framework is a coherent system of concepts flowing from an objective; without it, standard-setting devolves into personal, inconsistent judgments. The IASB's 2018 Conceptual Framework for Financial Reporting comprises eight chapters covering the objective through capital maintenance.
2. **Objective of Financial Reporting:** The primary objective is to provide financial information useful to present and potential equity investors, lenders, and other creditors in making resource-allocation decisions. Management stewardship—how well management uses company resources—is also explicitly part of this objective under the 2018 update.
3. **Fundamental Qualitative Characteristics:** Relevance (predictive value, confirmatory value, materiality) and faithful representation (completeness, neutrality, freedom from error) are the two qualities that make information decision-useful. Neither characteristic alone is sufficient; both must be present for information to be genuinely useful.
4. **Enhancing Qualitative Characteristics:** Comparability (including consistency across periods), verifiability, timeliness, and understandability complement the fundamental qualities without substituting for them. The cost constraint—benefits must exceed costs—is the pervasive limitation on all reporting.
5. **Elements of Financial Statements:** The Conceptual Framework defines five elements: assets (present economic resource controlled as a result of past events), liabilities (present obligation to transfer an economic resource), equity (residual interest), income (increases in equity from non-owner sources), and expenses (decreases in equity from non-owner sources). These elements articulate across statements so that period-of-time elements drive changes in moment-in-time elements.
6. **Basic Accounting Assumptions:** Five assumptions underpin financial accounting: economic entity (company activity is separate from owners), going concern (entity will continue indefinitely), monetary unit (money is a stable common denominator), periodicity (activities can be divided into artificial time periods), and accrual basis (events are recorded when they occur, not when cash changes hands).
7. **Measurement Principles—Historical Cost vs. Current Value:** IFRS employs a mixed-attribute system; historical cost provides verifiability and faithfulness at acquisition, while current value bases (fair value, value in use/fulfillment value, current cost) provide greater relevance for certain assets and liabilities. Fair value is defined as the exit price in an orderly market transaction and is organized into a three-level hierarchy from quoted prices (Level 1) to unobservable inputs (Level 3).

8. Revenue and Expense Recognition Principles: Revenue is recognized when a performance obligation is satisfied, following a five-step model; expenses are recognized by matching efforts to revenues (product costs) or immediate period charge-off (period costs). The full disclosure principle requires that all information material to informed decision-making appear in the financial statements, notes, or supplementary information.

Important Definitions

Term	Definition
Conceptual Framework	A coherent system of interrelated concepts that flows from an objective and provides guidance on identifying, recognizing, measuring, and reporting financial information.
Relevance	The quality of financial information that is capable of making a difference in a decision because it has predictive value, confirmatory value, or both.
Materiality	A company-specific threshold of relevance whereby omitting or misstating an item could influence the decisions of users; assessed on both quantitative magnitude and qualitative nature.
Faithful Representation	The quality that ensures reported numbers and descriptions match what really existed or happened; requires completeness, neutrality, and freedom from material error.
Prudence	The exercise of caution when making judgments under uncertainty so that assets and income are not overstated and liabilities and expenses are not understated; reintroduced in the 2018 Conceptual Framework as supportive of neutrality.
Going Concern Assumption	The presumption that a company will continue to operate long enough to fulfill its objectives, justifying historical cost accounting and depreciation policies.
Accrual Basis of Accounting	The method under which transactions are recorded in the periods in which economic events occur, regardless of when cash is received or paid; required under IFRS.
Historical Cost Principle	The measurement approach that records assets and liabilities at the amount paid or agreed upon at the date of acquisition, providing a verifiable and faithful benchmark.
Fair Value	The price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date; a market-based, exit-value measure.
Performance Obligation	A promise in a contract with a customer to transfer a distinct good or service; revenue is recognized when (or as) this obligation is satisfied.
Full Disclosure Principle	The practice of providing all information of sufficient importance to influence the judgment of an informed user, presented in financial statements, notes, or supplementary disclosures.
Cost Constraint	The pervasive limitation that the costs of providing financial information must not exceed the benefits derived from its use.

Accounting Treatments

- Revenue recognition—performance obligation satisfied: Record revenue in the period the performance obligation is met, not when cash is received. Example journal entry when Klinke Cleaners performs service on June 30 but collects in July: Dr. Accounts Receivable / Cr. Service Revenue on June 30; Dr. Cash / Cr. Accounts Receivable in July.
- Expense recognition—matching principle (product costs): Costs directly tied to generating revenue (materials, labor, overhead) are capitalized and expensed in the same period as the related revenue, not when cash is paid. Example: Dr. Inventory (or Work-in-Process) / Cr. Cash or Payables when incurred; Dr. Cost of Goods Sold / Cr. Inventory when revenue is recognized.
- Expense recognition—period costs: Administrative costs with no direct revenue linkage (officers' salaries, general overhead) are expensed immediately in the period incurred. Example: Dr. Salaries Expense / Cr. Salaries Payable.
- Historical cost measurement of assets: Assets are recorded at acquisition cost and carried at that cost less accumulated depreciation/amortization; price-level changes are ignored under the monetary unit assumption. Example: Equipment purchased for \$500,000: Dr. Equipment \$500,000 / Cr. Cash \$500,000.
- Fair value re-measurement (current value): When IFRS requires or permits fair value, assets (e.g., financial instruments, investment properties) are re-measured at each reporting date. A decline in fair value: Dr. Impairment Loss / Cr. Asset (or Fair Value Adjustment), while an increase: Dr. Asset / Cr. Other Comprehensive Income or Gain.
- Full disclosure in notes: Items that meet element definitions but are not recognized (e.g., contingent liabilities due to measurement uncertainty) must be disclosed in the notes. No debit/credit entry is made for unrecognized contingencies; instead, qualitative and quantitative narrative disclosure is required.

Critical Formulas / Calculations

Accounting Equation: $\text{Assets} = \text{Liabilities} + \text{Equity}$. Variables: Assets = economic resources controlled; Liabilities = present obligations; Equity = residual interest. Numerical example: If Assets = \$500,000 and Liabilities = \$200,000, then Equity = \$300,000.

Materiality Rule of Thumb: $\text{Item Amount} / \text{Net Income} \geq 5\%$ suggests potential materiality. Variables: Item Amount = amount of unusual or separately disclosed item; Net Income = income from operations for the period. Numerical example (Illustration 2.3, Company B): $\$5,000 / \$10,000 = 50\%$, clearly material; Company A: $\$20,000 / \$1,000,000 = 2\%$, likely immaterial.

Revenue Recognition Amount (five-step model): $\text{Revenue Recognized} = \text{Transaction Price Allocated to Satisfied Performance Obligation(s)}$. Variables: Transaction Price = total consideration expected; Allocated portion = amount assigned to the specific obligation satisfied. Numerical example: Airbus signs a €100 million contract; upon delivery of aircraft satisfying one performance obligation, it recognizes €100 million (assuming a single obligation).

Depreciation—Straight-Line (referenced in Practice Problem): $\text{Annual Depreciation} = (\text{Cost} - \text{Residual Value}) / \text{Useful Life}$. Numerical example from Practice Problem: Warehouse cost £1,000,000, residual value £0, life 10 years: Annual depreciation = $\text{£1,000,000} / 10 = \text{£100,000}$ (controller improperly doubled this to £200,000 to reflect replacement cost, which is not permitted under historical cost principle).

Illustrative Example

Klinke Cleaners—Revenue Recognition (Chapter text, p. 1003–1007): Klinke performs dry-cleaning services on June 30 (satisfies the performance obligation) but does not collect cash until the first week of July. Under the accrual basis and revenue recognition principle, Klinke records on June 30: Dr. Accounts Receivable \$X / Cr. Service Revenue \$X. When cash is collected in July: Dr. Cash \$X / Cr. Accounts Receivable \$X. If Klinke had incorrectly used the cash basis, no revenue would appear in June, understating June income and overstating July income—a violation of IFRS. Additionally, the Practice Problem (Meadow plc, item 4) illustrates that land purchased for £320,000 and improved for £80,000 cannot recognize £180,000 of revenue based on an appraisal when no sale has occurred, because the performance obligation (transfer of land to buyer) has not been satisfied.

Disclosures / Presentation Notes

- Fair value hierarchy disclosure: Companies must disclose the fair value measurement amount and the hierarchy level (Level 1, 2, or 3) for each major group of assets and liabilities measured at fair value; Level 3 measurements require additional disclosure about the valuation process and significant unobservable inputs.
- Accounting policy changes (consistency): When a company switches accounting methods (e.g., inventory valuation from FIFO to average cost), the nature and effect of the change and its justification must be disclosed in the financial statements for the period of change; the auditor's report will also reference the change in an explanatory paragraph.
- Contingencies and uncertainties not recognized on the face of statements: Items that meet element definitions but cannot be reliably measured—or whose existence is uncertain—are disclosed in the notes to financial statements with qualitative description and, where possible, quantitative ranges.
- Supplementary information: Information of high relevance but lower reliability (e.g., oil and gas companies' proven reserve estimates and discounted cash flows) is presented as supplementary data outside the primary financial statements to provide additional decision-useful context without distorting recognized amounts.

Potential Pitfalls

- Confusing prudence with aggressive conservatism: Prudence under the 2018 Conceptual Framework means cautious judgment under uncertainty, not a license to systematically understate assets or overstate liabilities. Deliberate understatement in one period inflates income in future periods, violating neutrality and producing results that are neither prudent nor faithfully representative.
- Recognizing revenue or gains before a performance obligation is satisfied: As illustrated by the Meadow plc land example, companies sometimes recognize revenue based on appreciation in value (appraisal gains) rather than on transfer of the good or service. Under IFRS, revenue is only recognized when the performance obligation is satisfied—not when market value increases—so recording unrealized gains as revenue is a direct violation of the revenue recognition principle.
- Applying the cost constraint or materiality as a blanket excuse to omit information: Materiality is both quantitative and qualitative; items below a 5% net income threshold may still be material if they convert a loss to a profit, mask an illegal transaction, or affect management compensation. Similarly, the cost constraint does not justify omitting disclosures when the benefits to users clearly exceed preparation costs, and disclosure in notes is never a substitute for correct accounting treatment on the face of the statements.

Notes: The chapter text was complete and fully readable in a single pass (1,860 lines). No figures (Illustrations 2.1 through 2.7) or tables were rendered as images in the extracted text, so visual hierarchies (e.g., the Hierarchy of Accounting Qualities diagram and the Fair Value Hierarchy diagram) are described textually but cannot be reproduced numerically. Illustration 2.3 (Materiality Comparison) contained a partially extracted table; figures for Company A's revenue/expenses were not fully printed in the text extraction, so the example relies on the narrative description provided. Appendices and online practice materials referenced in the chapter were not included in the extracted file.

Chapter 3: The Accounting Information System

Executive Summary

Chapter 3 explains the complete accounting information system, walking through every step of the accounting cycle from identifying transactions to preparing closing entries. Using the service company Yazici Advertising A.Ş. and the merchandising company Ciner Cabinets as extended examples, the chapter demonstrates how double-entry bookkeeping, adjusting entries, and financial statement preparation work together under accrual-basis accounting in conformity with IFRS.

Key Topics

1. **Basic Terminology and Double-Entry System:** The chapter establishes 11 foundational terms—event, transaction, account, real/nominal accounts, ledger, journal, posting, trial balance, adjusting entries, financial statements, and closing entries—and explains that every transaction requires equal debits and credits, maintaining the accounting equation ($\text{Assets} = \text{Liabilities} + \text{Equity}$).
2. **The Accounting Cycle:** An eight-step sequential process—journalize, post, prepare unadjusted trial balance, adjust, prepare adjusted trial balance, prepare financial statements, close, and optionally prepare a post-closing trial balance—provides the structured workflow companies follow each fiscal period.
3. **Journalizing and Posting:** Transactions are first recorded chronologically in a general journal (or special journals) with dates, account titles, debit/credit amounts, and explanations, then transferred (posted) to the general ledger; a chart of accounts assigns numbered codes to every account.
4. **Adjusting Entries — Deferrals:** Prepaid expenses (e.g., supplies, insurance, depreciation) and unearned revenues require end-of-period adjustments that reduce overstated balance-sheet accounts and recognize the proper period's expense or revenue; e.g., Yazici adjusted supplies from 25,000 to 10,000, recognizing 15,000 of Supplies Expense.
5. **Adjusting Entries — Accruals:** Accrued revenues (services performed but not yet billed) and accrued expenses (expenses incurred but not yet paid, such as interest and salaries) require adjustments that increase both income-statement and balance-sheet accounts simultaneously; e.g., Yazici accrued 6,000 of unpaid salaries for three working days.
6. **Financial Statements and Closing Process:** Financial statements are prepared directly from the adjusted trial balance; revenue and expense (nominal/temporary) accounts are then closed to Income Summary, which is transferred to Retained Earnings, and Dividends is also closed to Retained Earnings, leaving only permanent (real) accounts with balances.
7. **Merchandising vs. Service Company Statements:** A merchandising company (e.g., Ciner Cabinets) must separately report net sales, cost of goods sold, and gross profit on sales, as well as selling and administrative expense categories, whereas a service company reports only service revenue and operating expenses.
8. **Cash Basis vs. Accrual Basis (Appendix 3A):** Cash-basis accounting records revenues when cash is received and expenses when cash is paid, violating the revenue recognition and expense recognition principles; it is not permitted under IFRS. The accrual basis is theoretically superior because receivables and payables forecast future cash flows.

Important Definitions

Term	Definition
Accounting Information System	A system that collects and processes transaction data and disseminates financial information to interested parties.
Double-Entry Accounting	A recording system in which every transaction is recorded with equal debits and credits, ensuring the accounting equation always remains in balance.
Adjusting Entries	Journal entries made at the end of an accounting period to update all accounts to an accrual basis so that correct financial statements can be prepared.
Deferrals	Adjustments for cash that was received or paid before the related revenue is earned or expense is incurred; subdivided into prepaid expenses and unearned revenues.
Accruals	Adjustments for revenues earned or expenses incurred before cash is received or paid; subdivided into accrued revenues and accrued expenses.
Contra Asset Account	An account that offsets a related asset account on the statement of financial position and carries a normal credit balance (e.g., Accumulated Depreciation, Allowance for Doubtful Accounts).
Book Value (Carrying Value)	The difference between an asset's original cost and its related accumulated depreciation; represents the net reported amount on the statement of financial position.
Closing Entries	Journal entries made at the end of a period that reduce all nominal (temporary) accounts to zero and transfer net income or net loss to Retained Earnings.
Nominal (Temporary) Accounts	Revenue, expense, and dividend accounts that are closed to zero at the end of each accounting period; they appear on the income statement.
Real (Permanent) Accounts	Asset, liability, and equity accounts that are carried forward from period to period and appear on the statement of financial position.
Trial Balance	A list of all open ledger accounts and their balances, used to prove the mathematical equality of debits and credits after posting.
Unearned Revenue	A liability representing cash received from customers before the related service has been performed; recognized as revenue when the performance obligation is satisfied.

Accounting Treatments

- **Prepaid Expense Adjustment:** When a prepaid asset (e.g., supplies, insurance) is consumed, debit the expense account and credit the asset account for the amount used. Example — Supplies Expense 15,000 / Supplies 15,000 (to record supplies consumed during October).
- **Depreciation Adjustment:** Debit Depreciation Expense and credit Accumulated Depreciation for the allocated portion of the asset's cost. Example — Yazici: Depreciation Expense 400 / Accumulated Depreciation—Equipment 400 (monthly allocation of $(50,000 - 2,000) / 10 \text{ years} / 12 \text{ months}$).
- **Unearned Revenue Adjustment:** When earned, debit the liability (Unearned Revenue) and credit the revenue account. Example — Unearned Service Revenue 4,000 / Service Revenue 4,000.

- **Accrued Expense Adjustment:** Debit the expense account and credit the related payable. Example — Interest Expense 500 / Interest Payable 500 ($50,000 \times 12\% \times 1/12$); Salaries and Wages Expense 6,000 / Salaries and Wages Payable 6,000.
- **Accrued Revenue Adjustment:** Debit the receivable and credit the revenue account. Example — Accounts Receivable 2,000 / Service Revenue 2,000 (for services performed but not yet billed).
- **Bad Debt Adjustment:** Debit Bad Debt Expense and credit Allowance for Doubtful Accounts for estimated uncollectible receivables. Example — Bad Debt Expense 1,600 / Allowance for Doubtful Accounts 1,600.

Critical Formulas / Calculations

Accounting Equation: Assets = Liabilities + Equity. All transactions must keep this equation in balance. Example — issuing shares for 40,000 cash: Cash +40,000 = Share Capital +40,000.

Interest Formula: Interest = Face Value $\times$ Annual Interest Rate $\times$ Time (fraction of year). Variables: Face Value = principal of the note; Annual Interest Rate = stated yearly rate; Time = portion of year outstanding. Example — $50,000 \times 12\% \times 1/12 = 500$ interest for one month.

Depreciation (Straight-Line): Annual Depreciation = (Cost - Residual Value) / Useful Life. Monthly = Annual / 12. Example — $(50,000 - 2,000) / 10 \text{ years} = 4,800$ per year; $4,800 / 12 = 400$ per month.

Cash-to-Accrual Revenue Conversion: Accrual Revenue = Cash Receipts - Beginning Accounts Receivable + Ending Accounts Receivable + Beginning Unearned Revenue - Ending Unearned Revenue. Example — $300,000 - 30,000 + 45,000 + 0 - 12,000 = 303,000$.

Cash-to-Accrual Expense Conversion: Accrual Expenses = Cash Paid + Beginning Prepaid - Ending Prepaid - Beginning Accrued Liabilities + Ending Accrued Liabilities. Example — $170,000 + 8,000 - 11,000 - 4,000 + 6,000 = 169,000$.

Illustrative Example

Yazici Advertising A.Ş. — October Adjusting Entries (partial): (1) Supplies purchased 25,000; inventory count shows 10,000 on hand; adjust: Dr Supplies Expense 15,000 / Cr Supplies 15,000. (2) Insurance paid 6,000 for 12 months; one month expires: Dr Insurance Expense 500 / Cr Prepaid Insurance 500. (3) Note payable 50,000 at 12% for 3 months; one month accrued: Dr Interest Expense 500 / Cr Interest Payable 500 ($50,000 \times 0.12 \times 1/12$). (4) Salaries paid through Oct 26; 3 days unpaid at 2,000/day: Dr Salaries and Wages Expense 6,000 / Cr Salaries and Wages Payable 6,000. After these entries the adjusted trial balance confirms total debits equal total credits, and financial statements are prepared directly from it.

Disclosures / Presentation Notes

- **Contra Accounts on the Statement of Financial Position:** Accumulated Depreciation is deducted from the related asset's cost to show the asset's carrying (book) value; Allowance for Doubtful Accounts is deducted from gross receivables to show the net realizable amount. Both contra balances and the original cost must be disclosed.
- **Classified Statement of Financial Position:** Assets are segregated into non-current (property, plant, and equipment) and current (inventory, prepaid items, receivables, cash); liabilities are split into non-current (e.g., bonds payable due beyond one year) and current (payables due within one year). Under IFRS, the statement is called the 'statement of financial position.'

- **Merchandising Income Statement Structure:** Under IFRS, a merchandising company's income statement must present net sales, cost of goods sold, gross profit, selling expenses, administrative expenses, income from operations, other revenues/gains, other expenses/losses, income before taxes, income tax expense, and net income as separate line items.
- **IFRS vs. U.S. GAAP Accounting Systems:** The double-entry system, debit/credit rules, journal, ledger, and chart of accounts are identical under IFRS and U.S. GAAP. A key difference is that IFRS allows more fair-value measurements than U.S. GAAP, which relies more heavily on historical cost. SOX internal-control attestation requirements apply to U.S.-listed companies but not universally to non-U.S. filers.

Potential Pitfalls

- **Confusing Debits/Credits with Increases/Decreases:** Debit means 'left side' and credit means 'right side' — they do not inherently mean increase or decrease. Asset and expense accounts increase with debits; liability, revenue, and equity accounts increase with credits. Misapplying this rule is the most common journalizing error.
- **Trial Balance Limitations — It Does Not Prove Accuracy:** A balanced trial balance only confirms that total debits equal total credits; it does not detect errors such as omitted journal entries, duplicate postings, wrong account used (e.g., debiting the wrong expense), or offsetting errors of equal amount. Students often assume a balanced trial balance means the books are error-free.
- **Closing Dividends Through Income Summary:** Dividends must be closed directly to Retained Earnings — never through Income Summary. Routing dividends through Income Summary would incorrectly reduce reported net income, because dividends are distributions of earnings, not operating expenses. A separate closing entry (Dr Retained Earnings / Cr Dividends) is required.

Notes: The chapter text was complete and read in full (lines 0–1999). Numerical values in several illustrations (e.g., the exact dollar amounts in Illustration 3.20 unadjusted trial balance, Illustration 3.41 Ciner income statement, and Illustration 3A.5 Dr. Liwan financial data) appear as labels only without their corresponding numeric cells, as the PDF-extracted text omitted the actual table figures. Appendix 3C worksheet illustration (Illustration 3C.1) figures were referenced but not reproduced in the text extraction. All conceptual content, formulas, and journal entry structures are fully captured.

Chapter 4: Income Statement and Related Information

Executive Summary

Chapter 4 examines the income statement — its purpose, format, content, and limitations — under IFRS, covering how revenues, expenses, gains, losses, and discontinued operations are classified and reported. The chapter also addresses accounting changes and error corrections, intraperiod tax allocation, and the related equity statements (retained earnings, comprehensive income, and statement of changes in equity). A central theme is that consistent, transparent income reporting is essential for investors and creditors to predict the amounts, timing, and uncertainty of future cash flows.

Key Topics

1. Uses and limitations of the income statement: The income statement helps predict future cash flows by revealing past performance trends, but is limited because companies omit unmeasurable items, income numbers vary with accounting method choices, and measurement involves significant judgment.
2. Income statement format and sections: A multi-step income statement typically presents gross profit, income from operations, income before income tax, and net income, with separate sections for financing costs, income tax, discontinued operations, non-controlling interest, and earnings per share.
3. Expense classification — nature vs. function: Companies may classify expenses by their nature (e.g., salaries, depreciation) or by their function (e.g., cost of goods sold, selling expenses); the function-of-expense method is most common in practice.
4. Gains and losses reporting: Under IFRS, all gains and losses — including write-downs, restructuring charges, and asset disposals — are reported within income from operations (often in 'Other income and expense'), rather than being excluded as non-recurring; material items are separately disclosed.
5. Discontinued operations: When a component of an entity is disposed of or classified as held-for-sale and represents a major line of business, the related results and disposal gain or loss are reported net of tax in a separate section after income from continuing operations.
6. Intraperiod tax allocation: Tax expense is allocated within a period to specific income statement categories — income from continuing operations and discontinued operations — so that each section reflects its own tax effect, following the principle 'let the tax follow the income.'
7. Accounting changes and error corrections: Changes in accounting principle and error corrections are accounted for retrospectively (restating prior-period statements and adjusting beginning retained earnings); changes in accounting estimates are handled prospectively.
8. Comprehensive income and equity statements: Comprehensive income includes net income plus other comprehensive income (OCI) items — such as unrealized gains/losses on non-trading equity securities — that bypass the income statement; companies must present OCI in either a one-statement or two-statement format, and also prepare a statement of changes in equity.

Important Definitions

Term	Definition
Income (IFRS)	Increases in assets, or decreases in liabilities, that result in increases in equity, other than those relating to contributions from holders of equity claims; includes both revenues and gains.
Expenses (IFRS)	Decreases in assets, or increases in liabilities, that result in decreases in equity, other than those relating to distributions to holders of equity claims; includes both expenses and losses.
Earnings per Share (EPS)	Net income minus preference dividends, divided by the weighted average number of ordinary shares outstanding during the period; must be disclosed on the face of the income statement.
Discontinued Operation	A component of an entity that has been disposed of or is classified as held-for-sale, and represents a major line of business or geographic area, or a subsidiary acquired exclusively for resale.
Intraperiod Tax Allocation	The allocation of income tax expense within a single fiscal period to the specific income statement categories — continuing operations and discontinued operations — that give rise to the tax.
Earnings Management	The planned timing of revenues, expenses, gains, and losses to smooth or manipulate reported income, typically to meet earnings targets or reduce perceived earnings risk.
Quality of Earnings	The degree to which reported earnings are sustainable, backed by actual cash flows, and free from distortions introduced by earnings management or aggressive accounting choices.
Comprehensive Income	All changes in equity during a period except those resulting from investments by owners and distributions to owners; equals net income plus other comprehensive income.
Other Comprehensive Income (OCI)	Non-owner changes in equity that bypass the income statement, such as unrealized gains and losses on non-trading equity securities, foreign currency translation adjustments, and revaluation surpluses.
Retrospective Adjustment	Recasting prior-period financial statements to reflect a newly adopted accounting principle or an error correction, with the cumulative effect recorded as an adjustment to beginning retained earnings.

Accounting Treatments

- Revenue and expense recognition (transaction approach): All revenues, expenses, gains, and losses are recognized on the income statement for the period in which they occur; the statement uses a multi-step format with subtotals for gross profit, income from operations, income before income tax, and net income.
- Discontinued operations — net-of-tax presentation: Results from operations of a discontinued component and any gain or loss on disposal are each reported net of their respective tax effects in a separate section of the income statement below 'Income from continuing operations.' Journal entries record the tax effect separately: e.g., Gain on Discontinued Operations DR, Income Tax Payable CR,

Net Gain on Discontinued Operations (net) shown on statement.

- Change in accounting principle — retrospective: The company restates prior-year comparative income statements under the new method and debits or credits beginning retained earnings for the cumulative after-tax effect of the change. Example entry: Retained Earnings DR (or CR) / Inventory (or related asset/liability) CR (or DR).
- Correction of prior-period error — retrospective: The error is corrected in the year of discovery by adjusting the beginning balance of retained earnings and restating prior comparative statements. Example: Retained Earnings DR \$X / Accounts Receivable CR \$X (for an overstated prior-year receivable).
- Change in accounting estimate — prospective: No adjustment to prior periods; the revised estimate is applied in the current and future periods and disclosed by nature of the change. Example: Bad Debt Expense DR \$240,000 / Allowance for Doubtful Accounts CR \$240,000.
- Other comprehensive income — equity recording: OCI items such as unrealized holding gains on non-trading equity securities are recorded directly to equity (bypassing net income): Unrealized Holding Gain DR / Accumulated Other Comprehensive Income (AOCI) CR, net of deferred tax.

Critical Formulas / Calculations

Earnings Per Share (EPS) = (Net Income - Preference Dividends) / Weighted Average Ordinary Shares Outstanding. Variables: Net Income = total profit for the period; Preference Dividends = dividends declared (or accrued if cumulative) on preference shares; Weighted Average Shares = shares weighted by portion of year outstanding. Example: $(\$350,000 - \$50,000) / 100,000 \text{ shares} = \3.00 per share .

Gross Profit = Net Sales - Cost of Goods Sold. Variables: Net Sales = gross sales revenue minus sales discounts and returns/allowances; Cost of Goods Sold = cost of inventory sold. Example: $\$500,000 \text{ net sales} - \$320,000 \text{ COGS} = \$180,000 \text{ gross profit}$.

Income from Operations = Gross Profit - Selling Expenses - Administrative Expenses $\pm$ Other Income and Expense. Example: $\$180,000 - \$60,000 - \$40,000 + \$5,000 \text{ gain} = \$85,000 \text{ income from operations}$.

Intraperiod Tax on Discontinued Operations = Gain (or Loss) on Discontinued Operations $\times$ Tax Rate. Example (Schindler AG): $\text{€}100,000 \text{ gain} \times 30\% = \text{€}30,000 \text{ tax}$; net gain reported = $\text{€}70,000$.

Comprehensive Income = Net Income $\pm$ Other Comprehensive Income Items (net of tax). Example (Gill SpA): $\text{Net income €}110,000 + \text{unrealized holding gain €}30,000 \text{ (net of tax)} = \text{Comprehensive Income €}140,000$.

Illustrative Example

Schindler AG intraperiod tax allocation (Illustrations 4.12 and 4.13): Income before income tax = $\text{€}250,000$; tax rate = 30%. Tax on continuing operations = $\text{€}250,000 \times 30\% = \text{€}75,000$; Income from continuing operations = $\text{€}175,000$. Discontinued operations gain of $\text{€}100,000 \rightarrow \text{tax} = \text{€}100,000 \times 30\% = \text{€}30,000$; net gain = $\text{€}70,000$. Net income = $\text{€}175,000 + \text{€}70,000 = \text{€}245,000$. Alternatively, with a $\text{€}100,000$ discontinued operations loss: tax benefit = $\text{€}30,000$; net loss = $\text{€}70,000$; Net income = $\text{€}175,000 - \text{€}70,000 = \text{€}105,000$.

Disclosures / Presentation Notes

- IFRS minimum income statement line items: Companies must present at minimum revenue, financing costs, share of profit/loss of associates accounted for using the equity method, income tax expense, post-tax results of discontinued operations, and profit or loss for the period, with net income allocated between controlling and non-controlling interests.
- Earnings per share on the face of the statement: EPS for income from continuing operations and net income must be disclosed on the face of the income statement; per share amounts for discontinued operations may be shown on the face or in the notes.
- Comprehensive income display — one or two statements: Companies must present OCI components in either (1) a single statement of comprehensive income (net income as a subtotal) or (2) two consecutive statements (income statement followed by a separate comprehensive income statement); EPS is not required for comprehensive income.
- Statement of changes in equity: Required under IFRS, this statement must disclose total comprehensive income for the period, contributions from and distributions to owners (share issuances and dividends), and a reconciliation of the opening to closing balance of each equity component including accumulated OCI.

Potential Pitfalls

- Confusing prospective vs. retrospective treatment: A common error is applying retrospective restatement to changes in accounting estimates (which should be prospective) or applying prospective treatment to changes in accounting principle or error corrections (which require retrospective restatement of prior-period statements and adjustment to beginning retained earnings).
- Misclassifying discontinued operations: A component qualifies as a discontinued operation only if it represents a major line of business or geographic area and is disposed of or classified as held-for-sale; disposing of individual brands or products that cannot be separately identified with distinct cash flows does NOT qualify, and reporting it as discontinued operations overstates the separation of operating from non-operating results.
- Omitting intraperiod tax allocation: Presenting discontinued operations or other separately reported items before tax (rather than net of tax) understates the transparency of the income statement; IFRS requires each separately reported category — continuing operations and discontinued operations — to carry its own tax effect so that users see after-tax impacts and can properly forecast future cash flows.

Notes: The chapter text was read in full (1,978 lines). Actual dollar/currency figures in the Boc Hong, Braun Company, and Gaubert Inc. illustrations were partially absent from the extracted text (the PDF extraction process rendered numeric columns as blank spaces), so specific monetary amounts in those illustrative tables could not be verified. Illustration numbers (4.1–4.24) and all narrative content are complete. No appendices were present in this chapter.

Chapter 5: Statement of Financial Position

Executive Summary

Chapter 5 examines two core financial statements required under IFRS: the statement of financial position (balance sheet) and the statement of cash flows. It explains how each statement is structured, how items are classified and valued, and how analysts use the data to assess liquidity, solvency, and financial flexibility. The chapter concludes with disclosure requirements, presentation techniques, ratio analysis fundamentals, and a comparison of IFRS versus U.S. GAAP reporting conventions.

Key Topics

1. Usefulness and limitations of the statement of financial position: The balance sheet helps users assess liquidity, solvency, and financial flexibility, but is limited by the use of historical cost, management estimates, and omission of internally generated intangibles such as brand value and human capital.
2. Classification of assets: Assets are divided into non-current (long-term investments, property/plant/equipment, intangibles, other assets) and current assets (cash equivalents, short-term investments, receivables, inventories, prepaid expenses), with each sub-category carrying a specific valuation basis.
3. Equity section structure: Shareholders' equity comprises share capital, share premium, retained earnings, accumulated other comprehensive income, treasury shares, and non-controlling interest—each component requiring separate disclosure.
4. Non-current and current liabilities: Non-current liabilities cover bonds, long-term notes, pension obligations, and deferred taxes; current liabilities are obligations expected to be settled within the operating cycle or one year, and their excess over current assets is the working capital buffer.
5. Statement of cash flows — purpose and structure: This statement classifies all cash receipts and payments into operating, investing, and financing activities, and reconciles beginning and ending cash balances; it fills the gap left by accrual-based statements.
6. Preparation of the statement of cash flows: Starting from comparative balance sheets and the income statement, analysts adjust net income for non-cash items and changes in working capital to derive operating cash flow, then separately identify investing and financing cash flows and significant non-cash transactions.
7. Cash flow analysis ratios and free cash flow: The current cash debt coverage (operating cash flow / average current liabilities), cash debt coverage (operating cash flow / average total liabilities), and free cash flow (operating cash flow minus capital expenditures and dividends) measure liquidity, solvency, and discretionary financial flexibility respectively.
8. Notes, disclosure techniques, and IFRS presentation guidelines: IFRS requires a Summary of Significant Accounting Policies as the first note; additional notes include maturity analyses, roll-forward schedules, and disaggregated line-item detail. Disclosure techniques include parenthetical explanations, cross-references, contra accounts, and adjunct accounts. IAS 1 generally prohibits offsetting assets against liabilities.

Important Definitions

Term	Definition
Statement of Financial Position	A financial statement that reports an entity's assets, liabilities, and equity at a specific point in time; also called the balance sheet.
Liquidity	The amount of time expected to elapse until an asset is converted to cash or a liability must be paid; a measure of a company's ability to meet short-term obligations.
Solvency	A company's ability to pay its debts as they mature, reflecting the long-term relationship between debt levels and total assets.
Financial Flexibility	A company's ability to alter the amounts and timing of cash flows to respond to unexpected needs and opportunities, usually assessed via free cash flow.
Working Capital	The excess of total current assets over total current liabilities; it represents the net liquid buffer available to fund the operating cycle.
Current Assets	Cash and other assets expected to be converted to cash, sold, or consumed within one year or the operating cycle, whichever is longer.
Operating Activities	Cash flows arising from transactions that enter into the determination of net income, such as cash receipts from customers and cash payments to suppliers.
Free Cash Flow	Net cash provided by operating activities minus capital expenditures minus cash dividends; represents discretionary cash available for investment, debt retirement, or share repurchases.
Contra Account	An account that reduces an associated asset, liability, or equity balance on the balance sheet (e.g., Accumulated Depreciation reduces Property, Plant and Equipment; Allowance for Doubtful Accounts reduces Receivables).
Accounting Policies	The specific principles, bases, conventions, rules, and practices applied by a company in preparing and presenting financial information, disclosed in the first note to the financial statements.

Accounting Treatments

- Property, plant, and equipment (PPE) is recorded at cost and subsequently depreciated (straight-line or reducing balance) over estimated useful life, except land which is not depreciated. Journal entry at acquisition: Dr PPE / Cr Cash (or Payables). Depreciation: Dr Depreciation Expense / Cr Accumulated Depreciation.
- Inventories are stated at the lower of cost or net realizable value (LCNRV), using FIFO or weighted-average cost flow assumptions. Write-downs: Dr Inventory Write-down Expense / Cr Inventory (or Allowance for Inventory Obsolescence).
- Held-for-collection debt securities are reported at amortized cost; trading securities (debt and equity) are reported at fair value through profit or loss; non-trading equity securities are reported at fair value, with changes recorded as required under IFRS 9.
- Receivables are presented net of an allowance for doubtful accounts. Entry for estimated bad debts: Dr Bad Debt Expense / Cr Allowance for Doubtful Accounts. Write-off: Dr Allowance / Cr Accounts Receivable.

- Non-cash investing and financing transactions (e.g., issuing shares to acquire assets, converting bonds to equity) are excluded from the body of the cash flow statement but disclosed in a separate note to satisfy the full disclosure principle.
- Current portion of long-term debt is reclassified from non-current to current liabilities at each balance sheet date, unless the entity has refinanced on a long-term basis before the reporting period end—in which case it remains non-current.

Critical Formulas / Calculations

Working Capital = Current Assets - Current Liabilities. Example: Current Assets = \$500,000; Current Liabilities = \$300,000 → Working Capital = \$200,000.

Current Cash Debt Coverage = Net Cash Provided by Operating Activities / Average Current Liabilities. Example: Operating CF = €1,200,000; Avg Current Liabilities = €1,000,000 → Ratio = 1.2 (strong liquidity).

Cash Debt Coverage = Net Cash Provided by Operating Activities / Average Total Liabilities. Example: Operating CF = €1,200,000; Avg Total Liabilities = €1,500,000 → Ratio = 0.80.

Free Cash Flow = Net Cash Provided by Operating Activities - Capital Expenditures - Cash Dividends. Example (Nestor SA): €411,750 - €252,500 - €48,000 = €111,250 in free cash flow.

Current Ratio = Current Assets / Current Liabilities. Example (Acer Group): NT\$186,391M / NT\$149,315M = 1.25 (current assets are 125% of current liabilities).

Altman Z-Score (abbreviated): $Z = (\text{Working Capital}/\text{Total Assets}) \times 1.2 + (\text{Retained Earnings}/\text{Total Assets}) \times 1.4 + (\text{EBIT}/\text{Total Assets}) \times 3.3 + (\text{Sales}/\text{Total Assets}) \times [\text{coefficient}]$. $Z > 3.0$ signals low bankruptcy risk; $Z < 1.81$ signals high risk.

Illustrative Example

Telemarketing Inc. (first year of operations, 2022): Issued 50,000 shares at \$1 par for \$50,000 cash; earned revenues generating \$39,000 net income (after tax); purchased land for \$15,000; paid \$14,000 dividends. Cash flow from operations = Net Income \$39,000 - Increase in A/R \$41,000 + Increase in A/P \$12,000 = \$10,000. Cash from investing = -\$15,000 (land purchase). Cash from financing = +\$50,000 (share issuance) - \$14,000 (dividends) = +\$36,000. Net increase in cash = \$10,000 - \$15,000 + \$36,000 = \$31,000, reconciling to the ending cash balance on the comparative balance sheet.

Disclosures / Presentation Notes

- IFRS (IAS 1) requires a complete set of financial statements annually with comparative prior-period information; the set must include the statement of financial position, statement of comprehensive income, statement of changes in equity, statement of cash flows, and notes including a Summary of Significant Accounting Policies.
- Specific note disclosures required for the balance sheet include: disaggregation of PPE into classes with accumulated depreciation roll-forwards; aging/maturity analyses for receivables and financial liabilities; disaggregation of inventories by stage; and reconciliation (roll-forward) schedules for PPE, intangibles, and provisions.
- Offsetting of assets and liabilities is generally prohibited under IFRS unless a legally enforceable right of offset exists and the entity intends to settle net or simultaneously; however, valuation allowances

(e.g., allowance for doubtful accounts, inventory impairment) are not considered offsetting.

- Companies must apply accounting policies consistently from period to period (comparability principle); significant changes require disclosure and retrospective application. Parenthetical explanations and cross-referencing on the face of the statement are acceptable supplemental disclosure techniques.

Potential Pitfalls

- Confusing historical cost with fair value: Most balance sheet assets and liabilities are carried at historical cost, not current market value. Analysts who treat book values as proxies for fair value—especially for PPE, intangibles, or investments—can seriously misjudge a company's actual economic position.
- Misreading positive net income as a sign of healthy cash flow: A company can report net income while experiencing negative operating cash flows if receivables or inventories grow rapidly. The W. T. Grant bankruptcy case illustrates how consistent profits masked a deteriorating cash position for years before failure.
- Overlooking off-balance-sheet items and omitted intangibles: Significant obligations (e.g., operating leases before IFRS 16 adoption, contingent liabilities) and valuable internally generated assets (brand, workforce, customer relationships) may not appear on the balance sheet, causing users to underestimate both risk and economic value.

Notes: The chapter text was fully extracted across 2,099 lines with no truncation detected. Some exhibits (e.g., Illustrations 5.2–5.16, 5.19–5.29, 5.36, and Appendix 5A ratio tables) contained partially extracted numerical data because the original PDF tables were rendered as text stubs without all cell values. Specific currency amounts in several real-company illustrations (Christian Dior, AEON, RELX Group, Ahold Delhaize, adidas, AB InBev, Silver Fern Farms, Siemens AG) were not fully populated in the extracted text. The Altman Z-Score formula coefficients for the Sales/Total Assets term were not completely captured [VERIFY]. Appendix 5A ratio formulae for ratios 9–16 were partially rendered; verify exact ratio denominators (e.g., Return on Ordinary Share Capital denominator) against the original textbook table.

Chapter 6: Accounting and the Time Value of Money

Executive Summary

Chapter 6 develops the foundational time value of money (TVM) concepts that underpin modern accounting measurement, covering simple and compound interest, future and present values of single sums and annuities (ordinary and due), deferred annuities, bond valuation, and IFRS 13's expected cash flow approach. The chapter demonstrates that because a dollar received today is worth more than a dollar received in the future, accountants must discount or accumulate cash flows to produce fair, decision-relevant measurements for notes, leases, pensions, impairments, and other long-lived assets and liabilities.

Key Topics

1. Time Value of Money Fundamentals: A dollar today is worth more than a future dollar because of the opportunity to invest and earn interest; TVM underpins fair value measurements (IFRS 13 Level 3) when market prices are unavailable.
2. Simple vs. Compound Interest: Simple interest is computed only on principal ($I = p \times i \times n$); compound interest accrues on both principal and previously earned interest, producing significantly larger accumulations over multiple periods.
3. Future and Present Value of a Single Sum: $FV = PV(FVF_{n,i})$ accumulates a present amount forward; $PV = FV(PVF_{n,i})$ discounts a future amount back. Either formula can be rearranged to solve for an unknown interest rate or number of periods.
4. Ordinary Annuity vs. Annuity Due: An ordinary annuity makes rents at the end of each period; an annuity due makes rents at the beginning. Annuity due factors equal ordinary annuity factors multiplied by $(1 + i)$, reflecting one additional period of compounding.
5. Deferred Annuities: Rents do not begin until after a deferral period. Present value is computed by finding PV of an ordinary annuity for the full horizon and subtracting the PV of rents for the deferred periods (or by discounting the annuity PV as a single sum for the deferral length).
6. Bond Valuation: A bond's issue price equals the PV of periodic coupon interest payments (annuity) plus the PV of the principal (single sum), both discounted at the market rate. A coupon rate below the market rate produces a discount; above produces a premium.
7. Effective-Interest Amortization: Bond discount or premium is amortized each period by multiplying the carrying value by the effective (market) rate to get interest expense, then subtracting the cash coupon paid; the difference adjusts the carrying value.
8. Expected Cash Flow Approach (IFRS 13): Instead of using only the most likely cash flow, this method weights multiple possible cash flow scenarios by their probabilities. The resulting expected cash flows are discounted at the risk-free rate (pure rate + expected inflation), with credit risk captured in the probability weights.

Important Definitions

Term	Definition
Time Value of Money	The principle that a dollar received today is worth more than a dollar promised in the future, because today's dollar can be invested to earn interest.

Term	Definition
Simple Interest	Interest computed on the original principal only: $I = p \times i \times n$, regardless of any previously accrued but unpaid interest.
Compound Interest	Interest computed on both the outstanding principal and any interest earned in prior periods that has not been withdrawn; the typical basis for business finance.
Present Value (PV)	The current worth of a future cash amount or series, discounted at a specified compound interest rate; always less than the corresponding future value.
Future Value (FV)	The accumulated amount a present sum or series will grow to at a specified compound interest rate after n periods; always greater than the original present value.
Ordinary Annuity	A series of equal periodic rents (payments or receipts) made at the end of each period for n periods.
Annuity Due	A series of equal periodic rents made at the beginning of each period; its future and present value factors equal the corresponding ordinary annuity factors multiplied by $(1 + i)$.
Deferred Annuity	An annuity in which rents do not begin until two or more periods after the agreement date; the deferral period earns interest but produces no rents.
Effective Yield (Effective Interest Rate)	The actual annual rate of return earned on an investment when compounding occurs more than once per year; always exceeds the stated (nominal) rate when compounding frequency > 1 .
Expected Cash Flow Approach	An IFRS 13 present-value technique that weights multiple possible cash flow outcomes by their probabilities and discounts the resulting expected cash flows at the risk-free rate.
Risk-Free Rate of Return	Under IFRS 13, the discount rate for expected cash flows, equal to the pure rate of interest plus the expected inflation rate; credit risk is excluded because it is already embedded in the probability weights.
Effective-Interest Method	The preferred IFRS amortization method for bond discount/premium; interest expense each period equals the bond's carrying value multiplied by the market rate at issuance.

Accounting Treatments

- Bond Issuance at Discount: Debit Cash (proceeds), Debit Discount on Bonds Payable, Credit Bonds Payable (face value). Example: Alltech issues \$100,000, 5% bonds for \$95,787.80 — Debit Cash \$95,787.80, Debit Discount on Bonds Payable \$4,212.20, Credit Bonds Payable \$100,000.
- Effective-Interest Amortization of Discount: Each period, Debit Interest Expense (carrying value $\times$ effective rate), Credit Cash (face $\times$ coupon rate), Credit Discount on Bonds Payable (difference). This increases the carrying value toward par over the bond's life.
- Warranty Liability Using Expected Cash Flows: Debit Warranty Expense, Credit Warranty Liability for the present value of probability-weighted expected repair outflows, discounted at the risk-free rate (e.g., Juno's: PV of \$6,160 year-1 + \$6,900 year-2 at 5%).

- **Non-Current Notes / Leases with Below-Market Rates:** The asset (or lease right-of-use asset) and corresponding liability are recorded at the present value of future cash flows using the market rate, not the stated rate; the difference is recorded as a discount.
- **Pension Obligation (Deferred Annuity):** The pension liability is the present value of a deferred annuity — first compute PV of pension payments over the post-retirement period (ordinary annuity), then discount that lump sum back to today over the pre-retirement period (single-sum factor).

Critical Formulas / Calculations

Future Value of a Single Sum: $FV = PV \times (1 + i)^n$. Variables: PV = present value, i = interest rate per period, n = number of periods. Example: $€50,000 \times (1.06)^5 = €50,000 \times 1.33823 = €66,912$.

Present Value of a Single Sum: $PV = FV \times [1 \div (1 + i)^n]$. Variables as above. Example: $€73,466 \times [1 \div (1.08)^5] = €73,466 \times 0.68058 \approx €50,000$.

Future Value of an Ordinary Annuity: $FV-OA = R \times [(1 + i)^n - 1] \div i$. Variables: R = periodic rent. Example: $\$5,000 \times FVF-OA(5, 6\%) = \$5,000 \times 5.63709 = \$28,185.45$.

Present Value of an Ordinary Annuity: $PV-OA = R \times [1 - 1 \div (1 + i)^n] \div i$. Example: $\$6,000 \times PVF-OA(5, 6\%) = \$6,000 \times 4.21236 = \$25,274.16$.

Annuity Due Conversion: $FV-AD \text{ factor} = FV-OA \text{ factor} \times (1 + i)$; $PV-AD \text{ factor} = PV-OA \text{ factor} \times (1 + i)$. Example: $PV-OA(4, 5\%) = 3.54595$; $PV-AD(4, 5\%) = 3.54595 \times 1.05 = 3.72325$.

Expected Cash Flow: $E[CF] = \sum (\text{Cash Flow}_k \times \text{Probability}_k)$. Example: $\$100 \times 0.30 + \$200 \times 0.50 + \$300 \times 0.20 = \$30 + \$100 + \$60 = \$190$.

Effective-Interest Amortization: $\text{Amortization} = (\text{Carrying Value} \times \text{Effective Rate}) - (\text{Face Value} \times \text{Coupon Rate})$. Example (Year 1, Alltech): $\$95,787.80 \times 6\% - \$100,000 \times 5\% = \$5,747.27 - \$5,000.00 = \$747.27$ discount amortized.

Compounding Period Adjustments: Periodic rate = Annual rate $\div$ m; Total periods = Years $\times$ m (m = compounding periods per year). Example: 8% annual, quarterly, 6 years $\rightarrow i = 2\%$, $n = 24$.

Illustrative Example

Bond Valuation — Alltech Corporation (Illustration 6.40): Alltech issues \$100,000 of 5% bonds on January 1, 2022, maturing in 5 years, with annual coupons, when the market rate is 6%. Step 1 — PV of principal: $\$100,000 \times PVF(5, 6\%) = \$100,000 \times 0.74726 = \$74,726.00$. Step 2 — PV of coupons: $\$5,000 \times PVF-OA(5, 6\%) = \$5,000 \times 4.21236 = \$21,061.80$. Step 3 — Issue price: $\$74,726.00 + \$21,061.80 = \$95,787.80$ (discount of \$4,212.20). Year 1 interest expense under effective-interest method: $\$95,787.80 \times 6\% = \$5,747.27$; cash paid = \$5,000; discount amortized = \$747.27; new carrying value = \$96,535.07.

Disclosures / Presentation Notes

- IFRS 13 requires entities to disclose the valuation technique(s) and inputs used for Level 3 fair value measurements, including a description of the expected cash flow approach and the discount rate applied.
- For bonds and notes, the carrying value (face value adjusted for unamortized discount or premium) must be presented on the statement of financial position; the effective-interest amortization schedule supports the interest expense reported in profit or loss.

- Lease obligations measured at the present value of future payments must be disclosed with the discount rate used, maturity analysis of undiscounted cash flows, and reconciliation of opening to closing balances (IFRS 16).
- Pension and other post-employment benefit obligations require disclosure of the actuarial assumptions underlying the present value calculation, including the discount rate, expected salary growth, and mortality rates (IAS 19). [VERIFY exact IAS 19 disclosure checklist items against current standard]

Potential Pitfalls

- Ordinary Annuity vs. Annuity Due confusion: Failing to identify whether rents occur at the end (ordinary) or beginning (due) of each period leads to using the wrong table factor; annuity due values are always higher by a factor of $(1 + i)$, so the error is systematic and material.
- Incorrect period and rate conversion for sub-annual compounding: Using the annual interest rate and number of years directly when compounding is quarterly or monthly overstates present values and understates future values; the annual rate must be divided by m and years multiplied by m before entering the tables.
- Most-likely vs. expected cash flow under IFRS 13: Using the single most-probable cash flow estimate (traditional approach) instead of the probability-weighted expected cash flow approach required for Level 3 measurements can systematically understate or overstate fair value when cash flow distributions are skewed.

Notes: The chapter text was fully captured in a single read (1,381 lines). Numerical values inside certain illustrations (e.g., the precise figures for Illustrations 6.1, 6.3, 6.4, 6.5, 6.21, 6.28, 6.31, 6.44, and 6.45) were referenced but their table bodies were not rendered in the plain-text extraction — only surrounding narrative and row/column headers were present. The five compound interest tables (Tables 6.1–6.5) appended to the chapter were not included in the extracted text file. Practice problem solution values for parts (a), (c), and (d) of the Vesper Inc. problem contained blanks (indicated by '\$') in the source text, suggesting the answer key figures were in a separately formatted answer box not captured by the extractor. No appendices or Global Accounting Insights section were present, consistent with the chapter's own note that TVM concepts are universal.

Chapter 7: Cash and Receivables

Executive Summary

Chapter 7 covers the recognition, measurement, valuation, and presentation of cash and receivables under IFRS, two of the most liquid and important current assets on the statement of financial position. The chapter establishes that cash must be freely available and unrestricted, while receivables must be reported at cash realizable value using the allowance method to reflect expected credit losses. An appendix extends the discussion to internal cash controls, the imprest petty cash system, and bank reconciliation procedures.

Key Topics

1. Reporting Cash and Cash Equivalents: Cash includes coin, currency, bank deposits, and negotiable instruments; cash equivalents are short-term, highly liquid investments with original maturities of three months or less. Special reporting issues arise for restricted cash, bank overdrafts, and compensating balances.
2. Recognition of Accounts Receivable: Receivables are recognized when control of goods or services transfers to the customer. The transaction price is measured net of variable consideration such as trade discounts, cash (sales) discounts, and estimated sales returns.
3. Valuation of Accounts Receivable — Allowance Method: IFRS requires the allowance method when bad debts are material, recognizing estimated uncollectibles through Bad Debt Expense and Allowance for Doubtful Accounts. The percentage-of-receivables approach (including aging schedules) is used to estimate the required allowance balance.
4. Notes Receivable — Recognition and Valuation: Long-term notes are recorded at the present value of expected cash flows; when stated and market interest rates differ, a discount or premium is recorded and amortized using the effective-interest method over the note's life.
5. Derecognition of Receivables: Receivables are removed from the books when contractual rights expire or when substantially all risks and rewards of ownership are transferred. Transfers may be structured as a true sale without guarantee (factoring without recourse) or as a secured borrowing (factoring with recourse / failed sale).
6. Accounts Receivable Turnover Analysis: Analysts compute the accounts receivable turnover ratio (net sales divided by average net receivables) and the average collection period (365 days divided by turnover) to assess a company's liquidity and effectiveness of its credit and collection policies.
7. IFRS vs. U.S. GAAP Differences: Key differences include the classification of bank overdrafts (cash under IFRS; liability under U.S. GAAP), the criteria for derecognizing transferred receivables (risks-and-rewards plus control under IFRS; control only under U.S. GAAP), and the approach to estimating impairment on long-term receivables (12-month vs. lifetime expected loss model).
8. Cash Controls (Appendix 7A): Internal control techniques include using multiple bank accounts, lockbox systems, imprest bank accounts, the imprest petty cash system, physical protection of cash, and monthly bank reconciliations to reconcile book and bank balances to the correct cash balance.

Important Definitions

Term	Definition
Cash	Coin, currency, available funds on deposit at banks, and negotiable instruments (money orders, certified checks, bank drafts) that are freely available to meet current obligations.
Cash Equivalents	Short-term, highly liquid investments with original maturities of three months or less that are readily convertible to known amounts of cash and subject to insignificant risk of value changes (e.g., government bonds, commercial paper).
Restricted Cash	Cash set aside for a specific purpose (e.g., compensating balances, plant expansion, debt retirement) that is not freely available; classified as current or non-current depending on when it will be disbursed.
Compensating Balance	The minimum cash balance a borrower must maintain in a deposit account as support for an existing borrowing arrangement with a lending institution.
Cash Realizable Value	The net amount a company expects to receive in cash from its receivables, equal to gross receivables minus the allowance for doubtful accounts.
Allowance for Doubtful Accounts	A contra-asset account representing the estimated portion of accounts receivable that is not expected to be collected; paired with Bad Debt Expense in the adjusting entry.
Trade Discount	A percentage reduction from the list (catalog) price used to determine the actual billing price to a customer; not separately recorded in the accounts.
Cash (Sales) Discount	A reduction in the invoice price offered to customers as an incentive for early payment (e.g., 2/10, n/30); recorded using either the gross method or the preferred net method.
Imputed Interest Rate	An estimated market interest rate used to determine the present value of a note when no stated rate exists or the stated rate is unreasonable; determined at the note's inception and not revised for subsequent market changes.
Factoring Receivables	The sale of receivables to a finance company or bank (factor) that collects directly from customers; may be without guarantee (without recourse, true sale) or with guarantee (with recourse, treated as secured borrowing under IFRS).
Derecognition	The removal of a financial asset (receivable) from the statement of financial position when contractual rights to cash flows expire or when substantially all risks and rewards of ownership are transferred to another party.
Percentage-of-Receivables Approach	An allowance-method estimation technique that applies historical uncollectibility rates to the ending accounts receivable balance (possibly by age category in an aging schedule) to determine the required allowance balance.

Accounting Treatments

- Estimating bad debts (allowance method): Debit Bad Debt Expense / Credit Allowance for Doubtful Accounts for the amount needed to bring the allowance to the estimated required balance at period end. Example: if desired balance is €26,610 and existing credit balance is €800, record €25,810.

- Write-off of an uncollectible account: Debit Allowance for Doubtful Accounts / Credit Accounts Receivable for the specific amount written off. No impact on Bad Debt Expense or net realizable value.
- Recovery of a previously written-off account: Two entries — (1) Debit Accounts Receivable / Credit Allowance for Doubtful Accounts to reinstate; (2) Debit Cash / Credit Accounts Receivable to record collection.
- Zero-interest-bearing note received: Record the note at its present value (cash paid), with the discount representing unearned interest. Each period, Debit Notes Receivable / Credit Interest Revenue for the effective-interest amortization amount.
- Sale of receivables without guarantee (factoring without recourse): Debit Cash (net proceeds) + Due from Factor (retained %) / Credit Accounts Receivable (face) and record the difference as Loss on Sale of Receivables.
- Transfer of receivables with guarantee (with recourse / failed sale): Debit Cash (net proceeds) / Credit Notes Payable (or Recourse Liability) — the receivable remains on the transferor's books; transaction treated as a secured borrowing.

Critical Formulas / Calculations

Accounts Receivable Turnover = Net Sales / Average (Net) Accounts Receivable.

Variables: Net Sales = total sales less returns and allowances; Average

Accounts Receivable = (Beginning balance + Ending balance) / 2. Example: LVMH Net Sales €35,664M / [(€2,274M + €2,521M) / 2] = 14.9 times.

Average Collection Period (Days Outstanding) = 365 / Accounts Receivable Turnover. Variables: 365 = days in a year; AR Turnover from formula above. Example: 365 / 14.9 = 24.5 days.

Present Value of an Interest-Bearing Note = PV of Principal + PV of Interest Payments = Face Value × PVF_{n,i} + Annual Interest × PVF-OA_{n,i}. Example: €10,000 note, 10% stated, 12% market, 3 years: €10,000 × 0.71178 + €1,000 × 2.40183 = €7,118 + €2,402 = €9,520.

Effective-Interest Amortization of Discount = (Carrying Amount of Note × Market Rate) - Cash Received. Example (Year 1, Morgan Group): €9,520 × 12% = €1,142 interest revenue; €1,000 cash received; €142 discount amortized, new carrying amount €9,662.

Required Bad Debt Expense (Percentage-of-Receivables): Desired Allowance Balance - Existing Credit Balance (or + Existing Debit Balance). Example: Desired balance €26,610 - existing credit €800 = Bad Debt Expense of €25,810.

Net Cash Discount Annualized Rate = Discount % / ((Credit Period - Discount Period) / 365). Example: 1% discount, 10-day discount period, 30-day net period: 0.01 / (20/365) = 18.25% annualized cost of not taking the discount.

Illustrative Example

Factoring without recourse (Illustration 7.15): Crest Textiles factors €500,000 of accounts receivable with Commercial Factors. Finance charge = 3% × €500,000 = €15,000. Retained holdback = 5% × €500,000 = €25,000. Crest Textiles records: Debit Cash €460,000, Debit Due from Factor €25,000, Debit Loss on Sale of Receivables €15,000 / Credit Accounts Receivable €500,000. Commercial Factors records: Debit Accounts Receivable €500,000 / Credit Due to Customer (Crest Textiles) €25,000, Credit Interest Revenue €15,000, Credit Cash €460,000.

Disclosures / Presentation Notes

- Receivables must be segregated by category (trade vs. non-trade, current vs. non-current) on the face of the statement of financial position, with valuation allowances presented as a deduction from gross receivables to arrive at net carrying amount.
- Companies must disclose the fair value of receivables, concentrations of credit risk (e.g., significant receivables from one industry or geographic region), and any receivables pledged as collateral for borrowings.
- Restricted cash must be disclosed separately — if restricted for short-term purposes it is presented in current assets; if restricted for long-term obligations (e.g., retirement of long-term debt, plant expansion) it is classified as a non-current asset.
- IFRS requires disclosure of an aging analysis for receivables that are past due or impaired, the carrying amount of receivables whose terms have been renegotiated, and quantitative and qualitative information to allow assessment of the credit risk inherent in the receivable portfolio.

Potential Pitfalls

- ❑ Confusing the write-off entry with bad debt expense recognition: Under the allowance method, writing off a specific account (Debit Allowance / Credit Accounts Receivable) does NOT record any new expense — the expense was already recognized in the prior period's adjusting entry. Debiting Bad Debt Expense at the time of write-off double-counts the loss.
- ❑ Misclassifying a transfer of receivables as a sale when it should be a secured borrowing: If the transferor provides a full guarantee (recourse) covering all credit losses, substantially all risks and rewards remain with the transferor under IFRS, and the transaction is a failed sale — both the receivable asset and the borrowing liability must remain on the books.
- ❑ Ignoring the existing balance in the Allowance for Doubtful Accounts when computing bad debt expense under the percentage-of-receivables approach: Because this method targets a required ending balance in the allowance account, the bad debt expense journal entry must be adjusted upward (if the allowance has a debit balance) or downward (if it has a credit balance) relative to the gross estimated uncollectible amount.

Notes: The chapter text was fully readable across all chunks with no truncation. Certain numerical figures within aging schedule tables (Illustration 7.6) and journal entry amounts in some illustrations appeared as placeholders in the extracted text (values stripped during PDF conversion), so specific dollar amounts for those entries could not be verified from the source file. All formulas and examples cited are drawn from complete narrative passages. Appendix 7A (Cash Controls and Bank Reconciliation) was fully included in the source file.

Chapter 8: Valuation of Inventories: A Cost-Basis Approach

Executive Summary

Chapter 8 establishes the cost-basis approach to inventory valuation, covering how companies classify inventories, determine which goods and costs belong in inventory, and choose among allowable cost flow assumptions (specific identification, FIFO, and average-cost under IFRS). The chapter concludes that correct inventory measurement is critical because errors directly distort cost of goods sold, net income, working capital, and the current ratio across multiple periods.

Key Topics

1. Inventory classifications: Merchandising companies carry a single Inventory account, while manufacturers maintain three accounts — Raw Materials, Work in Process (WIP), and Finished Goods — each representing a different stage of the production cycle.
2. Inventory systems: A perpetual system continuously records every purchase and sale in the Inventory account and supports a Cost of Goods Sold account updated in real time; a periodic system defers cost determination to period-end through a physical count, computing COGS as a residual.
3. Goods included in inventory: Ownership (control/legal title) governs inclusion — goods in transit under f.o.b. shipping point belong to the buyer, consigned goods remain with the consignor, and product financing ('parking') arrangements keep inventory on the financing party's books.
4. Costs included in inventory: Product costs (purchase price, import duties, freight-in, conversion costs) are capitalised to inventory; period costs (selling, G&A, most interest) are expensed as incurred.
5. Cost flow assumptions: IFRS requires specific identification for non-interchangeable items and permits FIFO or average-cost (weighted-average for periodic; moving-average for perpetual) for all others. LIFO is prohibited under IFRS but discussed in Appendix 8A for completeness.
6. Comparative effects of cost flow methods: In a rising-price environment, FIFO produces the highest ending inventory and net income (but lowest cash retained after taxes), while average-cost falls in between; LIFO (Appendix) produces the lowest net income but highest cash retention.
7. Inventory errors: Understating ending inventory overstates COGS and understates net income in the current period; the error self-corrects (counterbalances) in the following period. Simultaneously omitting a purchase and its related ending inventory leaves COGS and net income unaffected but understates accounts payable and inventory, distorting the current ratio.
8. Treatment of purchase discounts: IFRS requires discounts to reduce inventory cost; under the gross method a separate Purchase Discounts account is used, while the preferred net method records inventory net of available discounts and isolates 'Purchase Discounts Lost' as a financial expense.

Important Definitions

Term	Definition
Inventories	Asset items held for sale in the ordinary course of business, or goods to be used or consumed in producing goods for sale.
Raw Materials Inventory	Costs assigned to materials on hand that have not yet been placed into the production process.

Term	Definition
Work in Process (WIP) Inventory	The accumulated cost of raw materials, direct labour, and allocated manufacturing overhead for units that have been started but not yet completed.
Finished Goods Inventory	Costs identified with completed but unsold units on hand at the end of the fiscal period.
Perpetual Inventory System	A system that continuously records every purchase and sale in the Inventory account, maintaining an up-to-date balance and a running Cost of Goods Sold figure.
Periodic Inventory System	A system that records purchases in a Purchases account and determines both ending inventory and COGS only at period-end through a physical count.
Product Costs	Costs that attach to inventory — including purchase price, import duties, freight-in, and conversion costs — and are capitalised in the Inventory account until the goods are sold.
Period Costs	Costs indirectly related to the acquisition or production of goods (e.g., selling expense, G&A) that are expensed in the period incurred rather than capitalised to inventory.
FIFO (First-In, First-Out)	A cost flow assumption that the first goods purchased are the first sold, so ending inventory reflects the most recent purchase prices.
Weighted-Average Cost Method	A periodic inventory cost flow method that divides the total cost of goods available for sale by total units available to derive a single average unit cost applied to all sales and ending inventory.

Accounting Treatments

- Perpetual system purchase entry: Dr Inventory / Cr Accounts Payable (freight-in also debited to Inventory, not a separate account).
- Perpetual system sale entry (two parts): Dr Accounts Receivable / Cr Sales Revenue AND Dr Cost of Goods Sold / Cr Inventory.
- Perpetual inventory shrinkage adjustment when physical count is below book: Dr Inventory Over and Short / Cr Inventory (the debit flows to Cost of Goods Sold or Other expense).
- Periodic system — end-of-period COGS computation: Beginning Inventory + Purchases + Freight-In - Purchase Returns & Allowances - Purchase Discounts = Cost of Goods Available for Sale; subtract physical Ending Inventory to derive COGS.
- Purchase discounts — gross method: Purchase recorded at gross invoice; discount taken is credited to Purchase Discounts (contra-purchases). Net method: Purchase recorded net; failure to take discount is Dr Purchase Discounts Lost / Cr Cash (reported as financial expense).
- Sales with repurchase (parking) arrangements: Inventory and related liability remain on the seller's books because control has not transferred; no revenue is recognised until the repurchase agreement is resolved.

Critical Formulas / Calculations

Cost of Goods Sold (periodic) = Beginning Inventory + Cost of Goods Purchased - Ending Inventory. Example: €600 + €5,400 - €2,400 = €3,600 COGS.

Weighted-Average Unit Cost = Total Cost of Goods Available for Sale / Total Units Available. Example (Call-Mart): €43,900 / 10,000 units = €4.39 per unit; Ending Inventory = 6,000 × €4.39 = €26,340.

Moving-Average Unit Cost (perpetual) = Running Total Cost / Running Total Units after each purchase. Example: after March 15 purchase, €34,400 / 8,000 units = €4.30 per unit for the next withdrawal.

FIFO Ending Inventory = Units on hand priced from most recent purchase layers backward. Example (Call-Mart periodic): 2,000 @ €4.75 + 4,000 @ €4.40 = €9,500 + €17,600 = €27,100.

Current Ratio = Current Assets / Current Liabilities. Illustration 8.17: with understated payables/inventory (\$40,000 omitted) ratio = \$120,000 / \$40,000 = 3:1 vs. correct \$160,000 / \$80,000 = 2:1.

Illustrative Example

Call-Mart SpA (March): purchases 2,000 units @ €4.00, 6,000 @ €4.40, 2,000 @ €4.75; sells 4,000 units; ending inventory = 6,000 units. Under FIFO (periodic): ending inventory = 2,000 @ €4.75 + 4,000 @ €4.40 = €27,100; COGS = €43,900 - €27,100 = €16,800. Under weighted-average (periodic): avg cost = €43,900 / 10,000 = €4.39; ending inventory = 6,000 × €4.39 = €26,340; COGS = €43,900 - €26,340 = €17,560. Under moving-average (perpetual): after March 15 purchase, avg = €4.30; March 19 withdrawal cost = 4,000 × €4.30 = €17,200; after March 30 purchase, new avg = €4.45; ending inventory = 6,000 × €4.45 = €26,700.

Disclosures / Presentation Notes

- Inventory classification: Companies must disclose the composition of inventory (raw materials, WIP, finished goods for manufacturers; single merchandise inventory for retailers) on the face of the statement of financial position or in the notes.
- Cost flow method: The cost flow assumption (FIFO or average-cost) must be disclosed and applied consistently; any change in method requires explanation and disclosure of its financial statement effect.
- Consigned goods and special arrangements: Material amounts of consigned goods held by or sent to third parties, and the nature of any product financing (repurchase) arrangements, should be disclosed in the notes.
- Inventory errors and restatements: When an inventory error is discovered that affects prior periods, the financial statements for those periods must be restated and the nature and amount of the error disclosed.

Potential Pitfalls

- ❑ Confusing legal title with control for transit goods: Goods shipped f.o.b. destination are still owned by the seller while in transit and must NOT be included in the buyer's year-end inventory count; failing to check shipping terms leads to under- or overstatement of both inventory and purchases.
- ❑ Assuming inventory errors are immaterial or self-correcting in year one: Although an ending inventory understatement counterbalances in the next period, both years' income statements are individually misstated; management cannot dismiss the error simply because the two-year total net income is correct.

- Conflating FIFO's balance sheet strength with income statement accuracy: FIFO reports ending inventory close to current replacement cost, but it charges older (lower) costs against current revenue in a rising-price environment, overstating gross profit and creating a 'phantom profit' that may lead to inadequate pricing or dividend decisions.

Notes: The chapter text was fully extracted without apparent truncation. Appendix 8A (LIFO) was present in full. Numerical tables within Illustrations 8.7–8.13 and 8A.1–8A.4 were partially stripped of column values during PDF extraction (rows show labels but numeric cells are blank); formulas and conclusions stated in the narrative were used to reconstruct figures where needed. No additional appendices beyond 8A appear in the file.

Chapter 9: Inventories: Additional Valuation Issues

Executive Summary

Chapter 9 extends inventory valuation beyond historical cost by covering the lower-of-cost-or-net realizable value (LCNRV) rule, special valuation bases for agricultural and commodity inventories, purchase commitment accounting, and two widely used estimation techniques—the gross profit method and the retail inventory method. The chapter concludes that proper inventory measurement requires matching the measurement basis to the nature of the inventory and the reliability of available evidence, while disclosing significant judgments and policies to financial statement users.

Key Topics

1. Lower-of-Cost-or-Net Realizable Value (LCNRV): Inventories must be written down to net realizable value (estimated selling price less costs to complete and sell) whenever NRV falls below cost, recognizing the loss in the period of decline rather than the period of sale.
2. Methods of Applying LCNRV: The rule can be applied on an individual-item, similar-or-related-items, or total-inventory basis; the individual-item approach is preferred because it yields the most conservative balance sheet figure and prevents offsetting unrealized gains against unrealized losses.
3. Recording Write-Downs and Recoveries: Two methods exist—the cost-of-goods-sold method (buries the loss in COGS) and the loss method (discloses it separately); an allowance account is used so both cost and NRV are visible; write-downs may be reversed in later periods up to the original write-down amount under IFRS.
4. Special Valuation Bases—Agricultural and Commodity Inventories: Biological assets and agricultural produce at harvest are measured at fair value less costs to sell (net realizable value), with unrealized gains and losses recognized in income; commodity broker-traders similarly use NRV.
5. Basket (Lump-Sum) Purchases and Relative Standalone Sales Value: When a group of dissimilar assets is purchased for a single price, cost is allocated among individual units in proportion to their relative standalone sales values to accurately reflect each unit's cost.
6. Purchase Commitments: Non-cancelable purchase contracts are disclosed in notes but not recognized as assets or liabilities; if market price falls below the contract price before delivery, a loss (and corresponding liability) is recognized in the period of the price decline—IFRS calls such a contract onerous.
7. Gross Profit Method: An estimation technique that derives ending inventory by subtracting estimated cost of goods sold (sales multiplied by the cost percentage) from goods available for sale; used for interim reports and insurance claims but not acceptable as a substitute for the annual physical count under IFRS.
8. Retail Inventory Method: Retailers estimate ending inventory at cost by applying a cost-to-retail ratio to the ending inventory valued at retail prices; the conventional (LCNRV) version excludes markdowns from the ratio, while the cost version includes them, producing a higher ending inventory.

Important Definitions

Term	Definition
Net Realizable Value (NRV)	The estimated selling price of inventory in the normal course of business less the estimated costs of completion and the estimated costs necessary to make the sale (IAS 2).
Lower-of-Cost-or-Net Realizable Value (LCNRV)	The inventory valuation rule requiring that inventory be reported at the lower of its historical cost or its net realizable value at each reporting date.
Allowance to Reduce Inventory to Net Realizable Value	A contra-asset account credited when inventory is written down, so that both the cost and the net amount are visible on the statement of financial position.
Biological Asset	A living animal or plant (e.g., dairy cows, fruit trees) classified as a non-current asset and measured at fair value less costs to sell under IFRS.
Agricultural Produce	The harvested product of a biological asset (e.g., milk, wool, picked fruit); measured at NRV at the point of harvest, which then becomes its cost for subsequent accounting.
Relative Standalone Sales Value	The method of allocating a lump-sum purchase price among individual units in proportion to each unit's selling price relative to the total selling price of all units acquired.
Purchase Commitment	A formal, non-cancelable contract to buy inventory at a specified price in the future; an onerous purchase commitment (where market price < contract price) requires immediate loss recognition under IFRS.
Gross Profit Method	An inventory estimation technique that uses the historical gross profit percentage to estimate cost of goods sold and thereby derive ending inventory; useful for interim periods and casualty loss claims.
Retail Inventory Method	A procedure for estimating inventory cost used by high-volume retailers, converting ending inventory at retail to cost by applying a cost-to-retail ratio computed from goods available for sale.
Cost-to-Retail Ratio	In the retail inventory method, the ratio of goods available for sale at cost to goods available for sale at retail price; used to convert retail inventory to approximate cost.
Inventory Turnover	Cost of goods sold divided by average inventory; measures how many times per period a company sells and replaces its inventory stock.
Markdown	A reduction below the original retail selling price of merchandise, reflecting a decline in the utility or market value of the item.

Accounting Treatments

- Write-down using the loss method (preferred): Debit 'Loss Due to Decline of Inventory to NRV' and credit 'Allowance to Reduce Inventory to NRV' for the excess of cost over NRV; this loss appears separately in the income statement. Example for Ricardo SpA: Dr Loss Due to Decline 12,000 / Cr Allowance 12,000.

- Write-down using the cost-of-goods-sold method: Debit 'Cost of Goods Sold' and credit 'Inventory' (or Allowance) for the write-down amount; the loss is buried in COGS and not separately disclosed—permissible under IFRS on materiality grounds.
- Reversal of prior write-down: Debit 'Allowance to Reduce Inventory to NRV' and credit 'Recovery of Inventory Loss' (income); reversal is limited to the original write-down amount and inventory may never be reported above original cost. Example: Dr Allowance 4,000 / Cr Recovery of Inventory Loss 4,000.
- Biological assets at NRV: Debit 'Biological Assets' and credit 'Unrealized Holding Gain or Loss—Income' for increases in fair value less costs to sell; the reverse entry is made for decreases. Example for Bancroft Dairy: Dr Biological Assets 33,800 / Cr Unrealized Holding Gain 33,800.
- Purchase commitment loss: When contract price exceeds market price, debit 'Unrealized Holding Loss—Income' and credit 'Purchase Commitment Liability.' Upon delivery, debit Purchases/Inventory and Purchase Commitment Liability, then credit Cash for the full contract amount. Example: Dr Unrealized Holding Loss 3,000,000 / Cr Purchase Commitment Liability 3,000,000.
- Retail inventory method—ending inventory at cost: Compute ending inventory at retail (goods available at retail minus net sales), then multiply by the cost-to-retail ratio. Example: Ending inventory at retail £25,000 × 0.70 cost-to-retail ratio = £17,500 ending inventory at cost.

Critical Formulas / Calculations

NRV = Estimated Selling Price - Estimated Costs to Complete - Estimated Costs to Sell. Example: Selling price €1,000 - completion €50 - selling costs €200 = NRV €750.

Gross Profit Method—Ending Inventory: Ending Inventory = (Beginning Inventory + Purchases) - [Net Sales × (1 - Gross Profit %)]. Example: (€60,000 + €200,000) - [€280,000 × 0.70] = €260,000 - €196,000 = €64,000.

Conversion of markup on cost to gross profit on selling price: GP% on selling price = Markup on Cost % ÷ (100% + Markup on Cost %). Example: 25% markup on cost → 25% ÷ 125% = 20% gross profit on selling price.

Retail Inventory Method—Cost-to-Retail Ratio: Cost-to-Retail Ratio = Goods Available at Cost ÷ Goods Available at Retail. Example: £77,000 ÷ £110,000 = 0.70 (70%). Ending inventory at cost = Ending inventory at retail £25,000 × 0.70 = £17,500.

Conventional (LCNRV) Retail Ratio: Computed on goods available at cost divided by (goods available at retail + net markups), excluding net markdowns—this lowers the ratio and approximates LCNRV. Example (In-Fusion): cost ratio = 53.9%; ending inventory = €12,500 × 0.539 = €6,737.50.

Inventory Turnover = Cost of Goods Sold ÷ Average Inventory; Average Days to Sell = 365 ÷ Inventory Turnover. Example: Tate & Lyle: £1,621M ÷ [(£419M + £434M) ÷ 2] = 3.80 times; 365 ÷ 3.80 ≈ 96 days.

Illustrative Example

Cetus SE—Gross Profit Method: Beginning inventory €60,000 + Purchases €200,000 = Goods available €260,000. Net sales at selling price = €280,000; gross profit rate on selling price = 30%; estimated gross profit = €84,000; estimated COGS = €280,000 - €84,000 = €196,000. Ending inventory = €260,000 - €196,000 = €64,000. Ricardo SpA—LCNRV: Ending inventory at cost €82,000; NRV €70,000; write-down = €12,000. Loss method entry: Dr Loss Due to Decline of Inventory to NRV €12,000 / Cr Allowance to Reduce Inventory to NRV €12,000. Statement of financial position reports: Inventory at cost €82,000 less Allowance €12,000 = Inventory at NRV €70,000.

Disclosures / Presentation Notes

- IFRS (IAS 2) requires disclosure of: (1) the accounting policies and cost formulas used (FIFO, weighted-average); (2) the total carrying amount and amounts by classification (raw materials, WIP, finished goods, merchandise); (3) any inventories carried at fair value less costs to sell; (4) the amount of inventories recognized as expense (COGS) during the period; (5) the amount of any write-down and any reversal of write-downs, with the circumstances that led to the reversal; and (6) the carrying amount of inventories pledged as security.
- Significant or unusual financing arrangements relating to inventories—such as product financing arrangements, firm purchase commitments, pledging of inventories as collateral, and transactions with related parties—should be disclosed in the notes to financial statements.
- A change in inventory cost flow method (e.g., from weighted-average to FIFO) is a change in accounting principle requiring separate disclosure of the effect on income in current and prior periods and an explanatory paragraph in the auditor's report.
- Under IFRS, inventories carried at net realizable value (such as biological assets, agricultural produce, and broker-trader commodities) must be separately identified; the amount of unrealized gains and losses recognized in income from NRV remeasurement should also be disclosed.

Potential Pitfalls

- Confusing markup on cost with gross profit on selling price: These are different percentages on different bases. A 25% markup on cost equals only 20% gross profit on selling price, and mistakenly applying a cost-based percentage to sales in the gross profit method will overestimate COGS and understate ending inventory.
- Including markdowns in the cost-to-retail ratio under the conventional retail method: The conventional (LCNRV) retail method intentionally excludes net markdowns from the denominator of the cost-to-retail ratio. Including markdowns inflates the ratio, produces a higher ending inventory value, and fails to approximate LCNRV—the result is the cost method, not the conventional method.
- Failing to recognize a loss on a non-cancelable purchase commitment when market price falls: Companies sometimes treat all executory purchase contracts as unrecognized off-balance-sheet items. Under IFRS, an onerous purchase commitment—where the contract price exceeds the current market price—requires immediate recognition of a loss and a corresponding Purchase Commitment Liability in the period the price decline occurs, not when the goods are received.

Notes: The chapter text was complete and read in full (1,499 lines). Several tables and illustrations contained numerical data that was partially truncated or presented without column headers in the source text (e.g., Illustrations 9.3, 9.4, 9.10, 9.11, 9.18, 9.21, 9.22); specific figures from those tables could not always be confirmed and are described narratively. Appendices (e.g., Appendix 17A on hedging derivatives referenced in note 11) were not included in this chapter file. The GAAP comparison section at the end covers a sixth learning objective (LO 6—IFRS vs. U.S. GAAP) that does not appear in the chapter's stated five learning objectives; its content has been incorporated where relevant.

Chapter 10: Acquisition and Disposition of Property, Plant & Equipment

Executive Summary

Chapter 10 covers the full lifecycle accounting for property, plant, and equipment (PP&E) under IFRS, from initial recognition and measurement of acquisition costs—including borrowing cost capitalization during construction—through subsequent expenditure treatment (additions, improvements, repairs) and final disposal. The chapter establishes that historical cost, inclusive of all costs necessary to bring an asset to its intended location and condition, is the primary measurement basis, and that gains or losses on disposition must be recognised in income.

Key Topics

1. Definition and characteristics of PP&E: tangible assets held for use in production, supply of goods/services, rental, or administration; expected to be used over more than one period; not held for resale.
2. Initial cost measurement: historical cost includes purchase price plus all direct costs (freight, installation, trial runs, legal fees, taxes) required to bring the asset to its intended location and condition for use.
3. Borrowing cost capitalisation: IFRS requires capitalising actual borrowing costs (less any investment income on idle specific-debt proceeds) on qualifying assets during the capitalisation period, using either the specific-debt rate or a weighted-average capitalisation rate for general debt.
4. Special acquisition situations: assets purchased via deferred-payment contracts are recorded at present value; lump-sum purchases require allocation of total cost on a relative-fair-value basis; assets acquired by issuing shares use the market price of the shares (or fair value of property if share price is unavailable).
5. Non-monetary exchanges: when an exchange has commercial substance (future cash flows change), assets are recorded at fair value and gains/losses are recognised immediately; when commercial substance is absent, gains are deferred and reduce the basis of the acquired asset, while losses are still recognised.
6. Government grants: IFRS requires the income approach—grants are recognised in income on a systematic basis matching related costs, either as deferred grant revenue amortised over the asset's life or as a reduction in the asset's carrying amount reducing future depreciation.
7. Subsequent expenditures: additions and improvements that increase future economic benefits are capitalised; ordinary repairs and rearrangement/reorganisation costs are expensed as incurred; major overhauls are treated as separate components and capitalised.
8. Disposal of PP&E: depreciation is recorded to the disposal date, all related accounts are removed, and any resulting gain or loss is reported in the income statement; involuntary conversions are recognised in two stages—loss at the date of conversion and recovery gain when the insurance settlement is established.

Important Definitions

Term	Definition
Property, Plant, and Equipment (PP&E)	Tangible assets held for use in production, supply of goods or services, for rental to others, or for administrative purposes, expected to be used during more than one period.
Historical Cost	The cash or cash equivalent price of obtaining an asset and bringing it to the location and condition necessary for its intended use.
Borrowing Costs	Interest and other costs incurred by an entity in connection with the borrowing of funds, which may be capitalised into the cost of a qualifying asset under IFRS.
Qualifying Asset	An asset that necessarily takes a substantial period of time to get ready for its intended use or sale, thereby qualifying for borrowing cost capitalisation.
Capitalization Period	The period during which borrowing costs are capitalised; begins when expenditures are being incurred, construction activities are in progress, and borrowing costs are being incurred, and ends when the asset is substantially complete.
Capitalization Rate	The weighted-average rate of borrowing costs applicable to general (non-specific) debt outstanding during the period, used to determine the amount of borrowing costs eligible for capitalisation.
Commercial Substance	An exchange has commercial substance when the future cash flows of the entity change as a result of the transaction, i.e., the economic positions of the parties differ after the exchange.
Government Grant	Assistance from a government in the form of transfers of resources to a company in return for past or future compliance with conditions relating to the company's operating activities.
Component Depreciation	IFRS requirement to identify significant components of an asset separately, determine each component's depreciation base, and depreciate each component individually over its own useful life.
Involuntary Conversion	Termination of an asset's service through an uncontrollable event such as fire, flood, theft, or condemnation, typically followed by an insurance recovery recorded as a separate gain.

Accounting Treatments

- Initial recognition — capitalise all costs necessary to bring PP&E to its intended location and condition: purchase price (net of trade discounts), import duties, non-refundable taxes, freight, installation, and trial-run costs. Start-up costs, administrative overhead, and promotional costs are expensed.
- Borrowing costs during construction (specific debt): Dr Buildings (Capitalised Borrowing Cost) / Cr Cash for actual interest less any investment income earned on temporarily invested specific-debt proceeds.
- Borrowing costs during construction (general debt): multiply the average carrying amount of the project by the weighted-average capitalisation rate; amount capitalised cannot exceed total actual borrowing costs incurred. Journal entry: Dr Buildings / Dr Interest Expense (remainder) / Cr Cash.

- Non-monetary exchange with commercial substance — record acquired asset at fair value of asset given up (or asset received if more clearly evident); recognise gain or loss immediately: Dr New Asset + Dr Accumulated Depreciation / Cr Old Asset ± Gain/Loss on Disposal ± Cash.
- Non-monetary exchange without commercial substance — record acquired asset at book value of asset given up plus cash paid (gain deferred, reducing asset basis); losses still recognised immediately: Dr New Asset + Dr Accumulated Depreciation / Cr Old Asset / Cr Cash.
- Disposal by sale — update depreciation to sale date, then: Dr Cash + Dr Accumulated Depreciation / Cr Asset at cost ± Gain or Loss on Disposal of [Asset].

Critical Formulas / Calculations

Average Carrying Amount (general debt) = Sum of (Expenditure Amount × Months Remaining in Capitalisation Period / 12). Example: \$400,000 on Jan 1 × 12/12 = \$400,000; \$200,000 on Mar 1 × 10/12 = \$166,667; \$200,000 on May 1 × 8/12 = \$133,333 → Total average carrying amount = \$700,000 [VERIFY exact figures per Illustration 10.3].

Capitalization Rate (general debt) = Sum of (Interest Rate × Debt Amount) / Total General Debt Outstanding. Example from chapter: $(0.10 \times \$1,000,000 + 0.12 \times \$1,500,000) / \$2,500,000 = \$280,000 / \$2,500,000 = 11.2\%$.

Borrowing Costs to Capitalise (general debt) = Average Carrying Amount × Capitalisation Rate, subject to a ceiling of Actual Total Borrowing Costs. Example: $\$820,000 \times 11.2\% = \$91,840$, which is less than actual costs of \$280,000, so \$91,840 is capitalised.

Lump-Sum Purchase Allocation = (Individual Asset Fair Value / Total Fair Values of All Assets) × Total Purchase Price. Example from chapter: Land FV \$40,000 / Total FV \$160,000 × \$80,000 = \$20,000 allocated to land.

Present Value of Zero-Interest Note (deferred payment) = Future Payment × PV Factor. Example: €20,000 annual payment × PVF-OA(5, 10%) = €20,000 × 3.79079 = €75,816 recorded as equipment cost.

Gain or Loss on Disposal = Proceeds Received - Book Value of Asset. Example: machine sold for €7,000; book value €6,600 (cost €18,000 - accumulated depreciation €11,400) → Gain = €400.

Illustrative Example

Columbia SA (Practice Problem): Machine purchased for €12,000 + 6% sales tax (€720) + €2,800 platform = total capitalised cost €15,520. Warehouse constructed under specific €1,000,000 borrowing at 10% p.a.; capitalisation period is 9 months (March 1 to December 31), so interest = $\text{€}1,000,000 \times 10\% \times 9/12 = \text{€}75,000$; less investment income on idle funds of €13,000 → borrowing costs capitalised = €62,000. Machine sold July 1, 2024: accumulated depreciation = €1,690 (2022 half-year) + €3,380 (2023 full year) + €1,690 (2024 half-year) = €6,760; book value = €15,520 - €6,760 = €8,760; proceeds €7,000 → Loss on disposal = €1,760.

Disclosures / Presentation Notes

- Borrowing costs: entities must disclose the amount of borrowing costs capitalised during the period and the capitalisation rate used; see Royal Dutch Shell example (Illustration 10.6) reporting interest incurred, interest capitalised, and the applied rate (4% in 2018).
- Government grants: entities must disclose the accounting policy adopted (deferred income approach or asset-reduction approach), the nature and extent of grants recognised, and unfulfilled conditions or

contingencies; PetroChina (deferred income) and Daimler AG (asset reduction) illustrate both policies.

- Non-monetary exchanges: entities must disclose the nature of transactions, the method of accounting for the assets exchanged, and the amount of any gains or losses recognised on the exchanges during the period.
- Fully depreciated assets still in use: although IFRS does not explicitly prohibit omitting zero-carrying-amount assets, adequate disclosure of the cost and nature of such assets is generally required to avoid a misleading presentation (noted as an evaluative question in the chapter).

Potential Pitfalls

- Capitalising rearrangement, reorganisation, or start-up costs: IFRS expressly prohibits capitalising costs incurred after an asset is already in the location and condition intended by management, as well as initial operating losses and promotional expenses—these must be expensed immediately even though they may feel like necessary launch costs.
- Misapplying the borrowing-cost ceiling for general debt: companies must compare their computed capitalisation amount (average carrying amount $\times$ capitalisation rate) to actual total borrowing costs incurred and capitalise only the lower figure; failing to apply this cap can overstate asset costs and understate interest expense.
- Recording a profit on self-construction or using inflated trade-in allowances: a company must never record assets above fair value—self-constructed assets capped at what an outside producer would charge, and non-monetary exchange assets recorded at fair value of the asset given up (not the dealer's inflated trade-in allowance), otherwise assets and future depreciation are both misstated (illustrated by the WorldCom fraud where operating costs were improperly capitalised).

Notes: The chapter text was fully captured in a single read (1,519 lines). Numerical figures within tables in Illustrations 10.3, 10.4, 10.7, 10.9–10.13, 10.15–10.16 were partially missing or truncated in the source file (column values absent from extracted text); formulas and examples were reconstructed from surrounding narrative text and the Practice Problem solution where specific numbers were available. No appendices or global accounting insights (noted as deferred to Chapter 11) were included in this file.

Chapter 11: Depreciation, Impairments and Depletion

Executive Summary

Chapter 11 covers the systematic allocation of long-lived asset costs through depreciation, impairment testing and write-downs, and depletion of mineral resources under IFRS. It also addresses revaluation accounting—the IFRS option to carry property, plant, and equipment at fair value rather than historical cost—and explains how to present, disclose, and analyse these assets in financial statements.

Key Topics

1. Depreciation concepts and methods: Depreciation is a cost-allocation process (not a valuation process) that spreads the depreciable base (cost minus residual value) over an asset's useful life using the activity, straight-line, sum-of-the-years'-digits, or declining-balance method.
2. Component depreciation: IFRS requires each significant component of a PP&E item (e.g., airframe, engines, and other parts of an aircraft) to be depreciated separately over its own useful life.
3. Revision of depreciation estimates: Changes in estimated useful life or residual value are treated as changes in accounting estimates and applied prospectively by spreading the remaining book value over the revised remaining life.
4. Impairment recognition and measurement: An asset is impaired when its carrying amount exceeds its recoverable amount (the higher of fair value less costs to sell or value-in-use); the excess is recognised as an impairment loss and subsequent depreciation is recalculated on the reduced base.
5. Reversal of impairment losses: Under IFRS, a previously recognised impairment loss may be reversed in a later period if the recoverable amount rises, but the reversal cannot bring the carrying amount above what it would have been had no impairment ever been recorded.
6. Cash-generating units (CGUs): When individual assets cannot generate independent cash flows, impairment testing is performed at the CGU level—the smallest identifiable group of assets whose cash flows are largely independent.
7. Depletion of mineral resources: Natural resource costs are allocated using the units-of-production method (cost depletion); the depletion base includes exploration and evaluation (E&E) costs, intangible development costs, and restoration obligations, but typically excludes movable tangible equipment.
8. Revaluation accounting: IFRS permits (but does not require) companies to carry PP&E at fair value; revaluation increases above historical cost bypass net income and go to Other Comprehensive Income (revaluation surplus), while decreases below cost are recognised as impairment losses in net income.

Important Definitions

Term	Definition
Depreciation	The systematic allocation of the depreciable amount of a tangible asset to expense over its estimated useful life; a cost-allocation process, not a measure of fair value.
Depreciable Base	The cost of an asset less its residual (salvage) value; the total amount to be depreciated over the asset's useful life.

Term	Definition
Residual (Salvage) Value	The estimated net amount the entity expects to obtain from disposal of an asset at the end of its useful life.
Impairment	The condition that exists when a long-lived asset's carrying amount is not recoverable through either continued use or sale, resulting in a write-down to recoverable amount.
Recoverable Amount	The higher of an asset's fair value less costs to sell and its value-in-use; the amount used in the impairment test.
Value-in-Use	The present value of estimated future net cash flows expected to arise from the continued use and ultimate disposal of an asset, discounted at a pretax rate reflecting current market assessments.
Cash-Generating Unit (CGU)	The smallest identifiable group of assets that generates cash inflows that are largely independent of the cash inflows from other assets or groups.
Depletion	The systematic allocation of the cost of mineral resources (oil, gas, coal, etc.) to expense, typically using the units-of-production method.
Revaluation Surplus	The cumulative unrealized gain on revaluation of a long-lived asset above its historical cost, reported in Accumulated Other Comprehensive Income in the equity section.
Component Depreciation	The IFRS requirement to identify and depreciate separately each significant part of a PP&E item that has a different useful life or pattern of consumption.

Accounting Treatments

- Recording depreciation: Debit Depreciation Expense / Credit Accumulated Depreciation for each period; the asset is presented on the statement of financial position at cost less accumulated depreciation (carrying amount).
- Recording an impairment loss (asset in use): Debit Loss on Impairment / Credit Accumulated Depreciation—Equipment for the excess of carrying amount over recoverable amount; the loss appears in 'Other income and expense' in the income statement.
- Reversing an impairment loss: Debit Accumulated Depreciation / Credit Recovery of Impairment Loss; reversal is limited to the carrying amount that would have existed had no impairment been recorded.
- Recording depletion: Debit Inventory / Credit Accumulated Depletion for units extracted during the period; when inventory is sold, depletion cost becomes part of Cost of Goods Sold.
- Recording a revaluation increase: Debit the asset account / Credit Unrealized Gain on Revaluation (Other Comprehensive Income); does not pass through net income unless it reverses a prior impairment loss.
- Revision of depreciation estimate (prospective): No journal entry at the revision date; future depreciation = remaining book value (less any residual value) divided by remaining revised useful life, applied from the current period forward.

Critical Formulas / Calculations

Activity Method: Depreciation = $[(\text{Cost} - \text{Residual Value}) / \text{Total Estimated Hours}] \times \text{Hours Used in Period}$. Example: $(\$500,000 - \$50,000) / 30,000 \text{ hrs} \times 4,000 \text{ hrs} = \$60,000$.

Straight-Line Method: Annual Depreciation = $(\text{Cost} - \text{Residual Value}) / \text{Estimated Service Life}$. Example: $(\$500,000 - \$50,000) / 5 \text{ years} = \$90,000 \text{ per year}$.

Sum-of-the-Years'-Digits: Depreciation = $(\text{Remaining Life} / \text{Sum of Digits}) \times \text{Depreciable Base}$; SYD for 5-year asset = 15. Year 1: $(5/15) \times \$450,000 = \$150,000$; Year 2: $(4/15) \times \$450,000 = \$120,000$.

Double-Declining-Balance: Rate = $2 \times (1 / \text{Useful Life})$; applied to beginning book value (residual value not deducted from base). Example: Year 1 = $40\% \times \$500,000 = \$200,000$; Year 2 = $40\% \times \$300,000 = \$120,000$; depreciation stops when book value reaches residual value.

Depletion Rate: Depletion Cost per Unit = $(\text{Total Cost} - \text{Residual Value}) / \text{Total Estimated Units}$. Example: $\text{€}1,000,000 / 100,000 \text{ oz} = \text{€}10/\text{oz}$; 25,000 oz extracted $\times \text{€}10 = \text{€}250,000$ depletion.

Asset Turnover: Net Sales / Average Total Assets. Example (Siemens 2018): $\text{€}83,044 / [(\text{€}138,915 + \text{€}136,111) / 2] = 0.60 \text{ times}$.

Return on Assets (ROA): Net Income / Average Total Assets (or Profit Margin on Sales $\times$ Asset Turnover). Example (Siemens): $7.4\% \times 0.60 = 4.4\%$.

Prospective Depreciation after Revision: Remaining Book Value / Remaining Revised Life. Example (Nestlé): $\text{CHF}45,000 \text{ book value} / 20 \text{ years remaining} = \text{CHF}2,250 \text{ per year}$.

Illustrative Example

Norwel SA purchases a circuit board stamping machine for $\text{€}24,000 + 5\% \text{ tax } (\text{€}1,200) + \text{platform } (\text{€}1,800) = \text{total cost } \text{€}27,000$. Depreciable base = $\text{€}27,000 - \text{€}2,000 \text{ residual} = \text{€}25,000$. Annual straight-line depreciation = $\text{€}25,000 / 5 = \text{€}5,000$. Using half-year convention: 2022 depreciation = $\text{€}2,500$; 2023 = $\text{€}5,000$. Book value at December 31, 2023 = $\text{€}27,000 - \text{€}7,500 = \text{€}19,500$. At June 30, 2024, accumulated depreciation = $\text{€}2,500 + \text{€}5,000 + \text{€}2,500 \text{ (half-year 2024)} = \text{€}10,000$; book value = $\text{€}17,000$. Recoverable amount = higher of value-in-use ($\text{€}13,000$) and fair value less costs to sell ($\text{€}13,400$) = $\text{€}13,400$. Impairment loss = $\text{€}17,000 - \text{€}13,400 = \text{€}3,600$. Entry: Debit Loss on Impairment $\text{€}3,600$ / Credit Accumulated Depreciation—Machinery $\text{€}3,600$.

Disclosures / Presentation Notes

- Basis of valuation: Companies must disclose whether PP&E is carried at historical cost or under the revaluation model, along with the depreciation methods and useful lives or rates used for each class of assets.
- Carrying amount rollforward: Disclosures should show gross cost, accumulated depreciation, and impairment losses at the beginning and end of the period, including additions, disposals, and currency effects (as illustrated by the Nestlé Group disclosures).
- Impairment disclosures: Companies must disclose the amount of impairment losses recognised and reversed during the period, the events and circumstances leading to the impairment, and how recoverable amount was determined (fair value less costs to sell or value-in-use).
- Mineral resource (E&E) disclosures: Companies must disclose their accounting policies for E&E expenditures (e.g., full-cost vs. successful-efforts), and the amounts of E&E assets, liabilities, income/expense, and operating cash flows related to mineral resource exploration.

Potential Pitfalls

- Depreciation does not provide cash for asset replacement: A common misconception is that accumulated depreciation represents a cash fund set aside for replacing assets. In reality, depreciation is a non-cash expense that reduces book value and net income; replacement funds come only from revenues generated by the asset.
- Declining-balance method omits residual value from the base but still requires a stop: Unlike other methods, the declining-balance rate is applied to the full (gross) book value without first deducting residual value; however, depreciation must cease once book value equals the estimated residual value, and companies often switch to straight-line near end of life to ensure this.
- Revaluation increases bypass income but losses hit income—creating asymmetry: Under IFRS revaluation accounting, upward revaluations above historical cost go to Other Comprehensive Income (not net income), while downward revaluations below historical cost are recognised as impairment losses in net income, creating an asymmetric impact on reported earnings that users must understand when comparing companies.

Notes: The chapter file was read in full (1,999 lines) with no truncation. Some numeric cells in tables (e.g., the SYD schedule, the DDB schedule, and the Siemens financial data table) were partially extracted as text without all column values due to the text-extraction format of the source; values used in formulas and examples above were sourced from the surrounding narrative text and are consistent with the chapter's worked calculations. Appendix 11A (Revaluation of PP&E in a multi-year setting for land and depreciable assets) was fully captured in the file.

Chapter 12: Intangible Assets

Executive Summary

Chapter 12 examines the recognition, measurement, amortization, impairment, and disclosure of intangible assets under IFRS, covering all major categories from patents and trademarks to goodwill. The chapter distinguishes between limited-life intangibles (amortised over useful life) and indefinite-life intangibles (not amortised but tested annually for impairment), and contrasts purchased versus internally created assets. Research costs are always expensed; certain development costs are capitalised once economic viability is achieved.

Key Topics

1. Characteristics of intangible assets: identifiability (separable or arising from contractual/legal rights), lack of physical substance, and non-monetary nature — forming the basis for recognition under IFRS.
2. Valuation of purchased vs. internally created intangibles: purchased intangibles are capitalised at cost (including incidental acquisition costs), while most internally created intangibles are expensed, with narrow exceptions for development-phase costs meeting economic-viability criteria.
3. Amortisation of limited-life intangibles: cost less residual value is allocated over useful life using the pattern of consumption; if indeterminable, straight-line is used. Residual value is assumed zero unless reliable evidence exists.
4. Indefinite-life intangibles: not amortised; instead tested for impairment at least annually by comparing carrying amount to recoverable amount (higher of fair value less costs to sell or value-in-use).
5. Six categories of intangible assets: marketing-related (trademarks, domain names), customer-related (customer lists, order backlogs), artistic-related (copyrights), contract-related (franchises, licences), technology-related (patents), and goodwill.
6. Goodwill: recorded only in a business combination as the excess of purchase price over the fair value of identifiable net assets acquired (master valuation approach); never amortised; tested for impairment annually at the cash-generating unit (CGU) level; goodwill impairment losses cannot be reversed.
7. Impairment of intangibles: limited-life assets are reviewed for indicators and tested when indicators exist; impairment loss equals carrying amount minus recoverable amount; losses on assets other than goodwill may be reversed in a later period up to the would-have-been carrying amount.
8. Research and development costs: all research-phase costs and most development-phase costs are expensed; capitalisation of development costs begins only when all six economic-viability criteria are simultaneously met. Similar costs — start-up costs, initial operating losses, and advertising costs — are also expensed as incurred.

Important Definitions

Term	Definition
Intangible Asset	An identifiable, non-monetary asset without physical substance that provides future economic benefits, normally classified as non-current.
Amortisation	The systematic allocation of the cost (less residual value) of a limited-life intangible asset to expense over its useful life.

Term	Definition
Limited Useful Life	A finite period over which an intangible asset is expected to generate cash flows, determined by legal, contractual, competitive, or other economic factors.
Indefinite Useful Life	A condition where no foreseeable limit exists on the period an intangible will provide cash flows; such assets are not amortised.
Goodwill	The excess of the purchase price of an acquired business over the fair value of its identifiable net assets; recognised only in a business combination and not amortised.
Recoverable Amount	The higher of an asset's fair value less costs to sell and its value-in-use; the benchmark used in impairment testing under IFRS.
Value-in-Use	The present value of estimated future cash flows expected from the continued use and ultimate disposal of an asset.
Economic Viability	The point in the development phase at which it is probable that the future economic benefits of an R&D project will flow to the company, triggering capitalisation of subsequent development costs.
Bargain Purchase	A business combination in which the purchase price is less than the fair value of identifiable net assets acquired; the excess is recognised immediately as a gain.
Cash-Generating Unit (CGU)	The smallest identifiable group of assets that generates cash inflows largely independent of other assets; the unit at which goodwill impairment is tested.

Accounting Treatments

- Purchased intangibles — capitalise at cost (purchase price + legal fees + incidental costs); in a basket purchase allocate on relative fair values. Journal entry: Dr Intangible Asset / Cr Cash (or other consideration).
- Internally created intangibles — expense all research-phase costs immediately; capitalise development-phase costs only once all economic-viability criteria are met. Journal entry before viability: Dr R&D Expense / Cr Cash; after viability: Dr Intangible Asset / Cr Cash.
- Amortisation of limited-life intangibles — Dr Amortisation Expense / Cr Accumulated Amortisation (or the asset account directly) each period; base = cost minus residual value; method follows pattern of consumption or straight-line.
- Impairment loss — if carrying amount exceeds recoverable amount: Dr Loss on Impairment / Cr Intangible Asset (or Accumulated Impairment). Recoverable amount becomes the new carrying amount for future amortisation. Impairment on goodwill is never reversed; impairment on other intangibles may be reversed.
- Goodwill recognition in a business combination — Dr identifiable assets at FV, Dr Goodwill (residual) / Cr Liabilities at FV, Cr Cash. Goodwill is subsequently held at cost less accumulated impairment; no amortisation.
- Start-up costs, initial operating losses, advertising costs — Dr Expense (operating or R&D) / Cr Cash or Payable; no deferral or capitalisation permitted under IFRS.

Critical Formulas / Calculations

Amortisation Expense = (Cost - Residual Value) / Useful Life | Variables: Cost = acquisition cost; Residual Value = estimated net realisable value at end of useful life (often 0); Useful Life = years of expected benefit. | Example: Patent cost €180,000, residual 0, useful life 10 yrs → €180,000 / 10 = €18,000 per year.

Goodwill = Purchase Price - Fair Value of Identifiable Net Assets | Variables: Purchase Price = total consideration paid; Fair Value of Identifiable Net Assets = FV of all identifiable assets minus FV of liabilities assumed. | Example: Feng pays \$400,000; fair value of Tractorling's net assets = \$350,000 → Goodwill = \$50,000.

Impairment Loss = Carrying Amount - Recoverable Amount (only if Carrying Amount > Recoverable Amount) | Variables: Carrying Amount = cost minus accumulated amortisation; Recoverable Amount = max(FV less costs to sell, Value-in-Use). | Example: Patent carrying value €5,000,000; value-in-use €2,000,000 → Impairment loss = €3,000,000.

Amortisation Base with Residual Value = Cost - Residual Value | Example: Customer list cost €6,000,000; residual €60,000; useful life 3 yrs → Base = €5,940,000; annual amortisation = €1,980,000.

Illustrative Example

Sky ASA practice problem (from chapter end-of-chapter section): Sky purchases a patent on 2 January 2023 for €40,000 (5-year life) and files a second patent on 1 July 2023 for legal fees of €11,400 (remaining life = 4.5 yrs, i.e., 54 months). At 31 December 2023, patent amortisation = (€40,000 × 12/60) + (€11,400 × 6/54) = €8,000 + €1,267 = €9,267. Separately, a licence purchased for €20,000 (8-year life starting 1 July 2022) has a carrying amount at 31 December 2023 of €20,000 - (€20,000 × 18/96) = €16,250. With a recoverable amount of €7,000, the impairment loss = €16,250 - €7,000 = €9,250. Statement of financial position intangibles at 31 December 2023: Licences €6,750 (€16,250 - €9,250), Patents €42,133 (€51,400 - €9,267), Goodwill €90,000 = Total €138,883.

Disclosures / Presentation Notes

- Statement of financial position: all intangible assets other than goodwill are reported as a single separate line item; goodwill, if present, is reported as a further separate line item. Contra accounts (accumulated amortisation) need not be shown on the face of the statement.
- Income statement: amortisation expense and impairment losses (and reversals) for intangibles other than goodwill are presented within operating income as separate line items; goodwill impairment is also presented in operating income.
- Notes to financial statements: must disclose the amortisation expense for each class of intangible, a roll-forward (reconciliation) of changes in carrying amounts during the period (additions, disposals, amortisation, impairment, exchange differences), useful lives or amortisation rates used, and the method of amortisation for limited-life intangibles.
- R&D disclosures: the total R&D costs expensed in each period for which an income statement is presented must be disclosed in the notes, along with the accounting policy for distinguishing research from development and for capitalising development costs (e.g., GlaxoSmithKline discloses capitalisation begins when a regulatory filing has been made and approval is highly probable).

Potential Pitfalls

- Confusing internally created intangibles with purchased ones: IFRS prohibits capitalising internally developed brands, mastheads, customer lists, and most other internally generated intangibles (including goodwill) because they fail the identifiability criterion or cannot be measured reliably; only externally purchased or narrow development-phase costs that satisfy all six economic-viability criteria may be capitalised.
- Misapplying goodwill impairment rules: goodwill impairment is tested at the cash-generating unit level (not the individual asset level), the loss is first absorbed by goodwill and then allocated pro-rata to other CGU assets, and — critically — goodwill impairment losses can never be reversed, unlike impairment losses on other intangible assets.
- Treating all development costs as immediately expensable: a common error is to expense all R&D in the period incurred; under IFRS, once a project reaches economic viability (all six criteria simultaneously met), subsequent development costs must be capitalised. Failure to capitalise these costs understates assets and overstates expenses in the period and understates amortisation in future periods.

Notes: The chapter text is complete and was read in full (2,128 lines). Certain numerical tables embedded in illustrations (e.g., Illustration 12.4 Tractorling balance sheet totals, Illustration 12.13 R&D-to-sales percentages for named companies, Illustration 12.16 GlaxoSmithKline income statement figures) appear to have had their specific numbers omitted or truncated in the source text file, likely due to PDF extraction formatting issues. Values referenced in the practice problem and Lerch/Arcon/Kohlbuy worked examples were intact and used above. IFRS vs. U.S. GAAP comparison section (Learning Objective 6) was present and incorporated.

Chapter 13: Current Liabilities and Provisions

Executive Summary

Chapter 13 explains the recognition, measurement, and reporting of current liabilities, provisions, and contingencies under IFRS. It draws a clear line between definite current obligations (accounts payable, taxes, wages) and uncertain obligations (provisions), then further distinguishes provisions—which meet the criteria for recognition—from contingent liabilities and assets, which are generally disclosed but not recorded.

Key Topics

1. Definition and classification of current liabilities: A liability is a present obligation arising from past events that requires transfer of economic resources; it is classified as current if expected to be settled within the operating cycle or within 12 months of the reporting date.
2. Types of current liabilities: Includes accounts payable, notes payable (interest-bearing and zero-interest-bearing), current maturities of long-term debt, dividends payable, unearned revenues, sales and value-added taxes payable, income taxes payable, and employee-related liabilities (payroll deductions, compensated absences, bonuses).
3. Short-term obligations expected to be refinanced: A short-term debt may be reclassified as non-current only if, at the reporting date, the company already has the contractual right to defer settlement for at least 12 months; post-period refinancing agreements do not qualify.
4. Provisions—recognition and measurement: A provision is recorded only when (1) a present legal or constructive obligation exists from a past event, (2) outflow of resources is probable (>50%), and (3) a reliable estimate is possible. The amount is the 'best estimate' of the expenditure needed to settle at the reporting date.
5. Common types of provisions: Litigation provisions, assurance-type and service-type warranty provisions, consideration payable (premiums, coupons, rebates), environmental liabilities (including unwinding of discount), onerous contracts, and restructuring provisions.
6. Environmental liabilities: Recognised at fair value (or present value of future costs) and capitalised as part of the related long-lived asset; the discount is subsequently unwound through interest expense each period.
7. Contingent liabilities and assets: Contingent liabilities are disclosed (not recognised) when possible but not probable, and omitted entirely when remote. Contingent assets are disclosed when probable and recognised only when virtually certain ($\geq 90\%$).
8. Liquidity analysis: Two key ratios—the current ratio ($\text{Current Assets} \div \text{Current Liabilities}$) and the acid-test ratio ($[\text{Cash} + \text{Short-Term Investments} + \text{Net Receivables}] \div \text{Current Liabilities}$)—assess a company's ability to meet short-term obligations.

Important Definitions

Term	Definition
Current liability	An obligation expected to be settled within the entity's normal operating cycle or within 12 months after the reporting date.

Term	Definition
Provision	A liability of uncertain timing or amount, recognised when a present obligation (legal or constructive) from a past event makes a probable outflow of resources reliably estimable.
Legal obligation	An obligation arising from a contract, legislation, or other operation of law that binds the entity to a specific course of action.
Constructive obligation	An obligation that arises from an entity's established pattern of practice, published policies, or specific statements that create a valid expectation in third parties that the entity will discharge the responsibility.
Contingent liability	A possible obligation whose existence will be confirmed only by uncertain future events not wholly within the entity's control, or a present obligation where payment is not probable or the amount cannot be reliably estimated; not recognised on the statement of financial position.
Contingent asset	A possible asset arising from past events whose existence will be confirmed by uncertain future events not wholly in the entity's control; disclosed when probable, recognised only when virtually certain.
Onerous contract	A contract in which the unavoidable costs of fulfilling the obligations exceed the economic benefits expected to be received; requires recognition of a provision at the lower of fulfilment cost or exit penalty.
Assurance-type warranty	A warranty that guarantees a product meets agreed specifications at the point of sale; treated as a provision (expensed at sale) rather than a separate performance obligation.
Service-type warranty	An extended warranty sold separately that provides services beyond mere defect assurance; recorded as a separate performance obligation in Unearned Warranty Revenue and recognised over the warranty period.
Virtually certain	A probability threshold of approximately 90% or more, used to trigger recognition of a contingent asset as a full asset.
Operating cycle	The time elapsing between the acquisition of goods or services and the final cash collection from sales, used to determine the current/non-current boundary for certain liabilities.
Compensated absences	Paid time off (vacation, sick leave, etc.) for which a liability must be accrued when rights are vested or accumulating, based on expected future wage rates.

Accounting Treatments

- Interest-bearing note payable: Dr Cash / Cr Notes Payable at face value on issuance; at period-end, Dr Interest Expense / Cr Interest Payable for accrued interest (e.g., $€100,000 \times 6\% \times 4/12 = €2,000$); at maturity, Dr Notes Payable + Interest Payable / Cr Cash.
- Zero-interest-bearing note payable: Dr Cash (present value) / Cr Notes Payable (present value) at issuance; at period-end, Dr Interest Expense / Cr Notes Payable to accrete the discount; at maturity, Dr Notes Payable (face) / Cr Cash.

- Assurance-type warranty provision: At sale date, Dr Warranty Expense / Cr Warranty Liability for estimated future claims; as claims are settled, Dr Warranty Liability / Cr Cash (or Inventory/Payroll).
- Environmental liability: Dr Long-Lived Asset (e.g., Drilling Platform) / Cr Environmental Liability at present value of decommissioning costs; each period Dr Depreciation Expense / Cr Accumulated Depreciation (cost allocation), and Dr Interest Expense / Cr Environmental Liability (unwinding of discount).
- Restructuring provision: Recognised when a formal plan exists and affected parties have been notified; Dr Restructuring Expense / Cr Restructuring Liability. Only direct, incremental costs (e.g., employee termination, lease penalties) are included; future operating losses and retraining costs are excluded.
- Contingent liability disclosure only (not recorded): When probability is 'possible but not probable' (roughly 5–50%), disclose nature, estimated financial effect, and timing uncertainties in notes; no journal entry is made. When remote (<5%), no disclosure is required.

Critical Formulas / Calculations

Interest on note payable: $\text{Interest} = \text{Principal} \times \text{Rate} \times \text{Time (in years)}$.

Example: $\text{€}100,000 \times 6\% \times (4/12) = \text{€}2,000$ interest for a 4-month note.

Warranty liability (expected value method): $\text{Warranty Liability} = \sum (\text{Probability}_i \times \text{Cost}_i)$ across all outcome scenarios. Example: 100 machines $\times$ \$200 expected cost per unit = \$20,000 total warranty provision; after \$4,000 incurred in year 1, balance = \$16,000.

Environmental liability (present value): $\text{Environmental Liability} = \text{Future Dismantlement Cost} \times \text{PV Factor}$. Example: $\text{\$}1,000,000 \times 0.62092$ (10%, 5 years) = \$620,920 initial liability.

Annual interest (unwinding discount): $\text{Interest Expense} = \text{Carrying Amount of Liability} \times \text{Discount Rate}$. Example: $\text{\$}620,920 \times 10\% = \text{\$}62,092$ in year 1.

Current ratio: $\text{Current Ratio} = \text{Current Assets} \div \text{Current Liabilities}$. Example (Wilmar): $\text{Current Assets} \div \text{Current Liabilities} = 2.02\times$.

Acid-test ratio: $\text{Acid-Test Ratio} = (\text{Cash} + \text{Short-Term Investments} + \text{Net Receivables}) \div \text{Current Liabilities}$. Example (Wilmar): = 1.27.

Sales tax extraction: $\text{Sales (net)} = \text{Total Receipts} \div (1 + \text{Tax Rate})$; $\text{Tax Payable} = \text{Total Receipts} - \text{Sales (net)}$. Example: $\text{€}150,000 \div 1.04 = \text{€}144,231$ sales; €5,769 tax payable.

Illustrative Example

Denson Machinery (Assurance-Type Warranty): Sells 100 machines at \$5,000 each in 2022; total revenue = \$500,000. Estimated warranty cost = \$200/unit $\times$ 100 = \$20,000. In 2022, \$4,000 of warranty costs are incurred: Dr Warranty Expense \$4,000 / Cr Cash \$4,000. At Dec 31, 2022, adjusting entry for remaining estimated liability: Dr Warranty Expense \$16,000 / Cr Warranty Liability \$16,000. Balance sheet shows Warranty Liability (current) = \$16,000; income statement shows Warranty Expense = \$20,000. In 2023, remaining claims settled: Dr Warranty Liability \$16,000 / Cr Cash \$16,000.

Disclosures / Presentation Notes

- Provisions rollforward: IFRS requires a reconciliation of opening to closing balance for each major class of provision, showing additions, amounts used, reversals, and the effect of unwinding the discount and changes in discount rate (as illustrated by Nokia's provisions note).

- Contingent liabilities must be disclosed in notes (unless remote) with: a brief description of nature, an estimate of financial effect if practicable, an indication of timing uncertainties, and the possibility of any reimbursement from third parties.
- Current liabilities presentation: Reported at full maturity value on the statement of financial position (present-value adjustment deemed immaterial for short periods); listed after non-current liabilities and may be ordered by maturity, descending amount, or liquidation preference. Secured liabilities and pledged collateral must be clearly identified.
- Cumulative preference dividends in arrears are not recognised as a liability (no past obligating event) but must be disclosed in a note or parenthetically in the equity section; share dividends distributable are reported in equity, not as a current liability.

Potential Pitfalls

- Classifying refinanced short-term debt as non-current when the refinancing agreement was finalised after the reporting date: Under IFRS, the right to defer settlement must exist at the reporting date itself—a post-period agreement does not retroactively reclassify the debt as non-current.
- Recording a provision for general future risks (self-insurance, anticipated future losses, or expected economic downturns): IFRS prohibits recognising provisions for obligations that do not yet exist; a liability requires a past obligating event, so future-event contingencies cannot be accrued even if highly probable.
- Confusing assurance-type and service-type warranties: Assurance-type warranties are expensed at the point of sale as a provision; service-type warranties are recorded as a separate performance obligation (Unearned Warranty Revenue) and recognised as revenue on a straight-line basis over the extended coverage period—mixing the two methods misstates both revenue and liabilities.

Notes: The chapter file was complete at 1,707 lines and contained no apparent truncation. However, actual numerical amounts for journal entry debits/credits in several illustrations (e.g., Illustrations 13.5–13.9 entries for Landscape Co.) were represented as labels only without dollar amounts in the extracted text, likely because the original PDF used a columnar table format that was stripped during extraction. Where amounts could be derived from surrounding narrative they have been included. Appendices, if any, and end-of-chapter exercise/problem numbers beyond the practice problem were not summarised as they are drill exercises rather than substantive content.

Chapter 14: Non-Current Liabilities

Executive Summary

Chapter 14 covers the recognition, measurement, and presentation of non-current liabilities—primarily bonds payable and long-term notes payable—under IFRS. It explains how bonds and notes are valued at the present value of future cash flows, how discounts and premiums are amortized using the effective-interest method, and how gains or losses are recognized when debt is extinguished before maturity through cash, asset transfer, or modification of terms.

Key Topics

1. Types of bonds: The chapter classifies bonds as secured/unsecured, term/serial/callable, convertible, commodity-backed, deep-discount (zero-interest), registered/bearer, income, and revenue bonds—each designed to meet different investor and issuer needs.
2. Bond pricing and present value: A bond's issue price equals the present value of its future interest and principal cash flows discounted at the market (effective) rate; when the market rate exceeds the stated rate the bond sells at a discount, and when it is lower the bond sells at a premium.
3. Effective-interest amortization: IFRS requires the effective-interest method to amortize bond discount or premium, producing a constant effective interest rate applied to the changing carrying value of the bond each period.
4. Long-term notes payable: Notes are valued identically to bonds at present value; zero-interest-bearing notes, interest-bearing notes issued at off-market rates, and notes issued for property/services all require imputed interest rate analysis and discount/premium amortization.
5. Extinguishment of debt: Debt can be extinguished before maturity by paying cash (gain or loss = carrying amount minus reacquisition price), transferring non-cash assets or equity interests (measured at fair value), or through creditor-granted modification of terms (old note extinguished, new note recorded at fair value).
6. Fair value option: Companies may elect to measure bonds and notes payable at fair value each period; unrealized holding gains/losses from general interest-rate changes flow through net income, while those attributable to credit-risk changes are reported in other comprehensive income under IFRS.
7. Off-balance-sheet financing: Arrangements such as unconsolidated subsidiaries and special purpose entities (SPEs) allow companies to borrow without recording the liability; IFRS has tightened consolidation and disclosure rules to curb abuse.
8. Presentation and solvency analysis: Non-current debt due within one year is reclassified as current; key ratios—debt to assets and times interest earned—measure long-run solvency and interest-coverage ability.

Important Definitions

Term	Definition
Non-current liability	An obligation not expected to require payment within one year or the company's operating cycle, whichever is longer; examples include bonds payable, long-term notes, mortgages, and lease liabilities.

Term	Definition
Bond indenture	The formal legal contract between the bond issuer and bondholders specifying face value, coupon rate, maturity date, call provisions, collateral pledged, sinking fund requirements, and restrictive covenants.
Stated (coupon/nominal) rate	The contractual annual interest rate printed on the bond certificate, expressed as a percentage of face value and used to calculate periodic cash interest payments.
Effective (market) rate	The actual rate of return earned by bondholders given the issue price; if the bond sells at a discount the effective rate exceeds the stated rate, and if at a premium the effective rate is lower.
Bond discount	The amount by which the face (par) value of a bond exceeds its issue price, arising when the market rate exceeds the stated rate; amortization of discount increases interest expense each period.
Bond premium	The amount by which the issue price of a bond exceeds its face value, arising when the market rate is below the stated rate; amortization of premium decreases interest expense each period.
Carrying (book) value	The face amount of a bond or note adjusted for any unamortized discount (subtracted) or unamortized premium (added); equals the present value of remaining cash flows at the original effective rate.
Imputed interest rate	An estimated market interest rate applied when a note's stated rate is unreasonable or absent, used to determine the present value of the note and the implicit interest charge.
Extinguishment of debt	The retirement or settlement of a debt obligation before or at maturity; any difference between the reacquisition price and the carrying amount is recognized immediately in income as a gain or loss.
Off-balance-sheet financing	Borrowing arrangements—such as non-consolidated subsidiaries or special purpose entities—structured to avoid recording debt on the statement of financial position, thereby understating leverage.
Debt to assets ratio	Total liabilities divided by total assets; measures the proportion of assets financed by creditors and indicates default risk.
Times interest earned	(Net income + interest expense + income tax expense) divided by interest expense; measures how many times operating earnings cover interest obligations.

Accounting Treatments

- Bond issuance at a discount: Debit Cash (proceeds), credit Bonds Payable at face value net of discount under IFRS (i.e., record at carrying value directly). Each period: Debit Interest Expense (carrying value x effective rate), Credit Cash (face x stated rate x period), and the difference reduces the discount, increasing carrying value.
- Bond issuance at a premium: Record bonds payable at carrying value (face + premium net under IFRS). Each period: Debit Interest Expense (carrying value x effective rate), Credit Bonds Payable

(premium reduction), Credit Cash (face x stated rate x period), reducing carrying value toward face over time.

- Bonds issued between interest dates: The bond is recorded at its present value; accrued interest paid by the investor is debited to Interest Expense (or Interest Payable). On the first coupon date, full coupon cash is paid and amortization covers only the partial period from issue date to coupon date.
- Zero-interest-bearing notes: Record at present value (cash received or fair value of consideration); the difference from face value is a discount debited to Notes Payable. Annually: Debit Interest Expense, Credit Notes Payable for the effective-interest amount (no cash interest paid).
- Extinguishment before maturity (cash): Amortize discount or premium to the redemption date, then: Debit Bonds Payable (carrying amount), Debit or Credit Loss/Gain on Extinguishment of Debt, Credit Cash (reacquisition price).
- Debt modification as extinguishment: Debit Notes Payable—old (carrying value), Credit Notes Payable—new (fair value of restructured terms), Credit Gain on Extinguishment of Debt (difference); subsequent interest expense uses the new effective market rate.

Critical Formulas / Calculations

Bond issue price = PV of principal + PV of interest annuity. Variables: F = face value, r = effective (market) rate per period, n = number of periods, C = coupon payment per period. Formula: $\text{Price} = F \times \text{PVF}(r, n) + C \times \text{PVF-OA}(r, n)$. Example: F=€100,000, C=€4,000 (8%/2), r=5% (10%/2), n=10 → $\text{Price} = 100,000 \times 0.61391 + 4,000 \times 7.72173 = €61,391 + €30,887 = €92,278$ (discount bond).

Effective-interest expense per period = Carrying value at period start x Effective rate per period. Example: €92,278 x 0.05 = €4,614 interest expense; Cash paid = €4,000; Discount amortized = €614.

Gain or loss on extinguishment = Carrying amount of debt - Reacquisition price. If positive: gain; if negative: loss. Example: Carrying value €94,925, call price €101,000 (101% of €100,000) → Loss = €101,000 - €94,925 = €6,075.

Debt to assets ratio = Total Liabilities / Total Assets. Example: Novartis: \$66,871M / \$145,563M = 0.46 = 46%.

Times interest earned = (Net income + Interest expense + Income tax expense) / Interest expense. Example: Novartis: (\$12,614M + \$957M + \$1,221M) / \$957M = \$14,792M / \$957M = 15.5 times.

Illustrative Example

Evermaster AG issues €100,000 of 8% bonds (semiannual coupons of €4,000) on January 1, 2022, due January 1, 2027, with a market rate of 10%. Issue price = €92,278 (discount = €7,722). First period (Jul 1, 2022): Interest Expense = €92,278 x 5% = €4,614; Cash = €4,000; Discount amortized = €614; new carrying value = €92,892. On January 1, 2024 (after 4 semiannual periods) carrying value = €94,925. Company calls bonds at 101 (€101,000): Loss on extinguishment = €101,000 - €94,925 = €6,075 recognized in income.

Disclosures / Presentation Notes

- Non-current liabilities are presented on the statement of financial position, usually as a single line item supported by detailed note disclosures covering nature of liabilities, maturity dates, interest rates, call provisions, conversion features, assets pledged as collateral, and restrictive covenants.

- Long-term debt maturing within one year must be reclassified to current liabilities unless the company has completed a refinancing agreement by the reporting date (IFRS) or by the issuance date (U.S. GAAP); sinking fund requirements and scheduled maturities for the next five years must be disclosed.
- If the fair value option is elected, bonds and notes are measured at fair value at each reporting date; unrealized gains/losses from interest-rate changes are in net income, while those from credit-risk changes are reported in other comprehensive income under IFRS.
- Off-balance-sheet commitments and contractual obligations must be disclosed in tabular format (required under U.S. SEC rules), including future cash flows by year bucket; the fair value of all long-term debt must also be disclosed for comparison to carrying amounts.

Potential Pitfalls

- ❑ Confusing stated rate and effective rate: Students often use the stated rate to compute interest expense rather than applying the effective rate to the period-opening carrying value; this produces incorrect amortization amounts and a carrying value that never reaches face value at maturity.
- ❑ Misclassifying extinguishment gains/losses: Gains and losses on early debt retirement (including call premiums and unamortized discount/premium at redemption) must be fully recognized in income in the period of extinguishment—they cannot be deferred or amortized over the remaining original term.
- ❑ Treating off-balance-sheet arrangements as immaterial: Companies sometimes assume that non-consolidated subsidiaries or SPE liabilities need not be evaluated for consolidation; under IFRS control criteria (power + variable returns + ability to affect returns), these entities may require full consolidation regardless of ownership percentage, and failure to consolidate can materially understate reported leverage.

Notes: The source file is complete and was read in full across four segments (lines 0–1768). Numerical tables for the full amortization schedules (Illustrations 14.7, 14.9, 14.16, 14.20, 14.24) had many interior rows omitted from the text extract (only the first row and totals were legible); formula footnotes were intact. No figures (charts) from the opening story were reproduced in the text. Appendix/exercise answer tables were partially truncated but all core conceptual and numerical content was available.

Chapter 15: Equity

Executive Summary

Chapter 15 covers the accounting and reporting of corporate equity, examining the issuance of ordinary and preference shares, treasury share transactions, and the four types of dividends. It concludes by showing how to present the equity section of the statement of financial position and how to compute key analytical ratios such as return on equity, payout ratio, and book value per share.

Key Topics

1. Corporate form and share system: Corporations issue shares to raise capital; each share within a class carries equal rights including participation in profits, voting, asset claims on liquidation, and the preemptive right to maintain proportionate ownership.
2. Equity components: Equity is classified into share capital, share premium, retained earnings, accumulated other comprehensive income, treasury shares, and non-controlling interest, with a key distinction drawn between contributed (paid-in) capital and earned capital (retained earnings).
3. Issuance of shares: Shares may be issued at par, at no-par with a stated value, in lump-sum transactions (allocated using the proportional or incremental method), for non-cash consideration (at fair value of goods/services received), or at a cost charged against share premium.
4. Preference shares: A special share class that may carry dividend priority, liquidation priority, convertibility into ordinary shares, callability, and non-voting status; redeemable preference shares with mandatory redemption features must be classified as liabilities under IFRS.
5. Treasury shares: Shares repurchased but not retired; recorded under the cost method as a deduction from equity, not as an asset; gains or losses on reissuance are routed through Share Premium—Treasury (not income), with any excess deficit charged to Retained Earnings.
6. Types of dividends: Cash dividends create a liability on declaration; property dividends require remeasurement to fair value at declaration with any gain or loss recognized; liquidating dividends return contributed capital and must be disclosed; share dividends reclassify retained earnings to share capital at par value without changing total equity.
7. Share splits versus share dividends: A share split increases shares outstanding and reduces par value per share with only a memorandum entry; a share dividend transfers par value from retained earnings to share capital and does not change par value per share.
8. Equity analysis ratios: Return on ordinary share equity (net income minus preference dividends divided by average ordinary equity), payout ratio (cash dividends to net income), and book value per share (ordinary shareholders' equity divided by outstanding ordinary shares) are the primary ratios used to evaluate equity.

Important Definitions

Term	Definition
Equity	The residual interest in the assets of a company after deducting all liabilities; equals net assets and is also called shareholders' equity or corporate capital.
Contributed (Paid-in) Capital	The total amount paid in by shareholders for shares issued, comprising par value of outstanding shares plus share premium (net of any discounts).

Term	Definition
Retained Earnings	Earned capital accumulated from profitable operations that has not been distributed as dividends; represents undistributed income remaining invested in the company.
Ordinary Shares	The basic residual ownership interest in a corporation, bearing the ultimate risks of loss and benefits of success, with no guaranteed dividends or liquidation priority.
Preference Shares	A special class of shares that possess certain preferences over ordinary shares—typically priority as to dividends and liquidation proceeds—in exchange for sacrificing some ordinary rights such as voting.
Treasury Shares	A corporation's own shares that have been issued, fully paid, and subsequently reacquired but not yet retired; deducted from equity and carry no voting rights, dividend rights, or liquidation claims.
Preemptive Right	The inherent right of existing shareholders to purchase a proportionate share of any new issue of the same class of shares, protecting them from involuntary dilution of their ownership interest.
Liquidating Dividend	A dividend paid from contributed capital rather than retained earnings, representing a return of shareholders' investment rather than a distribution of profits.
Share Premium	Amounts paid by shareholders in excess of a share's par or stated value; reported separately in equity and not available for recognition as income.
Par Value	A nominal value assigned to a share in the corporate charter; has no relationship to fair value but establishes the minimum legal capital amount credited to the Share Capital account.

Accounting Treatments

- Issuance of par value shares: Debit Cash (or asset received), Credit Share Capital—Ordinary at par value, Credit Share Premium—Ordinary for any excess. Direct issue costs (underwriting, legal, printing) are debited to Share Premium; indirect costs (management salaries) are expensed.
- Treasury shares—purchase (cost method): Debit Treasury Shares at full repurchase cost; Credit Cash. The Share Capital and Share Premium accounts are not adjusted on purchase. Treasury Shares is reported as a deduction from total equity.
- Treasury shares—reissuance above cost: Debit Cash, Credit Treasury Shares at cost, Credit Share Premium—Treasury for the excess. No gain is recognised in income.
- Treasury shares—reissuance below cost: Debit Cash, Debit Share Premium—Treasury (to extent of existing credit balance), Debit Retained Earnings (any remaining shortfall), Credit Treasury Shares at cost.
- Cash dividend: At declaration — Debit Retained Earnings (or Cash Dividends Declared), Credit Dividends Payable (current liability). At payment — Debit Dividends Payable, Credit Cash.
- Share dividend: At declaration — Debit Retained Earnings at par value of shares distributable, Credit Ordinary Share Dividend Distributable (shown in equity, not as a liability). At distribution — Debit Ordinary Share Dividend Distributable, Credit Share Capital—Ordinary. No change in total equity.

Critical Formulas / Calculations

Return on Ordinary Share Equity = (Net Income - Preference Dividends) / Average Ordinary Shareholders' Equity. Variables: Net Income = total earnings for the period; Preference Dividends = dividends declared on preference shares; Average Ordinary Shareholders' Equity = average of opening and closing ordinary equity. Example: $(\$360,000 - \$54,000) / \$2,550,000 = 12\%$.

Payout Ratio = Cash Dividends (to Ordinary Shareholders) / (Net Income - Preference Dividends). Variables: Cash Dividends = dividends paid in cash to ordinary shareholders; Net Income = earnings for the period; Preference Dividends = dividends paid to preference shareholders. Example: $\text{€}100,000 / \text{€}500,000 = 20\%$.

Book Value per Share = Ordinary Shareholders' Equity / Outstanding Ordinary Shares. Variables: Ordinary Shareholders' Equity = total equity less preference share equity; Outstanding Ordinary Shares = issued shares minus treasury shares. Example: $\text{HK\$}1,000,000 / 100,000 \text{ shares} = \text{HK\$}10 \text{ per share}$.

Lump-Sum Allocation—Proportional Method: Proceeds allocated to each class = (Fair Value of Class / Aggregate Fair Value of All Classes) × Total Proceeds. Example: Ordinary class fair value \$20,000; preference \$12,000; aggregate \$32,000; total proceeds \$30,000. Ordinary gets $(\$20,000 / \$32,000) \times \$30,000 = \$18,750$; Preference gets \$11,250.

Illustrative Example

Pacific Company issued 100,000 shares of \$1 par ordinary shares at \$10 per share (equity: \$100,000 share capital + \$900,000 share premium + \$300,000 retained earnings = \$1,300,000). On Jan 20, 2022, Pacific repurchases 10,000 shares at \$11 each: Debit Treasury Shares \$110,000 / Credit Cash \$110,000. On Mar 10, 2022, Pacific resells 1,000 treasury shares at \$15 each: Debit Cash \$15,000 / Credit Treasury Shares \$11,000 / Credit Share Premium—Treasury \$4,000. On Mar 21, Pacific sells another 1,000 shares at \$8: Debit Cash \$8,000 / Debit Share Premium—Treasury \$3,000 / Credit Treasury Shares \$11,000. The \$4,000 credit minus the \$3,000 debit leaves a \$1,000 credit balance in Share Premium—Treasury before the next transaction.

Disclosures / Presentation Notes

- Equity section of the statement of financial position must disclose for each class of shares: number of shares authorized, issued, and outstanding; par or stated value; dividend and liquidation preferences; participation rights; call prices and dates; conversion or exercise prices and dates; sinking fund requirements; and unusual voting rights.
- Dividends in arrears on cumulative preference shares do not constitute a liability but must be disclosed in the notes to the financial statements, with the amount clearly stated (e.g., 'The Company is in arrears on preference shares in the amount of X').
- IFRS requires a statement of changes in equity showing: total comprehensive income (split between owners of the parent and non-controlling interests), effects of retrospective restatements, and a reconciliation of each equity component from opening to closing balance including profit/loss, other comprehensive income, and owner transactions.
- Restrictions on retained earnings (e.g., from bond covenants, treasury share purchases, legal reserves) are best disclosed by note, identifying the source, pertinent provisions, and the restricted or unrestricted amount; liquidation preferences must be disclosed in the equity section itself, not only in

the notes.

Potential Pitfalls

- Treating treasury shares as an asset: Treasury shares are not assets—repurchasing own shares reduces both assets and equity simultaneously. Classifying them as assets overstates total assets and misstates equity, violating the conceptual definition that an asset must provide probable future economic benefits to the entity.
- Recognising gains or losses on treasury share transactions in income: Any difference between the cost of treasury shares and their resale price must be routed through Share Premium—Treasury (credit for gains; debit up to the credit balance for losses) and then to Retained Earnings—never to the income statement, because corporations cannot realise a profit or loss on transactions with their own shareholders.
- Confusing share dividends with share splits: A share dividend requires a journal entry (debit Retained Earnings at par value, credit Share Dividend Distributable) and increases total par value of outstanding shares without changing par value per share. A share split requires only a memorandum entry, reduces par value per share, and changes no dollar amounts in the equity accounts; treating one like the other produces incorrect retained earnings balances and misclassified equity components.

Notes: The chapter text was fully readable with no truncation. Some illustration tables (e.g., Illustrations 15.1–15.18) contained partially extracted numerical data due to PDF-to-text conversion formatting; specific dollar amounts in columnar illustrations (particularly the BASF Statement of Equity in Illustration 15.18 and the D'Ouille practice problem solution) may be incomplete. Appendix 15A covering detailed dividend preference calculations and book value per share with arrears was included in the reading. No figures or appendices appear to be missing from the extracted text.

Chapter 16: Dilutive Securities and Earnings per Share

Executive Summary

Chapter 16 explains the accounting for dilutive securities—convertible bonds, convertible preference shares, share warrants, and share-based compensation plans—under IFRS, and then details the computation of both basic and diluted earnings per share (EPS). The central conclusion is that compound instruments must be bifurcated into their liability and equity components using the with-and-without method, share-based compensation must be expensed at fair value over the service period, and EPS disclosures must reflect the worst-case dilutive impact of all potentially dilutive securities.

Key Topics

1. **Convertible Debt as a Compound Instrument:** Under IFRS, convertible bonds are split into a liability component (present value of cash flows discounted at the market rate for similar non-convertible debt) and an equity component (the residual), using the with-and-without method. The equity piece is recorded in Share Premium—Conversion Equity and does not change over the bond's life.
2. **Settlement of Convertible Bonds:** Four scenarios are covered—repurchase at maturity (no gain/loss), conversion at maturity (book value method, no gain/loss), conversion before maturity (carrying value transferred, no gain/loss), and repurchase before maturity (gain/loss on the liability component; equity adjustment recognised in equity).
3. **Induced Conversions (Sweeteners):** When an issuer pays extra consideration to accelerate conversion, the additional payment is expensed in the current period as Conversion Expense, not deducted from equity, because the IASB views it as payment for a service.
4. **Share Warrants and Rights:** Proceeds from bonds issued with detachable warrants are allocated between the debt (PV of cash flows) and equity (residual) using the with-and-without method. Share rights issued to existing shareholders require only a memorandum entry; upon exercise, cash received determines the credit to share capital accounts.
5. **Share-Based Compensation—Fair Value Method:** IFRS mandates the fair value method for all share-option and restricted-share plans. Total compensation is measured at the grant date using an option-pricing model and recognised over the service (vesting) period. Expense is not reversed for market condition forfeitures but is reversed for service condition forfeitures.
6. **Restricted Shares and Employee Share-Purchase Plans:** Restricted-share plans record total compensation at grant-date fair value as Unearned Compensation (contra equity) and amortise it over the vesting period. All ESPPs under IFRS are compensatory; the discount from market price is recognised as Compensation Expense.
7. **Basic Earnings per Share:** Basic EPS equals income available to ordinary shareholders (net income minus current-year preference dividends) divided by weighted-average ordinary shares outstanding. Share splits and stock dividends require retroactive restatement of all prior periods presented.
8. **Diluted Earnings per Share:** Diluted EPS uses the if-converted method for convertible securities (add back after-tax interest to the numerator; add assumed shares to the denominator) and the treasury-share method for options/warrants (incremental shares = shares issued on assumed exercise minus treasury shares repurchasable with proceeds). Antidilutive securities are excluded.

Important Definitions

Term	Definition
Dilutive Securities	Financial instruments—such as convertible bonds, convertible preference shares, options, and warrants—that, upon conversion or exercise, reduce (dilute) earnings per ordinary share.
Compound Instrument	A financial instrument that contains both a liability component and an equity component, such as convertible debt; IFRS requires each component to be recognised separately.
With-and-Without Method	IFRS-required allocation technique for compound instruments: first determine the fair value of the liability component (PV of future cash flows at market rate for equivalent non-convertible debt), then assign the residual to equity.
Book Value Method	The approach used upon conversion of bonds or preference shares, whereby the carrying (book) value of the converted instrument determines the amount credited to ordinary share accounts, with no gain or loss recognised.
Grant Date	The date on which a share-option or restricted-share award is given to an employee; fair value is measured at this date and does not change in subsequent periods for accounting purposes.
Service Period (Vesting Period)	The time between the grant date and the vesting date during which the employee must remain in service; compensation cost is allocated ratably over this period.
Basic Earnings per Share (EPS)	Net income minus preference dividends, divided by the weighted-average number of ordinary shares outstanding; applicable to simple capital structures.
Diluted Earnings per Share (EPS)	A 'worst-case' EPS measure that incorporates the effect of all potentially dilutive ordinary shares (from convertible securities, options, and warrants) into both the numerator and denominator.
If-Converted Method	Technique for computing diluted EPS for convertible securities: assumes conversion at the beginning of the period (or issue date if later), adds back after-tax interest (or preference dividends) to the numerator, and adds assumed shares to the denominator.
Treasury-Share Method	Technique for computing diluted EPS for options and warrants: assumes exercise at the beginning of the period and that proceeds are used to repurchase shares at the average market price; only the net incremental shares are added to the denominator.
Antidilutive Securities	Securities that, if converted or exercised, would increase EPS (or reduce a loss per share); they are excluded from diluted EPS calculations.
Intrinsic-Value Method	A disallowed alternative that values options as the excess of market price over exercise price at the grant date; since most options are granted at-the-money, it often produces zero compensation expense and is not permitted under IFRS.

Accounting Treatments

- Issuance of convertible bonds (Roche example): Debit Cash for total proceeds; credit Bonds Payable for the PV of cash flows at the non-convertible market rate (liability component); credit Share Premium—Conversion Equity for the residual (equity component). The bond discount is amortised

using the effective-interest method over the life of the bond.

- Conversion of bonds (book value method): Debit Bonds Payable at carrying value and Share Premium—Conversion Equity; credit Share Capital—Ordinary at par and Share Premium—Ordinary for the remainder. No gain or loss is recognised.
- Induced conversion sweetener: Debit Conversion Expense for the fair value of the additional consideration paid; Debit Share Premium—Conversion Equity; Debit Bonds Payable; credit Share Capital—Ordinary and Share Premium—Ordinary at book value; credit Cash for the sweetener amount.
- Share-option compensation (fair value method): At each year-end during the service period, Debit Compensation Expense and credit Share Premium—Share Options for the ratable portion of grant-date fair value. At exercise, Debit Cash (exercise price × shares) and Share Premium—Share Options (proportionate balance); credit Share Capital—Ordinary (par) and Share Premium—Ordinary (excess). At expiration, reclassify Share Premium—Share Options to Share Premium—Expired Share Options.
- Restricted shares: On grant date, Debit Unearned Compensation (contra equity) and credit Share Capital—Ordinary and Share Premium—Ordinary. Each period, Debit Compensation Expense and credit Unearned Compensation. On forfeiture before vesting, reverse all previously recorded compensation expense and the share accounts.
- Employee share-purchase plan (IFRS): Debit Cash (purchase price × shares) and Compensation Expense (discount × shares); credit Share Capital—Ordinary (par × shares) and Share Premium—Ordinary (excess over par).

Critical Formulas / Calculations

Basic EPS = (Net Income - Preference Dividends) / Weighted-Average Ordinary Shares Outstanding. Example: Net income = €300,000; preference dividends = €0; weighted-average shares = 100,000 → Basic EPS = €3.00.

Weighted-Average Shares = $\sum$ (Shares Outstanding × Fraction of Year). Example: 90,000 shares for 3/12 + 81,000 for 4/12 + 141,000 for 2/12 = 22,500 + 27,000 + 23,500 = 73,000 weighted-average shares [VERIFY exact totals from Franks Inc. illustration].

Diluted EPS (if-converted, convertible bonds) = (Net Income + Interest Expense Net of Tax) / (Weighted-Average Shares + Assumed Converted Shares). Example: Net income = £210,000; interest net of tax on 6% bond = £37,200; assumed shares = 20,000 → Adjusted numerator = £247,200; adjusted denominator = 120,000 → Diluted EPS ≈ £2.06.

Treasury-Share Method Incremental Shares = Shares Issued on Exercise - (Proceeds / Average Market Price). Example: 1,500 options at exercise price €30, average market price €50 → Proceeds = €45,000; treasury shares repurchasable = 900; incremental shares = 1,500 - 900 = 600.

Equity Component of Convertible Bond = Total Proceeds - PV of Liability Component. Example: Roche bonds: total proceeds = €2,000,000; PV of liability = €1,805,626 → Equity component = €194,374.

SAR Compensation Expense (liability award, percentage approach) = (% of service period elapsed × Total Estimated Fair Value) - Cumulative Expense Previously Recognised. Example: Year 1 of 2-year plan; fair value = R\$30,000 total; expense = 50% × R\$30,000 = R\$15,000.

Illustrative Example

Roche Group issues 2,000 convertible bonds (4-year, 6% stated rate, €1,000 par, €2,000,000 proceeds) when the market rate on similar non-convertible debt is 9%. $PV \text{ of liability} = €2,000,000 \times 0.70843 + €120,000 \times 3.23972 = €1,416,860 + €388,766 = €1,805,626$. Equity component = $€2,000,000 - €1,805,626 = €194,374$. Journal entry at issuance: Debit Cash €2,000,000; Credit Bonds Payable €1,805,626; Credit Share Premium—Conversion Equity €194,374. At conversion on 12/31/2023 (carrying value €1,894,464 per amortisation schedule): Debit Bonds Payable €1,894,464 and Share Premium—Conversion Equity €194,374; Credit Share Capital—Ordinary €500,000 ($2,000 \times 250 \times €1 \text{ par}$) and Share Premium—Ordinary €1,588,838. No gain or loss is recognised.

Disclosures / Presentation Notes

- EPS on the income statement face: Public companies must present both basic and diluted EPS with equal prominence on the face of the income statement; when discontinued operations exist, per-share amounts for continuing operations, discontinued operations, and net income are each required either on the face or in the notes.
- EPS reconciliation disclosure: Companies must disclose the numerators and denominators used in basic and diluted EPS, reconcile them to net income, identify the individual effect of each class of dilutive instrument, and list antidilutive securities excluded from diluted EPS.
- Share-compensation disclosures: Entities with share-based payment arrangements must disclose (1) the nature and extent of plans in existence, (2) how the fair value of equity instruments or services was determined, and (3) the effect on net income and financial position for the period.
- Post-period share transactions: Any ordinary share or potential ordinary share transactions occurring after the reporting date but before financial statement authorisation that would materially change the share count must be disclosed.

Potential Pitfalls

- ❑ Confusing the with-and-without method with relative fair value: IFRS requires the liability component to be determined first (at market rate for non-convertible debt), with equity as the residual. Assigning value to equity first and deriving the liability is explicitly prohibited; it violates the definition of equity as a residual.
- ❑ Incorrectly including antidilutive securities in diluted EPS: Options whose exercise price exceeds the average market price, or convertible debt whose per-share income effect exceeds basic EPS, are antidilutive and must be excluded entirely—they cannot offset other dilutive securities. Failing to screen each security individually in a multi-security complex capital structure (using the ranking and step-down method from Appendix 16B) is a common error.
- ❑ Reversing share-option expense for market-condition forfeitures: When options are not exercised because the share price never exceeded the exercise price (a market condition), compensation expense already recognised must not be reversed. Reversal is only permitted for service-condition forfeitures (employee leaves before vesting). Confusing these two forfeiture types leads to understated compensation expense in the period of expiration.

Notes: The source file appears complete and covers all five main Learning Objectives plus Appendices 16A (Share-Appreciation Rights) and 16B (Comprehensive EPS with multiple securities) and the IFRS vs. U.S. GAAP comparison section. Several illustrations contain placeholder table data without numeric values (particularly Illustrations 16.1–16.8, 16A.1, and 16B.1 through 16B.12) because the PDF-extraction process did not capture the numeric cells of tables and figures—only labels and headings were preserved. Formulas and worked numbers referenced in the narrative were used to reconstruct examples where table data was absent.

Chapter 17: Investments

Executive Summary

Chapter 17 covers the classification, measurement, and reporting of debt and equity investments under IFRS 9, using a mixed-attribute model that values some instruments at amortized cost and others at fair value. The chapter concludes that while fair value provides the most relevant information for financial statement users, a business-model-based approach determines which measurement applies, with unrealized gains and losses routed either through net income or other comprehensive income depending on investment type and intent. Appendices extend the discussion to derivative financial instruments and the three-level fair value disclosure hierarchy.

Key Topics

1. Classification and measurement of debt investments: IFRS 9 requires companies to evaluate two criteria—the business model for managing the investment and whether contractual cash flows are solely payments of principal and interest—to determine if a debt investment is carried at amortized cost (held-for-collection) or fair value (held-for-collection and selling, or trading).
2. Held-for-collection (HFC) debt investments: These are measured at amortized cost using the effective-interest method to amortize any premium or discount; interest revenue is recognised each period, and gains/losses on early sale are reported in income.
3. Held-for-collection and selling (HFCS) debt investments: Reported at fair value on the statement of financial position, with unrealized holding gains and losses recorded in other comprehensive income (OCI) and accumulated in equity; realised gains/losses are recycled to the income statement upon sale.
4. Trading debt investments: Reported at fair value with all unrealized holding gains and losses recognised directly in net income each period, reflecting the intent to profit from short-term price changes.
5. Equity investments and ownership thresholds: Holdings below 20% are generally accounted for at fair value (trading = unrealized gains/losses through income; non-trading election = unrealized gains/losses through OCI); holdings of 20–50% use the equity method; holdings above 50% require consolidated financial statements.
6. Equity method of accounting: The investor records the investment at cost and subsequently adjusts the carrying amount for its proportionate share of the investee's net income or loss; dividends received reduce the carrying amount rather than being recorded as revenue.
7. Impairment of investments: For HFC debt investments, the expected credit loss (ECL) model is used; the impairment loss is measured as the difference between the carrying amount and the present value of revised expected cash flows discounted at the historical effective-interest rate, with reversal permitted under IFRS if credit quality improves.
8. Derivative financial instruments and hedging (Appendix 17A): Derivatives are recognised at fair value on the statement of financial position; gains/losses on speculative derivatives go through income; fair value hedges record both the derivative and the hedged item at fair value in income; cash flow hedges record derivative gains/losses in OCI until the hedged transaction affects earnings.

Important Definitions

Term	Definition
Amortized cost	The initial recognition amount of a financial asset minus principal repayments, plus or minus cumulative amortization of any premium or discount, net of any allowance for impairment.
Fair value	The amount for which an asset could be exchanged between knowledgeable, willing parties in an arm's length transaction; under IFRS, equivalent to the price received to sell an asset in an orderly market transaction at the measurement date.
Held-for-collection (HFC)	A debt investment category where the company's business model is to hold the asset to collect contractual cash flows (principal and interest) over the investment's life; measured at amortized cost.
Held-for-collection and selling (HFCS)	A debt investment category where the objective is both to collect contractual cash flows and to sell the securities; measured at fair value with unrealized gains/losses in OCI.
Trading investments	Debt or equity investments bought and held primarily for sale in the near term to generate profits from short-term price differences; reported at fair value with unrealized gains/losses in net income.
Equity method	An accounting approach used when an investor has significant influence (generally 20–50% ownership) over an investee; the investment is initially recorded at cost and adjusted each period for the investor's share of the investee's net income or loss, with dividends reducing the carrying amount.
Significant influence	The power to participate in the financial and operating policy decisions of an investee without controlling those decisions; generally presumed when an investor holds 20% or more of the voting shares.
Unrealized holding gain or loss	The net change in fair value of a security from one period to the next, exclusive of dividend or interest revenue recognized; reported in net income (trading) or OCI (HFCS and non-trading equity), depending on classification.
Expected credit loss (ECL) model	An impairment model under IFRS 9 that requires recognition of credit losses based on probability-weighted expected cash shortfalls, using either 12-month expected losses (no significant credit deterioration) or lifetime expected losses (significant credit deterioration).
Recycling adjustment	A reclassification entry that transfers previously recognised unrealized gains or losses out of accumulated OCI into net income when a security classified as HFCS is sold, preventing double-counting of gains/losses in comprehensive income.
Derivative financial instrument	A financial contract whose value is derived from an underlying variable (share price, interest rate, commodity price, etc.); characterised by an underlying, little or no initial investment, and future settlement.

Term	Definition
Fair value option	An irrevocable election available at initial recognition to measure most financial instruments at fair value, with all subsequent gains and losses reported in net income, typically used to eliminate an accounting mismatch between an asset and a related liability.

Accounting Treatments

- Debt investment—HFC (amortized cost): Debit Debt Investments at purchase price; each period debit Cash and credit Interest Revenue for cash received, and adjust the carrying amount for discount/premium amortization using the effective-interest method (e.g., Robinson SA: debit Debt Investments €92,278, credit Cash €92,278 at purchase; debit Cash €4,000 and Debt Investments €614, credit Interest Revenue €4,614 on July 1, 2022).
- Debt investment—HFCS (fair value through OCI): Record at purchase price; amortize discount or premium each period; at reporting date, adjust to fair value via a Fair Value Adjustment valuation account and record the unrealized gain or loss in Unrealized Holding Gain or Loss—Equity (e.g., Graff plc: Unrealized Holding Gain or Loss—Equity 1,732 / Fair Value Adjustment 1,732 at December 31, 2022). On sale, recognise the realised gain/loss in income and make a recycling adjustment to eliminate the prior OCI balance.
- Debt investment—Trading (fair value through income): At each reporting date, adjust the portfolio to fair value via the Fair Value Adjustment account with the offset to Unrealized Holding Gain or Loss—Income, which flows through net income (e.g., Western Publishing: Fair Value Adjustment 3,750 / Unrealized Holding Gain or Loss—Income 3,750).
- Equity investment—Trading (<20%, fair value through income): Record at cost (broker commissions expensed); recognise dividends as revenue when declared; at period-end adjust to fair value, debiting or crediting Unrealized Holding Gain or Loss—Income (e.g., Republic SA: Unrealized Holding Gain or Loss—Income 35,550 / Fair Value Adjustment 35,550 for net unrealized loss).
- Equity investment—Non-trading option (fair value through OCI): Same as trading but unrealized gains/losses are recorded in Unrealized Holding Gain or Loss—Equity (OCI); upon sale, the carrying amount equals fair value so no separate gain/loss is recognised in income, and the accumulated OCI balance may be transferred to retained earnings (not recycled to income).
- Equity method (20–50% ownership): Record investment at cost; each period debit Equity Investments and credit Investment Income for the investor's share of investee net income (e.g., Maxi: Investment Income = 20% × Mini's net income); dividends received debit Cash and credit Equity Investments; if losses exceed the carrying amount, stop recognising further losses unless the investor has committed to providing further support.

Critical Formulas / Calculations

Effective-Interest Revenue on Debt Investment: $\text{Interest Revenue} = \text{Beginning Carrying Amount} \times \text{Effective (Yield) Rate} \times \text{Time Fraction}$. Variables: Beginning Carrying Amount = amortized cost at start of period; Effective Rate = yield rate at acquisition; Time Fraction = portion of year. Example: Robinson SA, July 1, 2022 – $\text{Interest Revenue} = €92,278 \times 10\% \times 6/12 = €4,614$.

Unrealized Holding Gain or Loss (Portfolio Adjustment): $\text{Fair Value Adjustment (Debit or Credit)} = [\text{Aggregate Fair Value of Portfolio}] - [\text{Aggregate Amortized Cost of Portfolio}] - [\text{Existing Fair Value Adjustment Balance}]$. Example: Webb AG

2022 HFCS portfolio: aggregate unrealized loss = -€9,537; prior balance = 0; entry = Unrealized Holding Gain or Loss—Equity 9,537 / Fair Value Adjustment 9,537.

Gain or Loss on Sale of Debt Investment: Gain (Loss) = Selling Price (net of accrued interest) - Book Value (amortized cost) at date of sale. Example: Robinson SA sells Evermaster bonds Nov. 1, 2024: Selling price = €99,750; Book value ≈ €98,675 (after amortization to Nov. 1); Gain ≈ €1,075.

Impairment Loss on HFC Debt Investment: Loss = Carrying Amount - Present Value of Revised Expected Cash Flows (discounted at historical effective-interest rate). Example: Mayhew Ltd.: Carrying amount = ¥200,000; PV of revised cash flows = ¥187,320 (¥136,602 principal + ¥50,718 interest annuity); Impairment Loss = ¥200,000 - ¥187,320 = ¥12,680.

Equity Method Income Recognition: Investment Income = Investor's Ownership % × Investee's Net Income (or Net Loss). Example: Maxi Company (20% of Mini): Mini reports €100,000 net income; Maxi records Investment Income = 20% × €100,000 = €20,000; debit Equity Investments €20,000, credit Investment Income €20,000.

Illustrative Example

Robinson SA purchases €100,000, 8% bonds of Evermaster AG on January 1, 2022, for €92,278 (discount; yield 10%); classified as held-for-collection. Step 1 — Purchase: Debt Investments 92,278 / Cash 92,278. Step 2 — July 1, 2022 interest receipt: Cash 4,000 / Debt Investments 614 / Interest Revenue 4,614 (carrying amount rises to €92,892). Step 3 — December 31, 2022 accrual: Interest Receivable 4,000 / Debt Investments 645 / Interest Revenue 4,645. Step 4 — Income statement 2022: Interest Revenue = €4,614 + €4,645 = €9,259. Step 5 — Early sale (Nov. 1, 2024) at 99¼ plus accrued interest: Cash (€99,750 + €2,667) / Interest Revenue 2,667 / Debt Investments [book value] / Gain on Sale of Investments [difference]; gain reflects excess of selling price over amortized book value.

Disclosures / Presentation Notes

- Statement of financial position classification: Debt investments at amortized cost (HFC) and at fair value (HFCS, trading) are presented as long-term investments; related interest receivable appears as a current asset; the Fair Value Adjustment valuation account adjusts the carrying amount to fair value for HFCS and trading portfolios.
- OCI and equity presentation: Unrealized holding gains and losses on HFCS debt investments and non-trading equity investments are reported as a separate component of other comprehensive income and accumulated in equity as Accumulated Other Comprehensive Income (AOCI); the components must be disclosed, including any recycling (reclassification) adjustments.
- Fair value hierarchy disclosures (Appendix 17B): Companies must disclose fair values and the corresponding hierarchy level—Level 1 (quoted prices in active markets), Level 2 (observable inputs for similar instruments), and Level 3 (unobservable/internal inputs); for Level 3 instruments, a roll-forward reconciliation of beginning to ending balances and a description of valuation techniques and significant unobservable inputs are required.
- Impairment disclosures: Companies must disclose the methodology used to identify and measure impairments, including use of the ECL model; for HFCS investments, the portion of a fair value decline attributable to credit risk (reported in income) must be separated from fair value changes due to market interest rate movements (reported in OCI).

Potential Pitfalls

- Confusing the routing of unrealized gains and losses: Trading debt/equity investments route unrealized gains and losses through net income, while HFCS debt investments and non-trading equity investments route them through OCI. A common error is applying income treatment to HFCS securities or failing to use the separate Fair Value Adjustment account (instead directly crediting the investment account), which destroys the amortized cost record needed for effective-interest calculations.
- Forgetting the recycling adjustment for HFCS securities upon sale: When an HFCS bond is sold, the realised gain/loss enters net income, but the same amount was previously recognised in OCI. Without a reclassification (recycling) entry or note disclosure, the gain/loss is double-counted in comprehensive income across periods. Non-trading equity investments are explicitly not recycled under IFRS—this asymmetry with HFCS debt is another source of confusion.
- Misapplying the equity method versus fair value when ownership is near 20%: The 20% threshold is a rebuttable presumption, not a hard rule; evidence of significant influence (board representation, policy participation, material intercompany transactions) can require equity method accounting even below 20%, while lack of influence can preclude it above 20%. Additionally, equity method investors must stop recognising further losses once the carrying amount reaches zero (unless they have guaranteed the investee's obligations), a step frequently overlooked.

*Notes: The chapter text was read in full across approximately 4,236 lines. Numerical values in several illustrations (Illustrations 17.7, 17.11, 17.14–17.16, 17.20, 17.22–17.23, 17.29–17.39, and practice problem tables) were partially presented in the extracted text without all cell values populated, as the original PDF tables were incompletely parsed during extraction. Specific dollar/euro amounts in those tables should be verified against the printed textbook. Appendices 17A (Derivatives) and 17B (Fair Value Disclosures) are summarised above but are marked optional topics (asterisked learning objectives *5–*8) and may not be required in all course syllabi.*

Chapter 18: Revenue Recognition

Executive Summary

Chapter 18 establishes the converged IASB/FASB standard 'Revenue from Contracts with Customers,' presenting a comprehensive five-step asset-liability model for recognizing revenue. The chapter demonstrates that revenue must be recognized when (or as) a performance obligation is satisfied—i.e., when control of a good or service transfers to the customer—replacing older 'risks and rewards' criteria. It also covers practical application to complex situations such as returns, repurchase agreements, consignments, warranties, long-term contracts (Appendix 18A), and franchises (Appendix 18B).

Key Topics

1. **The Five-Step Revenue Recognition Process:** The core framework requires companies to (1) identify the contract, (2) identify separate performance obligations, (3) determine the transaction price, (4) allocate the transaction price, and (5) recognize revenue when each obligation is satisfied by transfer of control.
2. **Asset-Liability Approach:** Revenue recognition is grounded in changes in contract assets (rights to consideration) and contract liabilities (obligations to transfer goods/services), providing a discipline-based alternative to the prior 'risks and rewards' model.
3. **Transaction Price Adjustments:** The transaction price must account for variable consideration (estimated via expected value or most likely amount), the time value of money for significant financing components, non-cash consideration at fair value, and consideration paid or payable to customers.
4. **Standalone Selling Price Allocation:** When multiple performance obligations exist, the transaction price is allocated based on relative standalone selling prices, using observable prices, adjusted market assessment, expected cost-plus-margin, or residual approaches.
5. **Special Revenue Recognition Issues:** The chapter applies the five-step model to sales returns and allowances, repurchase agreements (typically financing transactions), bill-and-hold arrangements, principal-agent/consignment relationships, assurance-type vs. service-type warranties, and non-refundable upfront fees.
6. **Contract Assets and Liabilities (Presentation):** Companies present unconditional rights to receive consideration as receivables and conditional rights as contract assets; advance receipts are contract liabilities (unearned revenue). Contract modifications are accounted for either as new contracts or prospectively via a blended price.
7. **Long-Term Construction Contracts (Appendix 18A):** Revenue is recognized over time using the percentage-of-completion method (cost-to-cost basis) when performance criteria are met, or using the cost-recovery (zero-profit) method when reliable estimates are unavailable. Expected losses on unprofitable contracts must be recognized immediately in full.
8. **Franchise Revenue Recognition (Appendix 18B):** Franchisors recognize initial franchise fee revenue when control of franchise rights transfers (typically at store opening) or over time when ongoing access and support obligations exist; sales-based royalties are recognized only when the uncertainty over variable consideration is resolved.

Important Definitions

Term	Definition
Revenue Recognition Principle	Revenue is recognized when (or as) a performance obligation is satisfied, i.e., when the customer obtains control of the good or service.
Performance Obligation	A promise in a contract to transfer a distinct good or service (or a bundle of goods/services) to a customer; can be explicit, implicit, or based on customary business practice.
Transaction Price	The amount of consideration a company expects to receive from a customer in exchange for transferring goods or services, adjusted for variable consideration, financing, non-cash items, and payments to customers.
Standalone Selling Price	The price at which a company would sell a promised good or service separately to a customer; the primary basis for allocating the transaction price among multiple performance obligations.
Contract Asset	A company's right to consideration in exchange for goods or services already transferred to a customer, where that right is conditional on satisfying another performance obligation before billing.
Contract Liability	A company's obligation to transfer goods or services to a customer for which consideration has already been received (e.g., unearned revenue, deferred revenue).
Variable Consideration	Consideration that depends on future events such as rebates, bonuses, volume discounts, or royalties; estimated using the expected value (probability-weighted) or most likely amount method, subject to a constraint against significant revenue reversal.
Asset-Liability Approach	The conceptual basis for the revenue standard that recognizes and measures revenue through changes in contract assets and liabilities arising from contracts with customers.
Percentage-of-Completion Method	A long-term contract accounting method (Appendix 18A) that recognizes revenue, costs, and gross profit progressively as construction progresses, typically measured on a cost-to-cost basis.
Cost-Recovery (Zero-Profit) Method	A long-term contract method (Appendix 18A) that recognizes revenue only to the extent of recoverable costs incurred until completion, deferring all gross profit recognition to the period of contract completion.
Assurance-Type Warranty	A warranty guaranteeing a product meets agreed-upon specifications at the time of sale; treated as part of the product's performance obligation with a related liability accrual, not a separate obligation.
Service-Type Warranty	An extended warranty providing services beyond defect assurance at an additional price; treated as a separate performance obligation with revenue allocated from the transaction price and recognized over the warranty period.

Accounting Treatments

- Revenue recognition at a point in time: Debit Accounts Receivable (or Cash), Credit Sales Revenue when all five control indicators are met. Simultaneously, debit Cost of Goods Sold, credit Inventory for

the cost side.

- Revenue recognition over time (long-term contracts, percentage-of-completion): Debit Construction in Process (gross profit earned), debit Construction Expenses (costs incurred), credit Revenue from Long-Term Contracts. Progress billings: Debit Accounts Receivable, credit Billings on Construction in Process.
- Sales returns and allowances: Debit Sales Returns and Allowances (contra-revenue) and credit Refund Liability for estimated returns; debit Estimated Inventory Returns (asset) and credit Cost of Goods Sold for the expected returned inventory cost.
- Repurchase agreement (financing transaction): No sale is recorded. Debit Cash, credit Liability to Customer at proceeds; subsequently debit Interest Expense, credit the liability for imputed interest each period until repurchase.
- Service-type warranty (separate performance obligation): At sale, debit Cash/Receivable, credit Sales Revenue and credit Unearned Warranty Revenue for the allocated portion. Recognize Unearned Warranty Revenue as Warranty Revenue on a straight-line basis over the service warranty period.
- Contract liability (advance payment): Debit Cash, credit Unearned Sales Revenue on receipt. Upon delivery and satisfaction of the performance obligation, debit Unearned Sales Revenue, credit Sales Revenue.

Critical Formulas / Calculations

Percentage Completion (Cost-to-Cost Basis): $\% \text{ Complete} = \text{Costs Incurred to Date} / \text{Most Recent Estimate of Total Costs}$. Example: Costs to date £1,000,000; total estimated costs £4,000,000 → 25% complete.

Revenue Recognized to Date (Percentage-of-Completion): $\text{Revenue to Date} = \% \text{ Complete} \times \text{Total Contract Revenue}$. Example: $25\% \times £4,500,000 = £1,125,000$.

Current-Period Revenue: $\text{Current Revenue} = \text{Revenue Recognized to Date} - \text{Revenue Recognized in Prior Periods}$. Example: Year 2 to-date revenue £3,240,000 – Year 1 £1,125,000 = £2,115,000 in Year 2.

Relative Standalone Selling Price Allocation: $\text{Allocated Amount} = (\text{Product's Standalone Price} / \text{Sum of All Standalone Prices}) \times \text{Total Transaction Price}$. Example: Motor Moka beans standalone \$12, coffee standalone \$3, total \$15, transaction price \$13 → Beans allocated $\$13 \times (12/15) = \10.40 .

Variable Consideration (Expected Value): $\text{Transaction Price} = \sum (\text{Probability} \times \text{Outcome Amount})$. Example: $60\% \times \$150,000 + 30\% \times \$145,000 + 10\% \times \$140,000 = \$147,500$.

Imputed Interest Revenue (Financing Component): $\text{Interest Revenue} = \text{Carrying Value of Note} \times \text{Imputed Rate} \times \text{Time Fraction}$. Example: $\text{R\$}900,000 \times 12\% \times 6/12 = \text{R\$}54,000$.

Illustrative Example

Handler Company sells equipment to Chai Company for \$2,000,000, which includes installation (standalone \$20,000) and training (standalone \$50,000). Total standalone value = \$2,070,000. Allocation: Equipment = $(\$2,000,000 / \$2,070,000) \times \$2,000,000 = \$1,932,367$; Installation = $(\$20,000 / \$2,070,000) \times \$2,000,000 = \$19,324$; Training = $(\$50,000 / \$2,070,000) \times \$2,000,000 = \$48,309$. On November 1, 2022, Handler records Cash \$2,000,000 / Service Revenue (installation) \$19,324 / Unearned Service Revenue \$48,309 / Sales Revenue \$1,932,367. Training revenue of \$4,026/month ($\$48,309 / 12$) is then recognized over 12 months; by December 31, 2022, \$8,052 is earned. Total revenue recognized by

year-end: \$1,932,367 + \$19,324 + \$8,052 = \$1,959,743.

Disclosures / Presentation Notes

- **Disaggregation of Revenue:** Companies must disclose revenue disaggregated into categories that reflect how economic factors (geography, product type, market, contract type, duration) affect the nature, amount, timing, and uncertainty of revenue and cash flows, reconciled to segment revenues.
- **Contract Balances Reconciliation:** Opening and closing balances of contract assets (unbilled receivables) and contract liabilities (deferred revenue) must be disclosed with a qualitative description of significant changes, including revenue recognized in the current period from obligations satisfied in prior periods.
- **Remaining Performance Obligations:** The aggregate transaction price allocated to unsatisfied performance obligations (not subject to significant variable consideration constraint) must be disclosed, along with expected timing of recognition, to inform users about future committed revenue.
- **Costs to Obtain or Fulfill Contracts:** Closing balances of capitalized incremental contract acquisition costs (e.g., commissions) and fulfillment costs, the amortization method, and amounts amortized in the period must be disclosed separately from other assets.

Potential Pitfalls

- **Misidentifying the number of performance obligations:** Combining interdependent goods/services into one obligation is correct (e.g., software license heavily customized for a client), but incorrectly bundling distinct, separately sold items—or conversely, splitting an integrated deliverable—leads to misstated revenue timing. The test is whether each element is distinct both on its own AND within the contract.
- **Recognizing constrained variable consideration too early:** Companies must limit recognition of variable amounts (bonuses, royalties, volume rebates) to amounts where a significant revenue reversal is not highly probable. Recording the full bonus before the outcome is reasonably assured overstates revenue; sales-based royalties in franchise and licensing arrangements cannot be recognized until the sale actually occurs regardless of estimates.
- **Treating repurchase agreements and consignments as completed sales:** When a seller retains a forward obligation or call option to repurchase an asset at or above selling price, control has not transferred and the transaction is a financing arrangement, not a sale. Similarly, consigned inventory remains on the consignor's books; recognizing revenue at shipment rather than upon notification of sale by the consignee inflates revenues prematurely.

Notes: The chapter text was complete and read in full (approximately 3,194 lines). Numerical table values embedded in illustrations (e.g., exact figures in Illustrations 18A.4, 18A.6, 18A.14, and the franchise Illustrations 18B.2/18B.3/18B.5) were partially truncated in the source text as the PDF extraction omitted some tabular cells—figures in those illustrations are marked [VERIFY] where the exact numeric output could not be confirmed from the raw text. Appendices 18A and 18B were included and covered in full. End-of-chapter brief exercises and problems beyond the practice problem solution were listed by heading only and are not summarized here.

Chapter 19: Accounting for Income Taxes

Executive Summary

Chapter 19 explains how companies account for the differences that inevitably arise between pretax financial income (determined under IFRS) and taxable income (determined under tax law), requiring recognition of deferred tax assets and liabilities on the statement of financial position. Using the asset-liability method, the chapter demonstrates that income tax expense equals current taxes payable plus or minus the change in deferred tax balances, and addresses special topics including net operating loss carryforwards, uncertain tax positions, and the required IFRS disclosures and tax-rate reconciliation.

Key Topics

1. Pretax financial income vs. taxable income: IFRS-based book income and tax-authority taxable income frequently differ due to different recognition rules (e.g., accrual vs. modified-cash basis), creating temporary and permanent differences that drive deferred tax accounting.
2. Deferred tax liabilities arise from taxable temporary differences—situations where book income currently exceeds taxable income, meaning higher taxes will be owed in future periods (e.g., revenue recognised for book but deferred for tax).
3. Deferred tax assets arise from deductible temporary differences—situations where book expenses currently exceed tax deductions, meaning taxes will be lower in future periods (e.g., warranty accruals expensed for book but deductible only when paid).
4. Non-recognition (partial impairment) of deferred tax assets: A deferred tax asset must be reduced when it is probable (more than 50% likely) that some or all of it will not be realised; this reduction increases income tax expense in the current period.
5. Temporary vs. permanent differences: Temporary differences reverse over time and generate deferred taxes; permanent differences (e.g., non-deductible fines, tax-exempt bond interest) never reverse and therefore cause no deferred tax consequences, only affecting the effective tax rate.
6. Tax rate considerations: Deferred taxes must be measured using the enacted (or substantially enacted) future tax rate; when tax rates change, the full effect on existing deferred tax balances is recognised immediately as an adjustment to income tax expense.
7. Net operating loss (NOL) carryforwards: A company with a tax loss may carry it forward to reduce future taxable income; the expected future tax saving is recorded as a deferred tax asset, subject to the same probability assessment as other deferred tax assets.
8. Financial statement presentation and disclosure: Deferred tax assets and liabilities are classified as non-current; they may be offset when legally enforceable; IFRS requires disclosure of components of income tax expense and a numerical reconciliation between the statutory rate and the effective tax rate (intraproduct tax allocation).

Important Definitions

Term	Definition
Pretax financial income	Income before income taxes as computed under IFRS, also called income for book purposes; the basis for reporting to investors and creditors.

Term	Definition
Taxable income	The excess of taxable revenues over tax-deductible expenses for a period as defined by the relevant tax authority; the base on which income taxes payable is calculated.
Temporary difference	The difference between the tax basis and the carrying (book) amount of an asset or liability that will result in taxable or deductible amounts in future years when the asset is recovered or the liability is settled.
Deferred tax liability	The deferred tax consequences attributable to taxable temporary differences; represents the increase in taxes payable in future years as a result of taxable temporary differences existing at the end of the current year.
Deferred tax asset	The deferred tax consequences attributable to deductible temporary differences and loss carryforwards; represents taxes that will be saved (refunded) in future years.
Permanent difference	A difference between pretax financial income and taxable income that arises from items recognised in one but never in the other (e.g., non-deductible fines, tax-exempt interest); creates no deferred tax balance.
Net operating loss (NOL)	The excess of tax-deductible expenses over taxable revenues in a year; may be carried forward (and sometimes back) to offset income in other years, generating a deferred tax asset.
Loss carryforward	The provision allowing a company to apply the current year's net operating loss against taxable income in future years to reduce future taxes payable.
Effective tax rate	Total income tax expense for the period divided by pretax financial income; differs from the statutory rate when permanent differences or other adjustments exist.
Intraperiod tax allocation	The process of allocating total income tax expense among continuing operations, discontinued operations, other comprehensive income, and prior period adjustments within a single reporting period.
Substantially enacted tax rate	Under IFRS, the tax rate used to measure deferred taxes when a future rate change has been enacted or is virtually certain to be enacted; used instead of the current rate for periods when it applies.
Asset-liability method	The IASB-preferred approach to deferred tax accounting that focuses on reporting accurate deferred tax assets and liabilities on the statement of financial position, measured at the enacted rate expected to apply when the temporary difference reverses.

Accounting Treatments

- Recording a deferred tax liability (taxable temporary difference): Debit Income Tax Expense and credit Deferred Tax Liability for the increase in the cumulative temporary difference multiplied by the enacted future tax rate; simultaneously debit Income Tax Expense and credit Income Taxes Payable for current taxes owed. Example — Chelsea 2022: Dr Income Tax Expense \$28,000 / Cr Income Taxes Payable \$16,000 / Cr Deferred Tax Liability \$12,000.

- Recording a deferred tax asset (deductible temporary difference): Debit Deferred Tax Asset and credit the deferred portion of Income Tax Expense (a benefit) for the increase in the deferred tax asset; debit Income Tax Expense and credit Income Taxes Payable for current taxes. Example — Hunt 2022: Dr Income Tax Expense \$180,000 / Dr Deferred Tax Asset \$20,000 / Cr Income Taxes Payable \$200,000.
- Partial non-recognition of a deferred tax asset: When it is probable that a portion of a deferred tax asset will not be realised, debit Income Tax Expense and credit Deferred Tax Asset for the unrecoverable portion. If evidence improves, reverse with Dr Deferred Tax Asset / Cr Income Tax Expense.
- Recording an NOL carryforward: In the loss year, if realisation is probable, debit Deferred Tax Asset and credit Income Tax Expense (Loss Carryforward) for $\text{NOL} \times \text{enacted future tax rate}$. In the profitable year of realisation, debit Income Tax Expense and credit Deferred Tax Asset (to eliminate it) and Income Taxes Payable (for reduced taxes owed).
- Tax rate change adjustment: When a new rate is enacted, remeasure all existing deferred tax balances at the new rate. Record the difference as Dr or Cr Deferred Tax Liability (or Asset) with the offset to Income Tax Expense in the period of enactment.
- Loss carryback (Appendix 19B): In the loss year, debit Income Tax Refund Receivable (current asset) and credit Income Tax Expense (Loss Carryback) for the tax refund recoverable from prior years' taxes paid; any residual loss is treated as a carryforward deferred tax asset.

Critical Formulas / Calculations

Income Tax Expense = Income Taxes Payable (or Refundable) +/- Change in Deferred Income Taxes. Variables: Income Taxes Payable = taxable income $\times$ enacted rate; Change in Deferred Income Taxes = ending deferred balance - beginning deferred balance (positive = expense; negative = benefit). Example: Payable \$16,000 + Deferred increase \$12,000 = Tax Expense \$28,000 (Chelsea 2022).

Deferred Tax Liability = Cumulative Taxable Temporary Difference $\times$ Enacted Future Tax Rate. Variables: Cumulative difference = book basis of asset minus tax basis (or tax basis of liability minus book basis). Example: \$30,000 receivable temporary difference $\times$ 40% = \$12,000 DTL (Chelsea end of 2022).

Deferred Tax Asset = Cumulative Deductible Temporary Difference $\times$ Enacted Future Tax Rate. Variables: Cumulative difference = tax basis of liability minus book basis (or book basis of asset minus tax basis). Example: \$50,000 litigation liability difference $\times$ 40% = \$20,000 DTA (Hunt end of 2022).

Deferred Tax Expense (Benefit) = Ending Deferred Tax Liability - Beginning Deferred Tax Liability (for liabilities) OR Ending Deferred Tax Asset - Beginning Deferred Tax Asset (for assets, sign reversed for expense). Example: \$12,000 DTL end - \$0 DTL beginning = \$12,000 deferred tax expense (Chelsea 2022).

Effective Tax Rate = Total Income Tax Expense / Pretax Financial Income. Variables: Total expense includes both current and deferred components; pretax income per IFRS books. Example: Bio-Tech 2021: \$60,000 / \$200,000 = 30%; 2022: \$61,500 / \$200,000 = 30.75% (due to permanent non-deductible insurance premium).

Illustrative Example

Akai Ltd. (Appendix 19A, 2021): Pretax financial income ¥412,000. Temporary differences: installment sales +¥448,000 taxable future amounts (DTL at 40% = ¥179,200); excess tax depreciation net ¥18,000 taxable (DTL = ¥7,200); warranty accruals ¥156,000 deductible (DTA at 40% = ¥62,400). Permanent differences: non-taxable bond interest -¥28,000; non-deductible fines +¥26,000. Taxable income = ¥100,000; income taxes payable (50% rate) = ¥50,000. Net deferred tax liability = ¥186,400 - ¥62,400 = ¥124,000 (non-current). Total income tax expense = ¥50,000 current + ¥124,000 deferred = ¥174,000. Journal entry: Dr Income Tax Expense ¥174,000 / Dr Deferred Tax Asset ¥62,400 / Cr Income Taxes Payable ¥50,000 / Cr Deferred Tax Liability ¥186,400. Net income = ¥412,000 - ¥174,000 = ¥238,000.

Disclosures / Presentation Notes

- Statement of financial position: Current income taxes payable or receivable are classified as current liabilities or assets; all deferred tax assets and liabilities are classified as non-current and reported net (offset) when they relate to the same tax jurisdiction and there is a legally enforceable right of offset.
- Income statement — intraperiod tax allocation: Income tax expense is allocated to continuing operations, discontinued operations, other comprehensive income, and prior period adjustments; both the current component and the deferred component must be disclosed either on the face of the income statement or in the notes.
- Tax reconciliation disclosure (IAS 12): Companies must present a numerical reconciliation between reported income tax expense and the product of pretax accounting income multiplied by the applicable statutory tax rate, or alternatively a reconciliation between the average effective tax rate and the applicable statutory rate, explaining the reasons for any differences.
- Loss carryforward disclosure: The amounts and expiration dates of operating loss carryforwards for tax purposes must be disclosed; this allows analysts to assess future periods in which the company will pay reduced or no income taxes and to evaluate the appropriateness of related deferred tax asset recognition.

Potential Pitfalls

- ❑ Confusing temporary and permanent differences: Permanent differences (e.g., non-deductible fines, tax-exempt interest) never create deferred tax assets or liabilities — they only affect the current year's effective tax rate. Erroneously recording a deferred tax balance for a permanent difference overstates assets or liabilities and distorts tax expense.
- ❑ Using the wrong tax rate for deferred tax measurement: Deferred taxes must be measured at the enacted (or substantially enacted) rate expected to apply when the temporary difference reverses, not the current year's rate. When different rates apply to different future years, each reversal tranche must be multiplied by its applicable enacted rate (as illustrated by Wang Group). Using only the current rate when future rates differ will misstate deferred balances.
- ❑ Incorrectly applying the probability threshold for deferred tax asset recognition: Under IFRS, a deferred tax asset (including for NOL carryforwards) is recognised only to the extent that realisation is 'probable' (more than 50% likely). Because this assessment is highly judgmental and relies on forecasts of future taxable income, companies may recognise an excessive asset during optimistic periods or fail to write it down quickly enough when negative evidence accumulates, creating earnings-management opportunities.

Notes: The chapter text was fully readable with no truncation. Numerical tables and schedule illustrations (e.g., Illustrations 19.1–19.49 and Appendix schedules 19A.1–19A.19, 19B.1–19B.3) were present in text form but many column headers and specific cell values from the tabular illustrations were partially omitted in the source text (columns listed without all corresponding figures). Figures and graphical illustrations referenced in the chapter could not be

reproduced. Appendix 19A (Comprehensive Example) and Appendix 19B (Loss Carrybacks) were included and summarised. The IFRS vs. U.S. GAAP comparison section (Learning Objective 7, end-of-chapter material) was also read and incorporated.

Chapter 20: Accounting for Pensions and Postretirement Benefits

Executive Summary

Chapter 20 addresses the accounting and financial reporting of defined benefit pension plans and other postretirement benefits under IAS 19, focusing on how employers measure and recognise the pension obligation (defined benefit obligation), pension expense (service cost and net interest), and remeasurements. The chapter concludes that full and immediate recognition of all changes in the net defined benefit obligation—with remeasurements routed through other comprehensive income rather than net income—provides the most transparent and decision-useful information for financial statement users.

Key Topics

1. **Defined Contribution vs. Defined Benefit Plans:** A defined contribution plan fixes only the employer's periodic contribution; the employee bears investment risk. A defined benefit plan specifies the retirement benefit formula, leaving the employer responsible for any funding shortfall and making accounting significantly more complex.
2. **Measures of the Pension Obligation:** Three alternative measures exist—vested benefit obligation (vested benefits at current salaries), accumulated benefit obligation (all service at current salaries), and defined benefit obligation (all service at projected future salaries); IAS 19 mandates use of the defined benefit obligation as the most representationally faithful measure.
3. **Net Defined Benefit Obligation (Asset) and Funded Status:** The funded status equals the defined benefit obligation minus the fair value of plan assets; a positive result is a pension liability and a negative result is a pension asset, both reported on the statement of financial position under the net approach.
4. **Components of Pension Expense:** Pension expense has two income-statement components—current service cost (increase in the DBO from employee service this period) and net interest (discount rate multiplied by the net funded status at the start of the period); both affect net income.
5. **Past Service Cost:** When a plan is amended to grant additional retroactive benefits, or curtailed, the resulting change in the DBO (past service cost) is recognised immediately as pension expense in the period of the amendment or curtailment, with no deferral or amortisation.
6. **Remeasurements (Asset and Liability Gains/Losses):** The difference between actual return on plan assets and assumed interest revenue is an asset gain or loss; changes in the DBO from revised actuarial assumptions are liability gains or losses. Both are recognised immediately in other comprehensive income (OCI) and are never recycled to net income under IFRS.
7. **Pension Worksheet:** A working tool (not part of the general ledger) that tracks formal journal entry columns alongside memo columns for the DBO and plan assets, ensuring the ending Pension Asset/Liability balance always reconciles to the net funded status.
8. **IFRS vs. U.S. GAAP Differences:** Under U.S. GAAP, past service cost is amortised over remaining service lives, and remeasurements accumulate in AOCI and are amortised using the corridor approach; under IFRS, both are recognised immediately (PSC in net income; remeasurements in OCI with no recycling).

Important Definitions

Term	Definition
Defined Benefit Obligation (DBO)	The present value of expected future benefit payments earned by employees for service in current and prior periods, calculated using projected future salary levels; the measure of pension liability required by IAS 19.
Defined Benefit Plan	A pension arrangement that specifies the retirement benefit formula (typically a function of years of service and final salary), placing investment and longevity risk on the employer.
Defined Contribution Plan	A pension arrangement in which the employer agrees only to contribute a specified amount each period; the ultimate benefit depends on amounts contributed and investment performance, with risk borne by the employee.
Net Defined Benefit Obligation (Asset)	The funded status of the plan, calculated as DBO minus the fair value of plan assets; reported as a pension liability (if positive) or pension asset (if negative) on the statement of financial position.
Service Cost	The increase in the present value of the DBO resulting from employee service; current service cost relates to the current period, while past service cost relates to prior periods arising from a plan amendment or curtailment.
Net Interest	The financing component of pension expense, calculated by multiplying the discount rate by the net defined benefit obligation (asset) at the beginning of the period; reported in the financing section of the income statement.
Remeasurements	Gains and losses arising from (1) the difference between actual return on plan assets and assumed interest revenue, and (2) changes in actuarial assumptions affecting the DBO; recognised in OCI and not recycled to net income.
Past Service Cost (PSC)	The change in the DBO caused by a plan amendment (introducing or changing benefits) or curtailment (significant reduction in employees covered); recognised immediately as pension expense.
Projected Unit Credit Method	The actuarial valuation method required by IAS 19 that allocates projected pension benefits to periods of service and incorporates expected future salary levels in measuring the DBO; also called the benefits/years-of-service method.
Asset Ceiling	The upper limit on the pension asset that may be recognised on the statement of financial position, equal to the present value of economic benefits available to the employer through refunds or reductions in future contributions.
Curtailment	A significant reduction in the number of employees covered by a pension plan, treated the same as a plan amendment for accounting purposes and giving rise to past service cost.
Vested Benefits	Benefits that an employee is entitled to receive even if he or she renders no additional service to the employer; the basis for the most conservative measure of the pension obligation.

Accounting Treatments

- Recognition of pension liability/asset: At each reporting date, the net defined benefit obligation (DBO less fair value of plan assets) is recognised as a pension liability or pension asset on the statement of

financial position. If the plan is overfunded, the recognised asset is the lower of the surplus and the asset ceiling.

- Pension expense journal entry (annual): Dr. Pension Expense [service cost + net interest] / Dr. or Cr. Pension Asset/Liability [difference between expense and cash contributed] / Cr. Cash [employer contribution]. Example (Zarle AG 2022): Dr. Pension Expense €9,000 | Cr. Cash €8,000 | Cr. Pension Asset/Liability €1,000.
- Past service cost: Recognised entirely in the period of plan amendment or curtailment. Dr. Pension Expense [PSC amount] / Cr. Defined Benefit Obligation [same amount]. For Zarle AG 2023: Dr. Pension Expense €81,600 | Cr. DBO €81,600.
- Remeasurement—asset gain/loss: If actual return > assumed interest revenue, debit Plan Assets and credit OCI (G/L) for the gain; if actual return < assumed interest revenue, debit OCI (G/L) and credit Plan Assets for the loss. These entries never pass through net income.
- Remeasurement—liability gain/loss: If actuarial assumptions increase the DBO beyond the computed balance, debit OCI (G/L) and credit DBO for the liability loss; a decrease gives a liability gain in the opposite direction. For Zarle AG 2024 liability loss: Dr. OCI (G/L) €26,594 | Cr. DBO €26,594.
- Presentation in comprehensive income: Service cost and net interest are included in net income (operating and financing sections respectively). Remeasurements (asset and liability gains/losses) are included in other comprehensive income, accumulating in Accumulated OCI in equity; they are never reclassified (recycled) to net income in subsequent periods.

Critical Formulas / Calculations

Net Defined Benefit Obligation (Funded Status) = Defined Benefit Obligation - Fair Value of Plan Assets. Example: DBO €4,000,000 - Plan Assets €3,700,000 = Pension Liability €300,000.

Net Interest = (DBO × Discount Rate) - (Plan Assets × Discount Rate) = Net Funded Status × Discount Rate. Example (Zarle 2022, equal DBO and assets of €100,000 at 10%): Interest expense €10,000 - Interest revenue €10,000 = Net interest €0.

Pension Expense (income statement) = Current Service Cost + Past Service Cost (if any) + Net Interest. Example (Zarle 2023): €9,500 + €81,600 + (€193,600 × 10% - €111,000 × 10%) = €9,500 + €81,600 + €8,260 = €99,360.

Actual Return on Plan Assets = (Ending Plan Assets - Beginning Plan Assets) - (Contributions - Benefits Paid). Example (Hasbro SA): (€4,460,000 - €4,200,000) - (€300,000 - €250,000) = €260,000 - €50,000 = €210,000.

Asset Gain (Loss) = Actual Return on Plan Assets - Assumed Interest Revenue (Plan Assets × Discount Rate). Example (Shopbob SE, 2022): Actual return €8,000 - (€100,000 × 6%) = €8,000 - €6,000 = Asset Gain €2,000.

Liability Loss (Gain) = Actuarially Determined DBO - Computed DBO (before remeasurement). Example (Zarle 2024): Actuary-determined DBO €265,000 - Computed DBO €238,406 = Liability Loss €26,594.

Illustrative Example

Zarle AG 3-year pension illustration (2022–2024): Beginning 2022, DBO and plan assets are both €100,000 (funded status = €0). Service cost €9,000, net interest €0 (equal assets and liabilities), contributions €8,000 → Pension Expense €9,000, cash €8,000, Pension Liability €1,000. In 2023, plan amended granting PSC €81,600; service cost €9,500; net interest €8,260; contributions €20,000 →

Pension Expense €99,360, Pension Liability grows to €80,360. In 2024, service cost €13,000; net interest €8,036; actual return €12,000 vs. assumed €13,410 (asset loss €1,410); actuarial DBO adjustment €26,594 (liability loss) → Pension Expense €21,036 in net income, OCI charge €28,004 (remeasurements), Pension Liability €105,400 and Accumulated OCI €28,004 reported on the statement of financial position.

Disclosures / Presentation Notes

- Reconciliation schedules required: Companies must disclose a full roll-forward reconciliation of the DBO and plan assets from beginning to ending balance, showing service cost, interest, PSC, contributions, benefits paid, and remeasurements separately; the funded status (pension asset/liability) must also be reconciled.
- Sensitivity analysis: For each significant actuarial assumption (e.g., discount rate, salary growth rate, mortality), companies must disclose the effect on the DBO of a reasonably possible change in that assumption at the reporting date, along with the methods and limitations of those analyses.
- Future cash flow information: Required disclosures include (a) the funding policy and arrangements, (b) expected contributions for the next reporting period, and (c) a maturity profile or distribution of the timing of benefit payments.
- Remeasurements in OCI—no recycling: IFRS prohibits reclassifying (recycling) pension-related OCI items into net income in future periods; the cumulative balance in Accumulated OCI is reported in the equity section of the statement of financial position and may be transferred to other equity accounts but never back through the income statement.

Potential Pitfalls

- ❑ Confusing DBO with accumulated benefit obligation: IAS 19 requires the defined benefit obligation (using projected future salaries), not the accumulated benefit obligation (using current salaries). Using current salaries understates both the pension liability and annual service cost, particularly for younger workforces with significant expected wage growth.
- ❑ Treating remeasurements as pension expense: Asset gains/losses (actual vs. assumed return) and liability gains/losses (actuarial assumption changes) must go to OCI, not net income. Routing them through pension expense overstates or understates reported earnings and violates IAS 19's explicit prohibition on recycling; this is a common error when transitioning from U.S. GAAP, which uses the corridor amortisation approach.
- ❑ Applying the asset ceiling incorrectly when the plan is overfunded: When plan assets exceed the DBO, the pension asset recognised on the statement of financial position is capped at the lower of the surplus and the asset ceiling (present value of refunds or future contribution reductions available to the employer). Recognising the full surplus without testing the ceiling overstates assets and equity.

Notes: The source file appears complete, covering all six learning objectives and the IFRS vs. U.S. GAAP comparison section. Numerical data within illustration tables (Illustration 20.2 pension fund sizes, Illustrations 20.10/20.13/20.16 worksheet figures, and the practice problem solution amounts) are partially presented as labels without accompanying numbers due to the text-extraction format of the source file—specific amounts from these tables should be [VERIFY]ed against the original textbook exhibit images. No appendices or additional figures appear to have been omitted.

Chapter 21: Accounting for Leases

Executive Summary

Chapter 21 covers the accounting and reporting for lease transactions under IFRS 16, which mandates that lessees capitalize virtually all leases by recording a right-of-use (ROU) asset and a corresponding lease liability at the present value of future lease payments. The chapter explains lessee accounting (finance lease method for all leases), lessor accounting (classification into finance or operating leases using five tests), special adjustments such as residual values, executory costs, initial direct costs, and sale-leaseback transactions, and highlights the significant balance-sheet and financial-ratio impacts the new standard brings to companies in retail, aviation, and other lease-intensive industries.

Key Topics

1. The leasing environment: leasing has grown into the fastest form of capital investment globally, with top-50 companies reporting over \$1.3 trillion in leasing volume; lessors include banks (~43%), captive leasing companies (~33%), and independents (~24%).
2. Lessee accounting under IFRS 16: all leases (except low-value assets under \$5,000 and short-term leases of 12 months or less) are capitalized by recording a ROU asset and lease liability equal to the present value of lease payments, with subsequent interest expense (effective-interest method) and straight-line depreciation.
3. Lease liability measurement: the liability is the present value of fixed payments, variable payments tied to an index/rate at commencement, the probable shortfall under a guaranteed residual value, and the cost of purchase or termination options reasonably certain to be exercised; unguaranteed residual values are excluded.
4. Discount rates: lessees use the implicit rate of the lessor if known; otherwise, the lessee's incremental borrowing rate is used. Lessors always use the implicit rate.
5. Lessor classification tests: a lessor classifies a lease as a finance lease if it meets at least one of five tests — (1) transfer of ownership, (2) bargain purchase option, (3) lease term is a major part (75% guideline) of economic life, (4) present value of payments equals or exceeds substantially all (90% guideline) of the asset's fair value, or (5) the asset is so specialized it has no alternative use to the lessor.
6. Finance lease accounting for lessors: at commencement the lessor derecognizes the asset and records a Lease Receivable; for sales-type leases, selling profit (Sales Revenue minus Cost of Goods Sold) is recognized immediately; interest revenue accrues over the lease term using the effective-interest method.
7. Operating lease accounting: lessors retain the asset on their balance sheet, depreciate it, and recognize lease revenue on a straight-line basis; lessees also capitalize a ROU asset and liability but (under IFRS 16) still apply the finance lease expense pattern.
8. Special adjustments and sale-leasebacks: the ROU asset is adjusted for lease prepayments (increases), lessor incentive payments (decreases), and initial direct costs (increases); in a sale-leaseback, whether a true sale or a failed sale (financing) is determined by applying revenue recognition control criteria.

Important Definitions

Term	Definition
Lease	A contract that conveys the right to control the use of an identified property, plant, or equipment asset for a period of time in exchange for consideration (IFRS 16).
Right-of-Use (ROU) Asset	An asset recorded by the lessee representing its right to use the underlying leased asset over the lease term; initially measured equal to the lease liability, adjusted for prepayments, lessor incentives, and initial direct costs.
Lease Liability	The lessee's obligation to make lease payments, measured at the present value of those payments using the implicit rate or, if not known, the incremental borrowing rate.
Finance Lease	A lease that transfers substantially all the risks and rewards incidental to ownership of an underlying asset; meets at least one of five lessor classification tests under IFRS 16.
Operating Lease	A lease that does not meet any of the five finance lease classification tests; the lessor retains the asset and recognizes revenue on a straight-line basis.
Implicit Interest Rate	The discount rate that equates the present value of lease payments plus unguaranteed residual value to the fair value of the leased asset at lease commencement.
Incremental Borrowing Rate	The rate the lessee would pay to borrow, over a similar term, funds sufficient to purchase the leased asset; used when the implicit rate is not known.
Guaranteed Residual Value	A residual value whose realization at lease end is guaranteed by the lessee or a third party; included in the lessee's lease payments only to the extent the guaranteed amount is expected to exceed the asset's fair value at that date.
Unguaranteed Residual Value	The portion of the leased asset's residual value not guaranteed; excluded from the lessee's lease liability computation but included in the lessor's Lease Receivable.
Bargain Purchase Option	An option allowing the lessee to purchase the leased asset at a price substantially below expected fair value, making exercise reasonably certain; included in lease payments; lessee depreciates the ROU asset over the asset's economic life.
Initial Direct Costs	Incremental costs (e.g., commissions, document preparation fees) that would not have been incurred absent execution of the lease; added to the lessee's ROU asset; expensed immediately by the lessor in a sales-type lease.
Sale-Leaseback	A transaction in which a company sells an asset to another party and simultaneously leases it back; classified as a true sale or a failed sale (financing) based on whether control passes to the buyer-lessor.

Accounting Treatments

- Lessee at commencement — record ROU asset and lease liability at PV of lease payments: Dr Right-of-Use Asset / Cr Lease Liability. Immediately record first annuity-due payment: Dr Lease Liability / Cr Cash.

- Lessee ongoing — at each period end recognize interest on the liability (effective-interest method): Dr Interest Expense / Cr Lease Liability; and depreciate the ROU asset (straight-line over lease term, or economic life if ownership transfers or bargain purchase option exists): Dr Depreciation Expense / Cr Accumulated Depreciation—ROU Asset.
- Lessor — finance (sales-type) lease at commencement: Dr Lease Receivable (PV of all payments including residual values) / Cr Sales Revenue / Cr Inventory (or Equipment); also Dr Cost of Goods Sold / Cr Sales Revenue for the cost portion. Unguaranteed residual value reduces both Sales Revenue and COGS by its PV, leaving gross profit unchanged.
- Lessor — operating lease: retain asset on balance sheet; collect cash: Dr Cash / Cr Unearned Lease Revenue; recognize revenue ratably: Dr Unearned Lease Revenue / Cr Lease Revenue; record depreciation: Dr Depreciation Expense / Cr Accumulated Depreciation—Equipment.
- Residual value loss (lessee): if actual fair value at lease end is below guaranteed amount, record: Dr Loss on Lease / Cr Cash (or Lease Liability) for the deficiency.
- ROU asset adjustment formula: $\text{ROU Asset} = \text{Lease Liability} + \text{Prepaid Lease Payments} - \text{Lease Incentives Received} + \text{Initial Direct Costs}$.

Critical Formulas / Calculations

Lease Liability (PV) = Lease Payment $\times$ PVF-AD(n, r) [annuity-due] or $\times$ PVF-OA(n, r) [ordinary annuity]; variables: lease payment = periodic cash payment, n = number of periods, r = discount rate (implicit or incremental borrowing rate). Example: $\text{€}20,711.11 \times \text{PVF-AD}(5, 4\%) = \text{€}20,711.11 \times 4.62990 = \text{€}95,890.35$.

Lease Receivable (Lessor) = PV of Rental Payments + PV of Guaranteed Residual Value + PV of Unguaranteed Residual Value. Example: $\text{€}20,711.11 \times 4.62990 + \text{€}5,000 \times 0.82193 = \text{€}95,890.35 + \text{€}4,109.65 = \text{€}100,000.00$.

Lessor Lease Payment Derivation: Lease Payment = (Fair Value - PV of Residual Value) $\div$ PVF-AD(n, r). Example: $(\text{€}100,000 - \text{€}4,109.65) \div 4.62990 = \text{€}95,890.35 \div 4.62990 \approx \text{€}20,711.11$.

Guaranteed Residual Value Inclusion (Lessee): include PV of expected shortfall only = PV of (Guaranteed RV - Expected RV) if Guaranteed RV > Expected RV. Example: Expected RV = €3,000; Guaranteed RV = €5,000; shortfall = €2,000; PV = $\text{€}2,000 \times 0.82193 = \text{€}1,643.86$, added to lease liability.

ROU Asset at Commencement = Lease Liability + Prepayments - Incentives + Initial Direct Costs. Example: $\text{€}400,000 + \text{€}10,000 - \text{€}30,000 + \text{€}1,500 = \text{€}381,500$.

Illustrative Example

CNH Capital (lessor) leases a backhoe to Ivanhoe Mines (lessee) for 5 years; fair value €100,000, cost to CNH €85,000, guaranteed residual value €5,000, implicit rate 4%. Annual payment (annuity-due) = €20,711.11. Lessee PV of payments = $\text{€}20,711.11 \times 4.62990 = \text{€}95,890.35$. Jan 1, 2022 — Lessee: Dr ROU Asset €95,890.35 / Cr Lease Liability €95,890.35; then Dr Lease Liability €20,711.11 / Cr Cash €20,711.11. Dec 31, 2022 — Lessee: Dr Interest Expense €3,007.17 / Cr Lease Liability €3,007.17; Dr Depreciation Expense €19,178.07 / Cr Accumulated Depreciation €19,178.07. Lessor Jan 1, 2022: Dr Lease Receivable €100,000 / Dr COGS €85,000 / Cr Sales Revenue €100,000 / Cr Inventory €85,000 (gross profit = €15,000). Total lessee financing cost over 5 years = $\text{€}103,555.55 - \text{€}95,890.35 = \text{€}7,665.20$.

Disclosures / Presentation Notes

- Lessee statement of financial position: ROU assets and lease liabilities are presented separately from other assets/liabilities, or disclosed in notes; current portion of lease liability (due within one year) classified as current, remainder as non-current.
- Lessee income statement: interest expense on lease liabilities and depreciation expense on ROU assets are disclosed separately; under IFRS 16 these replace the single operating lease expense that existed under the old IAS 17.
- Qualitative disclosures required by both lessees and lessors include: general description of lease arrangements, how variable payments are determined, terms and conditions of renewal/termination options and residual value guarantees, and significant assumptions/judgments such as discount rates.
- Lessee quantitative disclosures include: total and component lease costs, weighted-average remaining lease term and discount rate, and a maturity analysis of finance lease liabilities showing undiscounted cash flows for each of the next five years plus thereafter. Lessor quantitative disclosures include: lease-related income, components of net investment in finance leases, and maturity analyses for both operating and finance lease receivables.

Potential Pitfalls

- Residual value guarantee misapplication: students often include the full guaranteed residual value in the lease liability, but IFRS 16 requires the lessee to include only the present value of the expected shortfall (i.e., the amount by which the guaranteed value exceeds the expected fair value at lease end). If the expected value meets or exceeds the guarantee, nothing is added.
- Discount rate selection: lessees must use the lessor's implicit rate when it is practicable to determine it; only when it is impracticable to determine the implicit rate may the lessee substitute its incremental borrowing rate. Using the incremental borrowing rate when the implicit rate is known overstates the liability and ROU asset.
- Depreciation period confusion: the ROU asset is depreciated over the lease term when ownership does not transfer and there is no bargain purchase option; if either condition is present the depreciation period is the full economic life of the underlying asset. Using the wrong period materially misstates depreciation expense and the carrying amount of the ROU asset.

Notes: The chapter text was complete and read in full across all chunks. However, the amortization schedule tables (Illustrations 21.7, 21.11, 21.14, 21.23, 21B.3, 21B.7, GAAP21.3–21.4) were partially truncated in the extracted text — specific numerical column values were omitted, though the schedule structure and key totals were recoverable from surrounding narrative. Appendix figures and some journal entry amounts in the Practice Problem section were also partially omitted. All key concepts, formulas, and accounting treatments are fully represented.

Chapter 22: Accounting Changes and Error Analysis

Executive Summary

Chapter 22 establishes the IFRS (IAS 8) framework for reporting three distinct categories of accounting changes: changes in accounting policy (requiring retrospective application), changes in accounting estimates (requiring prospective application), and corrections of errors (treated as prior period adjustments). The chapter demonstrates that consistent and transparent reporting of these changes is essential for financial statement comparability, using detailed worked examples with journal entries for inventory method changes, long-term contract accounting, and multiple error correction scenarios.

Key Topics

1. Changes in accounting policy: A change from one accepted accounting policy to another (e.g., average-cost to FIFO inventory, cost-recovery to percentage-of-completion for long-term contracts) must be applied retrospectively, restating all prior-period financial statements presented as if the new policy had always been in use.
2. Retrospective application mechanics: When a policy change is made, companies adjust the carrying amounts of affected assets and liabilities and record an offsetting cumulative adjustment to the opening retained earnings of the earliest year presented in the comparative statements.
3. Changes in accounting estimates: Revisions to estimates such as useful lives, residual values, bad debt allowances, and warranty liabilities are applied prospectively — affecting only the current and future periods — because they represent normal corrections arising from new information, not errors.
4. Differentiation between policy changes and estimate changes: When it is impossible to separate a policy change from an estimate change (e.g., switching depreciation methods), the change is treated entirely as a change in estimate and applied prospectively.
5. Correction of errors: Errors from prior periods (mathematical mistakes, misapplication of policies, oversights) are corrected as prior period adjustments to the opening balance of retained earnings; comparative statements are restated to reflect the correction.
6. Counterbalancing vs. non-counterbalancing errors: Counterbalancing errors self-correct over two accounting periods (e.g., failure to accrue wages), whereas non-counterbalancing errors persist for longer (e.g., expensing a capital asset), requiring correcting entries even after books are closed.
7. Direct and indirect effects of policy changes: Direct effects (e.g., adjustments to inventory balances and related deferred taxes) are applied retrospectively; indirect effects (e.g., changes to profit-sharing payments triggered by restated net income) are recognized in the current period only, not retroactively.
8. Impracticability exception: If it is not feasible to determine prior-period effects of a policy change (because required estimates cannot be objectively reconstructed), the company applies the new policy prospectively from the earliest practicable date and discloses the reason for omitting retrospective restatement.

Important Definitions

Term	Definition
Change in Accounting Policy	A change from one accepted accounting policy to another, such as switching inventory valuation from average-cost to FIFO; requires retrospective application under IFRS (IAS 8).
Change in Accounting Estimate	An adjustment to the carrying amount of an asset or liability, or to the periodic consumption of an asset, arising from new information or experience (e.g., revising a building's useful life from 15 to 25 years); applied prospectively.
Correction of an Error	The process of rectifying omissions or misstatements in prior-period financial statements caused by mathematical mistakes, misapplication of accounting policies, oversights, or fraud; treated as a prior period adjustment to retained earnings.
Retrospective Application	Applying a new accounting policy to all prior transactions and events as if that policy had always been used, restating prior-period financial statements accordingly.
Retrospective Restatement	Correcting the recognition, measurement, and disclosure of prior-period financial statement amounts as if a prior period error had never occurred; distinguishes error correction from policy changes.
Prospective Application	Applying a new accounting policy or recognizing the effect of a change in estimate only in the current period and future periods, without adjusting prior-period amounts.
Counterbalancing Error	An error that is automatically offset or corrected over two consecutive accounting periods (e.g., failure to record accrued wages at year-end), so that the combined two-year totals are correct.
Non-Counterbalancing Error	An error that is not offset within the following accounting period and takes longer than two periods to fully correct itself (e.g., expensing a capital asset that should be depreciated over five years).
Prior Period Adjustment	A correction recorded directly to the opening balance of retained earnings in the current period to account for material errors discovered after the prior-period financial statements were issued.
Impracticable	A standard is impracticable to apply when an entity cannot do so after making every reasonable effort — specifically when prior-period effects cannot be determined, require unverifiable assumptions about past management intent, or depend on estimates that cannot be objectively reconstructed.

Accounting Treatments

- Change in accounting policy — retrospective journal entry: Debit the affected asset/liability account for the cumulative difference (e.g., Debit Construction in Process €220,000), credit Deferred Tax Liability for the tax effect (€88,000), and credit Retained Earnings for the net-of-tax cumulative income difference (€132,000); then recast all comparative prior-period financial statements.
- Change in accounting estimate — prospective depreciation: Compute a new annual charge as (Book Value at Date of Change) / (Remaining Useful Life). For Lao Labs: ¥2,000,000 / (25 – 5) years = ¥100,000/year. Debit Depreciation Expense and credit Accumulated Depreciation for the revised

amount going forward only; no prior-period restatement.

- Error correction — prior period adjustment: Debit Retained Earnings for the after-tax net income overstatement, adjust any related balance sheet accounts (e.g., Debit Deferred Tax Liability, Credit Accumulated Depreciation—Buildings) to bring them to correct balances; restate comparative financial statements for the affected year.
- Counterbalancing error (books not yet closed): Debit or credit Retained Earnings for the prior-year income misstatement and adjust the offsetting income statement account in the current period. Example: for unrecorded accrued wages of \$1,500, Debit Retained Earnings \$1,500 and Credit Salaries and Wages Expense \$1,500.
- Non-counterbalancing error (equipment expensed in error, books open): Debit Equipment for original cost, Credit Retained Earnings for the understated net income from year one (net of proper prior-year depreciation), Debit Depreciation Expense for current-year amount, and Credit Accumulated Depreciation for total depreciation to date.
- Retained earnings statement presentation: Show opening retained earnings 'as reported,' then add (or subtract) the cumulative adjustment labeled 'Adjustment for retrospective change' or 'Correction of an error,' arriving at 'Retained earnings, January 1, as adjusted,' before adding current-year net income.

Critical Formulas / Calculations

Revised Depreciation After Change in Estimate = Book Value at Date of Change / Remaining Service Life. Example: Book value ¥2,000,000, original 15-year life, 5 years elapsed, revised total life 25 years → Remaining life = 20 years → Annual depreciation = ¥2,000,000 / 20 = ¥100,000.

Cumulative Effect on Retained Earnings (Policy Change, net of tax) = Sum of (Annual Income Difference under New Method - Annual Income Difference under Old Method) × (1 - Tax Rate) for all pre-presentation periods. Example (Denson SA): Cumulative pretax difference prior to 2022 = €220,000; tax effect at 40% = €88,000; net retained earnings adjustment = €132,000.

Error Impact on Net Income (single period) = Gross Error Amount × (1 - Tax Rate). Example (Selectro plc): £20,000 omitted depreciation × (1 - 0.40) = £12,000 net income overstatement; correcting entry debits Retained Earnings £12,000, Deferred Tax Liability £8,000, credits Accumulated Depreciation £20,000.

Retained Earnings Cumulative Difference (Inventory Method Change) = Ending Inventory (New Method) - Ending Inventory (Old Method) at the beginning of the earliest comparative period, adjusted for tax. Example (Lancer Company, no tax): FIFO inventory Jan 1, 2021 minus average-cost inventory Jan 1, 2021 = \$2,000 cumulative retained earnings adjustment.

Illustrative Example

Denson SA changes from cost-recovery to percentage-of-completion in 2022. Pretax income differences: prior to 2021 = €120,000 (net €72,000 after 40% tax); in 2021 = €20,000 (net €12,000). Total cumulative pretax difference entering 2022 = €140,000 (Construction in Process debit) plus 2022 difference €10,000 = €150,000... [VERIFY exact totals; illustration data partially truncated in source]. The journal entry at January 1, 2022 records: Dr Construction in Process €220,000 / Cr Deferred Tax Liability €88,000 / Cr Retained Earnings €132,000. Comparative 2021 income statements are restated to percentage-of-completion basis; Note A discloses the nature of the change, justification, per-share

impact, and the €120,000 increase in retained earnings at January 1, 2021.

Disclosures / Presentation Notes

- For a voluntary change in accounting policy, companies must disclose: (1) the nature of the change; (2) why the new policy provides more representationally faithful and relevant information; (3) the amount of adjustment for each financial statement line item affected and for basic and diluted EPS for the current and each prior period presented; and (4) the cumulative adjustment to retained earnings for periods before those presented.
- Error corrections require disclosure of: (1) the nature of the prior period error; (2) the effect on each financial statement line item and per-share amounts for each prior period presented; and (3) the cumulative effect on opening retained earnings or other equity components as of the beginning of the earliest period presented. Disclosures need not be repeated in subsequent periods.
- Changes in accounting estimates that have a material effect in the current period or are expected to affect future periods must disclose the nature of the change and the amount of the effect; estimates that are part of normal operations and are immaterial (e.g., routine bad debt adjustments) do not require separate disclosure.
- When retrospective application is impracticable, companies must disclose: (1) the fact that retrospective application was not used; (2) the reasons why it was impracticable; (3) the date from which the new policy was applied; and (4) the effect of the change on the current period only.

Potential Pitfalls

- Confusing a change in accounting policy with a change in estimate: When it is impossible to separate the two (e.g., switching depreciation methods), the entire change must be treated as a change in estimate and applied prospectively — not retrospectively. Misclassifying it as a policy change and restating prior periods is a common error that overstates the impact reported in the current period.
- Assuming that a counterbalancing error requires no correcting entry regardless of book closure status: If the books are not yet closed when a counterbalancing error is discovered, correcting entries are still required (adjusting Retained Earnings and the current-year income statement account) even though the two-year aggregate effect is zero. The counterbalancing 'no entry needed' rule applies only after books are closed and both error periods have passed.
- Treating indirect effects of a policy change the same as direct effects: Only direct effects (e.g., the inventory balance adjustment and its deferred tax consequence) are applied retrospectively. Indirect effects (e.g., higher profit-sharing expense triggered by restated income) are recognized only in the current period when the policy change is adopted; retroactive adjustment of prior periods for indirect effects is not permitted under IFRS and is also prohibited under U.S. GAAP.

Notes: The chapter text was complete and read in full (1,742 lines). Several illustration tables (e.g., Illustrations 22.1, 22.2, 22.6–22.9, 22.29, 22.30) contained numerical data columns that were not fully populated in the extracted text — specific dollar/euro/yen amounts within those tables may be partially truncated or missing. Formulas and conceptual descriptions were intact. No appendices were identified as missing. All four Learning Objectives and the IFRS vs. U.S. GAAP comparison section (Learning Objective 5) were fully present.

Chapter 23: Statement of Cash Flows

Executive Summary

Chapter 23 explains the purpose, format, and preparation of the statement of cash flows, which classifies all cash receipts and payments into operating, investing, and financing activities. The chapter demonstrates both the indirect method (reconciling net income to cash from operations) and the direct method (reporting gross operating cash receipts and disbursements), using a multi-year Tax Consultants Inc. illustration and a comprehensive Satellite Ltd. worksheet example. A central lesson is that accrual-based net income can diverge sharply from operating cash flow — a disconnect that can signal financial distress, as illustrated by the W. T. Grant bankruptcy.

Key Topics

1. Purpose and usefulness of the statement of cash flows: The statement reveals a company's ability to generate future cash, pay dividends and obligations, and explains the gap between net income and operating cash flows — information unavailable from the income statement alone.
2. Three-activity classification: Every cash flow is classified as operating (income-statement-related), investing (non-current asset transactions), or financing (liability and equity transactions), with IFRS allowing some flexibility on interest/dividend placement.
3. Indirect method for operating activities: Starts with accrual-based net income and adds back non-cash expenses (depreciation, amortization) while adjusting for working capital changes (receivables, payables, inventory) and non-operating gains/losses to arrive at net cash from operations.
4. Direct method for operating activities: Converts each income statement line to a cash basis, reporting major classes such as cash collected from customers and cash paid to suppliers; IASB encourages this method but most companies use the indirect method in practice.
5. Special adjustments: Non-cash items such as deferred taxes, share-based compensation expense, bond discount/premium amortization, equity-method income, and unrealized gains/losses on investments each require specific adjustments to reconcile net income to operating cash flow.
6. Significant non-cash transactions: Material investing/financing events that involve no cash (e.g., acquiring land via a note payable, converting bonds to shares) must be disclosed in a supplemental note or schedule rather than on the face of the statement.
7. Worksheet approach: A structured worksheet is recommended for complex entities; it lists beginning and ending balance-sheet balances, reconciles every account change with a corresponding cash flow effect, and ensures the totals agree before the formal statement is prepared.
8. IFRS vs. U.S. GAAP differences: IFRS permits more classification flexibility for interest paid/received and dividends paid/received and requires non-cash activities to appear only in notes, while U.S. GAAP mandates operating treatment for most interest/dividend items and may show non-cash activities in a separate cash flow schedule.

Important Definitions

Term	Definition
Statement of Cash Flows	A required financial statement that reports all cash receipts and cash payments during a period, classified into operating, investing, and financing activities, and reconciles the beginning and ending cash balances.
Operating Activities	Cash flows directly related to the revenues, expenses, gains, and losses that enter into the determination of net income, such as collections from customers and payments to suppliers and employees.
Investing Activities	Cash flows from acquiring and disposing of non-current assets — including property, plant, equipment, intangibles, and investments — and from making and collecting loans.
Financing Activities	Cash flows from transactions with debt and equity providers, including issuing or repurchasing shares, borrowing or repaying debt, and paying cash dividends.
Indirect Method	A presentation approach for operating activities that begins with accrual-based net income and adjusts for non-cash charges (e.g., depreciation), non-cash credits, and changes in working capital to compute net cash from operations; also called the reconciliation method.
Direct Method	A presentation approach for operating activities that reports the actual gross amounts of operating cash receipts (e.g., cash from customers) and cash disbursements (e.g., cash to suppliers), producing a condensed cash-basis income statement.
Cash Equivalents	Short-term, highly liquid investments with original maturities of three months or less that are readily convertible to known amounts of cash and carry insignificant risk of value change (e.g., Treasury bills, commercial paper, money market funds).
Significant Non-Cash Transactions	Material investing or financing events that do not involve cash flows — such as acquiring assets by issuing shares or assuming liabilities — and must be disclosed in a supplemental note rather than within the body of the statement.
Free Cash Flow	Operating cash flow minus capital expenditures and dividends; represents the discretionary cash available to management and is widely used by analysts to value shares.
COROA (Cash Operating Return on Assets)	An advanced profitability metric that divides pre-tax, pre-interest operating cash flow by total gross assets (including accumulated depreciation), intended to measure pure cash returns on all invested capital without distortion from leverage or tax rates. [VERIFY exact formula details from original source]

Accounting Treatments

- **Depreciation and amortization:** Because these non-cash charges reduce net income but not cash, they are added back to net income in the operating section under the indirect method. No separate cash flow entry is made; the actual cash outflow was recorded as an investing activity in the year the asset was purchased.
- **Gains and losses on asset disposals:** A gain on sale is deducted from net income in the operating section (to avoid double-counting with the full cash proceeds reported in investing); a loss on sale is

added back to net income (it is a non-cash income-statement charge). The complete cash proceeds from the sale appear as an investing inflow.

- Working capital changes: Increases in current assets (receivables, inventory, prepaid expenses) are deducted from net income; decreases are added. Increases in current liabilities (accounts payable, accrued liabilities) are added; decreases are deducted. These adjustments convert accrual revenues and expenses to a cash basis.
- Bond discount/premium amortization: Discount amortization increases interest expense without a cash outflow, so it is added back to net income; premium amortization decreases interest expense without a cash inflow, so it is deducted from net income in the operating section.
- Share-based compensation: The non-cash credit to Share Premium—Options is matched by a debit to Compensation Expense that reduces net income. The expense is added back to net income in operating activities; subsequent option exercises appear as a financing cash inflow.
- Non-cash investing/financing transactions (e.g., land acquired via note payable): These transactions are excluded from the body of the statement of cash flows but must be disclosed in a supplemental note or schedule. Example journal entry for disclosure: Dr. Land 60,000 / Cr. Notes Payable 60,000 — reported as a supplemental non-cash transaction.

Critical Formulas / Calculations

Net Cash from Operating Activities (Indirect) = Net Income + Non-Cash Expenses (depreciation, amortization) + Losses on Asset Sales - Gains on Asset Sales - Increases in Current Assets + Decreases in Current Assets + Increases in Current Liabilities - Decreases in Current Liabilities. Example: Net income \$134,000 + Depreciation \$21,000 + Decrease in A/R \$10,000 - Increase in prepaid \$6,000 + Increase in A/P \$35,000 = Net Cash from Operations \$194,000.

Cash Receipts from Customers (Direct) = Sales Revenue - Increase in Accounts Receivable (or + Decrease in Accounts Receivable). Example: Sales \$780,000 - Increase in A/R \$15,000 = Cash collected \$765,000.

Cash Payments to Suppliers (Direct) = Cost of Goods Sold + Increase in Inventory (or - Decrease in Inventory) - Increase in Accounts Payable (or + Decrease in Accounts Payable). Example: COGS \$450,000 + Inventory increase \$160,000 - A/P increase \$60,000 = Cash paid to suppliers \$550,000.

Cash Payments for Operating Expenses (Direct) = Operating Expenses + Increase in Prepaid Expenses (or - Decrease) - Increase in Accrued Expenses Payable (or + Decrease). Example: Op. expenses \$160,000 + Prepaid increase \$8,000 - Accrued payable increase \$20,000 = Cash paid for op. expenses \$148,000.

Change in Cash = Cash End of Period - Cash Beginning of Period = Net Cash from Operating + Investing + Financing Activities. Example: \$49,000 - \$0 = \$3,000 + \$0 + \$46,000.

Illustrative Example

Tax Consultants Inc. — Year 2 (2023): Net income \$134,000. Non-cash add-back: depreciation \$21,000. Working capital: A/R decreased \$10,000 (add), prepaid increased \$6,000 (deduct), A/P increased \$35,000 (add). Net cash from operations = \$134,000 + \$21,000 + \$10,000 - \$6,000 + \$35,000 = \$194,000. Investing: purchased land \$(70,000), building \$(200,000), equipment \$(68,000) = \$(338,000). Financing: issued bonds \$150,000, paid dividends \$(18,000) = \$132,000. Net change in cash = \$194,000 - \$338,000 + \$132,000 = -\$12,000 (cash decreased from \$49,000 to \$37,000).

Disclosures / Presentation Notes

- Significant non-cash investing and financing activities (e.g., acquiring land by issuing a long-term note, converting bonds to ordinary shares) must be disclosed in a separate supplemental note or schedule to the financial statements and excluded from the body of the cash flow statement under IFRS (IAS 7).
- Interest paid, interest received, dividends received, and dividends paid must each be separately disclosed and classified consistently from period to period; under the chapter's assignment conventions, the first three are classified as operating and dividends paid as financing (IFRS IAS 7 reference [9]).
- Income taxes paid must be disclosed separately as an operating cash flow unless they can be directly identified with a specific investing or financing activity; many entities present a supplemental line 'Income taxes paid: \$X' within or alongside the operating section.
- Companies must disclose the components of cash and cash equivalents each period; if cash equivalent definitions change or bank overdrafts are included in cash and cash equivalents (permitted under IFRS but not U.S. GAAP), that policy must be disclosed.

Potential Pitfalls

- ❑ Double-counting gains/losses: A common error is failing to remove a gain (or add back a loss) from net income when the full sale proceeds are already reported as an investing cash inflow. Leaving the gain in net income overstates operating cash flow, while leaving the loss in net income understates it.
- ❑ Netting accounts receivable incorrectly under the direct method: Using the net change in accounts receivable (net of the allowance for doubtful accounts) overstates cash received from customers and understates cash operating expenses; under the direct method each must be adjusted separately — gross receivables for cash collections and the allowance change for bad-debt expense.
- ❑ Treating all working capital changes as operating: Not all current-account movements belong in operating activities. The purchase of short-term non-trading equity investments is an investing outflow; issuance of a short-term non-trade note payable is a financing inflow; and a declared-but-unpaid dividend does not appear anywhere on the statement until actually paid.

Notes: The chapter text was fully readable through line 2200. Numeric values inside many illustration tables (e.g., specific dollar amounts in Illustrations 23.3–23.42 for Tax Consultants and Satellite Ltd.) were partially omitted from the extracted text because the PDF-to-text conversion rendered column headings but suppressed some table cell values. The Satellite Ltd. comprehensive illustration and its formal statement (Illustration 23.42) are described qualitatively but specific HK\$ subtotals beyond those explicitly stated in the text could not be verified from the extracted file. No figures or appendix graphics were recoverable from the plain-text extraction.

Chapter 24: Presentation and Disclosure in Financial Reporting

Executive Summary

Chapter 24 examines the full disclosure principle under IFRS, explaining that financial statements must report any facts significant enough to influence an informed reader's judgment. The chapter covers four primary disclosure areas—related-party transactions, subsequent events, segment reporting, and interim reporting—along with the auditor's report, management commentary, fraudulent financial reporting, and (in appendices) financial statement analysis techniques and first-time IFRS adoption procedures.

Key Topics

1. **Full Disclosure Principle:** Companies must disclose all material financial facts in the statements, notes, or supplementary information; the IASB has amended IAS 1 to stress that materiality governs what must be disclosed, and that immaterial information can actually reduce usefulness.
2. **Notes to Financial Statements:** Notes amplify and explain items in the primary statements; required disclosures include accounting policies (the first or prominent note), significant management judgments and estimation uncertainties, inventory methods, PP&E valuation and depreciation, creditor claims, equity-holder claims, contingencies, fair values, and deferred taxes.
3. **Related-Party Transactions:** Transactions where one party can significantly influence another must disclose the nature of the relationship, transaction amounts, outstanding balances and guarantees, provisions for doubtful debts, and bad-debt expenses recognized.
4. **Subsequent Events (Events After the Reporting Period):** Adjusted subsequent events (conditions existing at the statement date) require restatement of the financial statements; non-adjusted subsequent events (new conditions arising after the statement date, e.g., fire, acquisition) require footnote disclosure only, not adjustment.
5. **Segment Reporting (IFRS 8 Management Approach):** Operating segments are identified based on how management internally organizes and reviews results; a segment is reportable if it meets any of three 10% quantitative thresholds (revenue, profit/loss, assets), and reported segments must collectively cover at least 75% of consolidated external revenue.
6. **Interim Reporting (IAS 34):** IFRS requires the discrete approach—each interim period is treated as a separate accounting period using the same policies as annual reports; income taxes use an annualized effective-rate approach, and companies subject to material seasonality must supplement interim reports with 12-month rolling data.
7. **Auditor's Report and Types of Opinions:** The auditor issues an unmodified (clean) opinion when statements comply with IFRS; modified opinions include qualified (exception not material enough to invalidate the whole), adverse (exceptions so material that statements as a whole are misstated), and disclaimer (insufficient evidence to form an opinion); Key Audit Matters (KAMs) highlight the most significant judgments in the audit.
8. **Fraudulent Financial Reporting:** Defined as intentional or reckless conduct resulting in materially misleading statements; common causes include weak internal controls, management pressure for short-term results, and complex transactions subject to subjective judgment; auditing standards require explicit fraud-risk assessment.

Important Definitions

Term	Definition
Full Disclosure Principle	The requirement to report any financial facts significant enough to influence the judgment of an informed reader of the financial statements.
Accounting Policies	The specific principles, bases, conventions, rules, and practices applied by a company in preparing and presenting financial statements.
Adjusted Subsequent Event	An event occurring after the statement of financial position date but before authorization for issuance that provides additional evidence about conditions existing at the statement date; requires adjustment to the financial statements.
Non-Adjusted Subsequent Event	An event occurring after the statement of financial position date that provides evidence about conditions that did not exist at that date; requires footnote disclosure but not adjustment of the financial statements.
Operating Segment	A component of an enterprise that earns revenues and incurs expenses, whose results are regularly reviewed by the chief operating decision-maker, and for which discrete financial information is available.
Management Approach (Segment Reporting)	The IFRS method of identifying reportable segments based on how company management organizes the business internally for decision-making and resource allocation.
Discrete Approach (Interim Reporting)	The view that each interim period is a separate accounting period; expenses are recognized as incurred without anticipating or deferring costs based on expected annual results. Required by IFRS.
Integral Approach (Interim Reporting)	The view that an interim period is an integral part of the annual period; costs are allocated among periods based on expected annual activity. Required under U.S. GAAP.
Key Audit Matters (KAMs)	Matters that, in the auditor's professional judgment, were of most significance in the audit of the current period financial statements; must be described in the auditor's report.
Fraudulent Financial Reporting	Intentional or reckless conduct, whether act or omission, that results in materially misleading financial statements.
Safe Harbor Rule	A regulatory protection that shields a company from liability for an erroneous financial forecast, provided the forecast was prepared on a reasonable basis and disclosed in good faith.
Alternative (Management) Performance Measures	Financial metrics developed by management that are not defined solely by IFRS (e.g., constant-currency revenue growth, free cash flow); must be reconciled to the nearest IFRS measure when disclosed.

Accounting Treatments

- Related-party disclosures (IAS 24): No special measurement rule — amounts are reported at the values agreed between the parties; however, IFRS requires disclosure of the nature of the relationship, transaction amounts, outstanding balances, and any bad-debt provisions so users can assess whether arm's-length terms were absent.

- **Adjusted subsequent events:** The company adjusts the financial statements before issuance. Example — a customer bankruptcy discovered after year-end but stemming from pre-year-end insolvency: Dr Allowance for Doubtful Accounts / Cr Accounts Receivable for the estimated uncollectible balance.
- **Non-adjusted subsequent events:** No adjusting entry is made; instead, the company discloses in a footnote the nature and estimated financial effect (e.g., a post-year-end plant fire, a major acquisition, or a new share issue).
- **Segment reporting measurement:** Segment profit/loss and assets are measured on the same basis used in internal management reports, which need not conform to IFRS consolidation policies; companies reconcile total segment revenues, profits, and assets to the consolidated IFRS totals.
- **Interim income taxes:** Tax expense each quarter is based on the best estimate of the weighted-average annual effective tax rate applied to year-to-date pre-tax income; the accrual is adjusted in subsequent quarters if the estimated annual rate changes.
- **First-time IFRS adoption (IFRS 1):** Opening IFRS statement of financial position adjustments are recorded through retained earnings. Example — capitalizing previously expensed development costs of €500,000: Dr Intangible Assets €500,000 / Cr Retained Earnings €500,000. A warranty not recognized under prior GAAP: Dr Retained Earnings €100,000 / Cr Warranty Liability €100,000.

Critical Formulas / Calculations

10% Revenue Threshold for Reportable Segments: $\text{Segment Revenue} \geq 0.10 \times \text{Total Combined Revenue of ALL Segments}$. Example: Total combined revenue = €2,150k; threshold = €215k; any segment with revenue $\geq$ €215k is potentially reportable.

10% Profit/Loss Threshold: $|\text{Segment Profit or Loss}| \geq 0.10 \times \text{MAX}(|\text{Sum of all profitable segments}|, |\text{Sum of all loss segments}|)$. Example: Sum of profitable segments = €90k; threshold = €9k; a segment with operating profit of €10k or a loss of €10k meets this test.

10% Assets Threshold: $\text{Segment Identifiable Assets} \geq 0.10 \times \text{Total Combined Assets of ALL Segments}$. Example: Total combined assets = €970k; threshold = €97k; any segment with assets $\geq$ €97k is potentially reportable.

75% Adequacy Test: $\text{Sum of External Revenue of All Reportable Segments} \geq 0.75 \times \text{Total Consolidated External Revenue}$. Example: Reportable segments' external revenue = €2,000k; $0.75 \times €2,150k = €1,612.5k$; €2,000k $\geq$ €1,612.5k, so the test is met.

Annualized Interim Tax Rate: $\text{Interim Income Tax Expense} = (\text{Estimated Weighted-Average Annual Effective Tax Rate}) \times \text{Year-to-Date Pre-Tax Income}$. Example: Estimated annual rate = 30%; Q1 pre-tax income = €100k; Q1 tax accrual = $0.30 \times €100k = €30k$.

Illustrative Example

Konetzke AG (Practice Problem): Four segments — Badger (revenue €60k, profit €15k, assets €167k), Spartan (revenue €9k, profit €1.5k, assets €83k), Cornhusker (revenue €23k, loss €2k, assets €25k), Hawkeye (revenue €10k, profit €1k, assets €15k); total revenue = €102k. Revenue test threshold = €10.2k: Badger and Cornhusker qualify. Profit/loss test threshold = €1.75k (10% of €17.5k combined profitable segments): Badger (€15k) and Cornhusker (|−€2k|) qualify. Assets test threshold = €29k: Badger (€167k) and Spartan (€83k) qualify. Result: Three reportable segments — Badger, Spartan, Cornhusker. 75% test: €60k + €83k... [VERIFY exact segment sums match problem data since numerical cells were not fully rendered in source].

Disclosures / Presentation Notes

- Summary of Significant Accounting Policies must be presented as the first note (or in a prominent separate section) and must include: basis of preparation, all significant policies (depreciation methods, inventory costing, revenue recognition), key management judgments, and major sources of estimation uncertainty with the nature and carrying amount of affected assets/liabilities.
- Segment disclosures (IFRS 8) must include: general description of how segments were identified, segment revenues from external customers and intersegment transfers, segment profit/loss with interest, depreciation, unusual items, and tax; segment assets and liabilities; and a reconciliation of segment totals to consolidated IFRS totals. If a single customer accounts for 10% or more of company revenue, that fact and the revenue amount by segment must be disclosed.
- Subsequent events note must state the nature of each event and provide an estimate of its financial effect; if estimation is not practicable, that fact must be stated. The authorization date of the financial statements must also be disclosed so users know the end of the subsequent-events period.
- First-time IFRS adopters must present at minimum three statements of financial position, two of each other primary statement, and reconciliations of equity and total comprehensive income from previous GAAP to IFRS at both the transition date and the end of the comparative period, with explanatory notes for each material adjustment.

Potential Pitfalls

- ❑ Confusing adjusted vs. non-adjusted subsequent events: Students often adjust financial statements for all post-year-end events. Only events that provide additional evidence about conditions existing at the statement date require adjustment; events arising from entirely new conditions after year-end require disclosure only, not adjusting entries.
- ❑ Misapplying the segment 10% tests: The profit/loss test uses the greater of the absolute total of profitable segments OR the absolute total of loss segments as the denominator — NOT the net profit figure; ignoring this can cause loss segments (like a €2k loss in the example) to be incorrectly excluded when the company has large profitable segments.
- ❑ Treating interim periods as integral under IFRS: Under IFRS the discrete approach applies, meaning expenses (including fixed costs like advertising) are recognized when incurred, not allocated proportionally over the year based on annual sales projections; allocating fixed non-manufacturing costs to quarters based on expected annual sales activity is an integral approach and is not permitted under IAS 34.

Notes: The chapter file was read through offset 4500 (approximately lines 0–4500 out of the total file). This covered all four Learning Objectives and their narrative content, the GAAP comparison section, the practice problem with solution, the key terms review, and the beginning of Appendix A (Marks and Spencer specimen financial statements) and Appendix B (adidas AG specimen financial statements). The specimen financial statement tables in Appendices A and B (offsetting numerical data for M&S and adidas) were not fully captured as the figures were largely unpopulated in the extracted text (table cells showed only labels with missing values). Appendix 24A (Financial Statement Analysis) and Appendix 24B (First-Time Adoption of IFRS) were fully read and are reflected in the summary. No core conceptual content appears to have been missed.